CHAINSAW COURSES

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IIRSM Course Approval

Course Materials

Chainsaw Maintenance & Cross Cutting

Course	Chainsaw Maintenance & Cross Cutting
Provider	Chainsaw Courses Ltd · chainsawcourses.com
Delivery	Fully Online — E-Learning (self-paced)
Hours	4 hrs GLH · 2 hrs Assessment · 4 hrs Self-Study · 10 hrs TQT
CPD	5 CPD Points (IIRSM)
Threshold	Module quizzes: 100% · Final exam: 80% (45 questions)
Purpose	Training presentation & delegate materials

Contents

SECTION 1

How to Access the Live App & Admin Panel

All content unlocked for IIRSM assessors

The course is delivered through a custom-built e-learning platform. All video modules, quizzes, AI-assisted mock practice, summative examination, and certificate generation are accessible via the two routes below.

Admin / Assessor Access — All Content Unlocked

1. Go to app.chainsawcourses.com
2. Scroll to the bottom of the screen and click the 'Admin Panel' button.
3. Enter the admin credentials when prompted — Password: CHAIN26 — the full admin dashboard will appear.
4. Explore the dashboard tabs: Feedback, Backup, Storage, Policy Documents and more.
5. All Export buttons send data directly to protected Google Drive storage.
6. Click 'APP PREVIEW' in the top-right corner of the dashboard to enter the fully unlocked app.
7. All videos, features, certificates, and IIRSM-referenced content are fully accessible for review.

New Learner Journey — Standard Access Route

Experience the app exactly as a learner — sequential locking, quiz flow, and liability waiver. The app is under internal testing and has not been publicly released.

1. Log out of the admin panel (or open an incognito browser window).
2. Go to app.chainsawcourses.com and enter login code: CHAIN
3. Enter a name and email address of your choice.
4. You will be directed to the Liability Waiver — accept to enter the course.
5. Each module is locked until the previous video is fully watched and its quiz is passed at 100%.
6. On completion, the 45-question final exam unlocks. An 80% pass auto-generates the IIRSM certificate.

SECTION 2

A Note on the Training Manual

Published author · File size & access options

Published Author

Author of 'The Chainsaw Manual' (Version 1.1, July 2026) — currently sold as a standalone physical learning aid to various colleges and training providers across the UK. The manual underpins the theoretical content of this eLearning course. The first edition was published in September 2024.

The training manual — Chainsaw Maintenance & Cross Cutting: A Comprehensive Technical Manual (v1.1) — is the primary delegate learning resource. It is a richly illustrated 138-page document covering all Learning Outcomes and Assessment Criteria, including Advanced Workshop Extension modules.

File Size Notice

The manual PDF is approximately 71 MB due to high-resolution diagrams and photographs. This exceeds most standard upload limits and cannot be attached directly to this IIRSM submission.

How to obtain the manual:

Option A — Request a Digital Copy via WeTransfer (Recommended)

Email info@chainsawcourses.com to request a digital copy. A download link will be sent via WeTransfer or a similar large-file transfer service.

Option B — View within the App

The full manual is embedded within the course platform. After logging in via the Admin Access route in Section 1, click APP PREVIEW. The manual is viewable within the app at any point during the course. All supporting policy documents are also accessible via the Admin Dashboard.

Option C — Request a Physical Copy

A physical copy of the manual can be requested by contacting David Daniel at chainsawcourses.com.

SECTION 3

Learning Outcome Framework

3 Units · 6 Learning Outcomes · 22 Assessment Criteria

The course is structured across three units and six formal learning outcomes, each broken into specific assessment criteria (ACs). All examination questions in Section 7 are tagged to their LO and AC. The modules listed under each outcome are the primary digital activities through which the content is delivered.

LO1 Legal Framework & Personal Safety

Unit 1 — Occupational Standards, Health & Safety, and Risk Evaluation

Describe the statutory legal framework and personal safety requirements governing chainsaw operations.

AC 1.1 Identify employer and employee obligations under the Health and Safety at Work Act (HSWA).
AC 1.2 Explain PUWER operational parameters governing tool maintenance and operator competence.
AC 1.3 Summarise COSHH control tracking required for hazardous fuels, battery cells, lubricants, and toxic flora species.
AC 1.4 Detail CE/UKCA and global standard class markings for safety helmets, hearing protection, gloves, and Type A/C protective trousers.

Activities: PPE & First Aid · Law & Regulations

LO2 Hazard Evaluation & Emergency Protocols

Unit 1 — Occupational Standards, Health & Safety, and Risk Evaluation

Evaluate environmental hazards, risk metrics, and implement emergency protocols for chainsaw site operations.

AC 2.1 Execute a 5-step site-specific risk assessment documenting ground hazards, pedestrian proximity, and structural vulnerabilities.
AC 2.2 Formulate an emergency communication and extraction map with grid references, postcodes, access limitations, and trauma kit deployments.
AC 2.3 Identify bio-security cleaning controls necessary to stop the spread of invasive arboreal pathogens and pests.

Activities: 5 Steps To Risk Assessment · Emergency Planning Information

LO3 Chainsaw Architecture & Safety Features

Unit 2 — Power Unit Architecture, Mechanical Integrity, and Component Maintenance

Analyse the mechanical differences, design attributes, and safety features of internal combustion and battery-powered chainsaws.

AC 3.1 Describe the 2-stroke combustion cycle and determine fuel-to-oil lubrication ratios at a standard 50:1 mix.
AC 3.2 Compare the advantages and operational risks of battery-powered power units against internal combustion chainsaws.
AC 3.3 Map and explain the mechanical function of the 10 core safety features across the front, centre, and rear chainsaw architecture.

Activities: Chainsaw Safety Features · Battery Chainsaws · Chain Brake · Fuel & Oil Filters

LO4 Diagnostic, Servicing & Maintenance Procedures

Unit 2 — Power Unit Architecture, Mechanical Integrity, and Component Maintenance

Describe the diagnostic, servicing, and maintenance procedures required to sustain the structural integrity of the chainsaw cutting assembly.

AC 4.1 Detail air filter cleaning procedures and interpret spark plug electrode colour indicators (Brown / Black / White-Grey).
AC 4.2 Explain safe carburettor adjusting rules using factory Idle (LA/T), Low (L), and High (H) screw limit constraints. (Advanced Extension)
AC 4.3 Differentiate Rim and Spur drive sprockets and diagnose guidebar wear including burring, rail splaying, and thermal bluing.
AC 4.4 Identify chain pitch, gauge, and tooth shapes (Full-Chisel vs. Semi-Chisel) and calculate correct filing profile configurations.

Activities: Air Filter · Spark Plug · Cooling System · Exhaust · The Oiling System · Recoil Starter · Clutch Assembly · Sprocket · Guidebar · Chain Basics · Chain Tension · How to identify a chainsaw chain · Replacing The Chain · Chain Sharpening

LO5 Pre-Use Verification & Startup Methodologies

Unit 3 — System Startups, Operational Testing, and Processing Techniques

Implement safe pre-use verification protocols and startup methodologies for safe chainsaw deployment.

AC 5.1 Differentiate between safe cold start floor-anchor and upright knee-clamp warm start methods for Husqvarna and Stihl platforms.
AC 5.2 Perform a 4-point dynamic check assessing chain brake engagement, oil dispersion flow, chain creep at tick-over, and off-switch motor cut.

Activities: Pre-Start Checks · Starting The Chainsaw · Pre-Use Checks

LO6 Cutting Mechanics: Tension & Compression

Unit 3 — System Startups, Operational Testing, and Processing Techniques

Apply mechanical principles to resolve tension and compression forces during timber cross-cutting operations.

AC 6.1 Analyse a log setup to determine where tension and compression forces reside.
AC 6.2 Explain the physics behind pulling chains versus pushing chains during cross-cutting.
AC 6.3 Define the bar tip kickback zone and explain how to execute precise plunge bore entries safely.
AC 6.4 Differentiate cut sequences for standard logs, oversized timber, and timber under extreme tension (Toast Rack / reduction sink).
AC 6.5 Evaluate specialised branch removal, snedding, and de-limbing sequences along a felled stem. (Advanced Extension)
AC 6.6 Identify site threats with windblown windfalls, establishing escape routes and managing root plate movements. (Advanced Extension)

Activities: Kickback · Work Positioning · Cutting Basics · Tension & Compression · Releasing A Trapped Chainsaw · Bore Cutting · Oversized & Tensioned Timber · Stacking · Additional Cuts

SECTION 4

Syllabus Mapping Table

38 Modules · Learning Outcome & Assessment Criteria Alignment

The table below maps every active app module to its corresponding Learning Outcome (LO), Assessment Criteria (AC), and Assessment Method. This mirrors the alignment table at the back of the training manual.

Module Title	#	Category	LO	Assessment Criteria	Assessment Method	Manual pp.
UNIT 1 — OCCUPATIONAL STANDARDS, HEALTH & SAFETY & RISK EVALUATION						
PPE & First Aid	1	Under-pinning Knowledge	LO1	AC 1.4	Module Quiz & Final Exam	pp.10–21
5 Steps To Risk Assessment	2	Under-pinning Knowledge	LO2	AC 2.1, AC 2.3	Module Quiz & Final Exam	pp.14–15
Hazards & Risks	3	Under-pinning Knowledge	—	—	—	pp.17–23
Emergency Planning Information	4	Under-pinning Knowledge	LO2	AC 2.2	Module Quiz & Final Exam	p.16
Law & Legislation	5	Under-pinning Knowledge	LO1	AC 1.1, AC 1.2, AC 1.3	Module Quiz & Final Exam	p.18
Chainsaw Safety Features	6	Under-pinning Knowledge	LO3	AC 3.3	Module Quiz & Final Exam	pp.30–33
UNIT 2 — POWER UNIT ARCHITECTURE, MECHANICAL INTEGRITY & COMPONENT MAINTENANCE						
Battery Chainsaws	7	Maintenance	LO3	AC 3.2	Module Quiz & Final Exam	pp.34–35
Air Filter	8	Maintenance	LO4	AC 4.1	Module Quiz & Final Exam	pp.36–39
Spark Plug	9	Maintenance	LO4	AC 4.1	Module Quiz & Final Exam	pp.36–39
Cooling System	10	Maintenance	LO4	AC 4.1	Module Quiz & Final Exam	pp.36–39
Exhaust	11	Maintenance	LO4	AC 4.1	Module Quiz & Final Exam	pp.36–39
Fuel & Oil Filters	12	Maintenance	LO4	AC 4.1, AC 3.1	Module Quiz & Final Exam	pp.26–39
The Oiling System	13	Maintenance	LO4	AC 4.1	Module Quiz & Final Exam	pp.36–39
Recoil Starter	14	Maintenance	LO4	AC 4.1	Module Quiz & Final Exam	pp.36–39
Clutch Assembly	15	Maintenance	LO4	AC 4.1	Module Quiz & Final Exam	pp.36–39
Sprocket	16	Maintenance	LO4	AC 4.3	Module Quiz & Final Exam	pp.54–63
Chain Brake	17	Maintenance	LO3	AC 3.3	Module Quiz & Final Exam	pp.30–33
Guidebar	18	Maintenance	LO4	AC 4.3	Module Quiz & Final Exam	pp.54–63
Chain Basics	19	Maintenance	LO4	AC 4.4	Module Quiz & Final Exam	pp.64–81
Chain Tension	20	Maintenance	LO4	AC 4.4	Module Quiz & Final Exam	pp.64–81
How to identify a chainsaw chain	21	Maintenance	LO4	AC 4.4	Module Quiz & Final Exam	pp.64–81
Replacing The Chain	22	Maintenance	LO4	AC 4.4	Module Quiz & Final Exam	pp.64–81
Chain Sharpening	23	Maintenance	LO4	AC 4.4	Module Quiz & Final Exam	pp.64–81
UNIT 3 — SYSTEM STARTUPS, OPERATIONAL TESTING & PROCESSING TECHNIQUES						
Kickback	24	Cross Cutting	LO6	AC 6.3	Module Quiz & Final Exam	pp.82–83
Work Positioning	25	Cross Cutting	LO6	AC 6.1	Module Quiz & Final Exam	pp.104–107
Pre-Start Checks	26	Cross Cutting	LO5	AC 5.2	Module Quiz & Final Exam	pp.100–101
Starting The Chainsaw	27	Cross Cutting	LO5	AC 5.1	Module Quiz & Final Exam	pp.96–99
Pre-Use Checks	28	Cross Cutting	LO5	AC 5.2	Module Quiz & Final Exam	pp.100–101
Cutting Basics	29	Cross Cutting	LO6	AC 6.1	Module Quiz & Final Exam	pp.104–107
Tension & Compression	30	Cross Cutting	LO6	AC 6.1, AC 6.2	Module Quiz & Final Exam	pp.104–107
Releasing A Trapped Chainsaw	31	Cross Cutting	LO6	AC 6.2	Module Quiz & Final Exam	p.104
Bore Cutting	32	Cross Cutting	LO6	AC 6.3	Module Quiz & Final Exam	pp.110–111
Oversized & Tensioned Timber	33	Cross Cutting	LO6	AC 6.4	Module Quiz & Final Exam	pp.108–121
Stacking	34	Cross Cutting	LO6	AC 6.4	Module Quiz & Final Exam	pp.108–121

Module Title	#	Category	LO	Assessment Criteria	Assessment Method	Manual pp.
Additional Cuts	35	Cross Cutting	LO6	AC 6.4	Module Quiz & Final Exam	pp.108–121
ADVANCED WORKSHOP EXTENSION — SUPPLEMENTARY ASSESSMENT CRITERIA						
Carburettor Adjustment	36	Maintenance	LO4	AC 4.2	Not Assessed	pp. 40–41
Snedding & De-Limbing	37	Cross Cutting	LO6	AC 6.5	Not Assessed	pp. 118–119
Windfall & Root Plates	38	Cross Cutting	LO6	AC 6.6	Not Assessed	pp. 122–127

SECTION 5

Compliance Tools & Practical Worksheets

Available within the course app · PPE Verification · Risk Assessment · Emergency Action Plan · Bio-Security · CPD News

All practical compliance worksheets are embedded directly within the course app and are completed digitally by the learner. This approach ensures records are date-stamped, stored securely, and immediately exportable as a PDF or plain-text report for portfolio submission or regulatory inspection.

PPE Verification Checklist

Accessible from the app Inspection Checklist. The learner confirms each item of PPE — helmet, hearing protection, gloves, trousers, boots, hi-vis, and first aid kit — against its UK/international standard (EN 397, EN 352, EN ISO 11393, EN ISO 17249, EN ISO 20471, BS 8599-1). Mapped to LO1 / AC 1.4 and LO2 / AC 2.2.

Pre-Start & Pre-Use Chainsaw Inspection

Also within the app Inspection Checklist. Covers chain tension, sharpness, brake function, bar condition, oiler, fuel mix, air filter, handguards, chain catcher, exhaust, anti-vibration mounts, and controls. A 4-point dynamic check confirms brake engagement, oil spray, chain creep at idle, and off-switch cut. Mapped to LO5 / AC 5.2.

Risk Assessment, Emergency Action Plan & Bio-Security Log

A 5-step site risk assessment template, Emergency Action Plan (site grid reference, nearest A&E, first aider, muster point), and bio-security cleaning log are available via the app dashboard. Each record is retained and can be exported for IIRSM portfolio use. Mapped to LO2 / AC 2.1, 2.2, and 2.3.

News & CPD Updates

Industry news is published regularly through the app and tagged to its relevant Learning Outcome and AC. Engaging with updates directly supports the IIRSM RM&CF Ongoing Competence Assurance pillar. Accessible via the Admin Dashboard under the Updates tab.

SECTION 6

Video Transcripts

34 Modules · LO & AC Aligned · Timecoded

The transcripts below correspond to each video module in the course. They are provided for assessor and IIRSM review purposes only and are not accessible to learners within the platform. Each transcript is mapped to its Learning Outcome (LO) and Assessment Criteria (AC) and includes timecodes from the original recording.

Module 1: PPE & First Aid

LO1 · AC 1.4

Timecode	Transcript
0:01 – 0:09	In this video, we'll cover the personal protective equipment, PPE for short, and some first aid provisions you'll need for operating a chainsaw.
0:10 – 0:14	Let's start with essential personal protective equipment.
0:14 – 0:19	You'll need chainsaw protective boots with steel or composite toe caps
0:20 – 0:24	chainsaw trousers,
0:24 – 0:28	gloves
0:28 – 0:34	a suitable helmet with ear and eye protection.
0:34 – 0:42	In addition to the face shield, supplementary eye protection is recommended.
0:42 – 0:46	Chainsaw PPE can be identified with this symbol
0:46 – 0:49	and is available in four main categories.
0:49 – 1:17	Class zero 16m a second. Class one 20m a second class two 24m a second and class three 28m a second. Chainsaw protection is made up of layers of ballistic nylon, Kevlar and other synthetic materials. It is designed to clog up the sprocket to stop the chain moving, but do not assume this will always work due to the difference in chainsaw power.
1:17 – 1:30	Carry out a risk assessment to decide what level of chainsaw protection you need. For example, using more powerful chainsaws will require a higher level of protection.
1:30 – 1:33	You also need a personal first aid kit
1:34 – 1:38	with a minimum of a large trauma, wound dressing,
1:42 – 1:46	Ensure you have received training in how to use your first aid kit.
1:47 – 1:51	Have a larger, more comprehensive first aid kit nearby
1:51 – 1:56	a nominated first aider.
1:56 – 2:00	This concludes the video on PPE and first aid requirements.

Module 2: 5 Steps To Risk Assessment

LO2 · AC 2.1, AC 2.3

Timecode	Transcript
0:00 – 0:02	In this video, we'll cover
0:02 – 0:04	the five steps to risk assessment
0:04 – 0:05	Hazards, the risk
0:05 – 0:06	they pose,
0:06 – 0:08	and how to deal with them.
0:08 – 0:09	so what's the difference
0:09 – 0:11	between a hazard and a risk?
0:11 – 0:13	Well, a hazard is something
0:13 – 0:15	that causes harm.

0:15 – 0:16	the risk is how likely
0:16 – 0:18	it is to cause harm
0:19 – 0:20	For example,
0:20 – 0:21	if you're cross-cutting near
0:21 – 0:23	overhead power cables,
0:23 – 0:25	the hazard is electrocution.
0:25 – 0:26	But the risk of you
0:26 – 0:27	coming into contact with
0:27 – 0:29	those cables is very low.
0:30 – 0:32	If the cables were running on
0:32 – 0:33	or close to the floor,
0:33 – 0:35	then the risk changes to very high
0:37 – 0:38	to minimise the risk
0:38 – 0:39	of these hazards.
0:39 – 0:40	We use the five
0:40 – 0:42	steps to risk assessment.
0:42 – 0:45	Step one identify the hazard.
0:47 – 0:49	who might be harmed and how.
0:50 – 0:52	the risk of the hazard
0:52 – 0:53	and decide on precautions
0:53 – 0:55	to minimise the harm.
0:57 – 0:59	and write it down.
1:01 – 1:02	and update the risk assessment
1:02 – 1:04	if conditions change.
1:05 – 1:06	It is always advised
1:06 – 1:08	to record and discuss
1:08 – 1:10	any findings with work colleagues,
1:10 – 1:12	not only improves site safety
1:13 – 1:15	litigation issues.
1:17 – 1:19	the video on the five
1:19 – 1:20	Steps to Risk Assessment.

Module 4: Emergency Planning Information

LO2 · AC 2.2

Timecode	Transcript
0:01 – 0:04	This video is about emergency planning and information.
0:05 – 0:24	Emergency planning is vital for everyone's safety on site. In an event of a major incident occurring, you need to be prepared for the worst. The following list is not exhaustive, but highlights the more important information that needs to be recorded. The location with street address and postcode. A grid reference or what
0:24 – 0:28	three words. The nearest hospital with its phone number.
0:28 – 0:31	Site access and extraction point
0:31 – 0:35	a rendezvous point to meet the emergency services.

0:35 – 0:41	The nearest phone with signal. Emergency contact details of all personnel on site.
0:41 – 0:53	The site manager's contact. Potential helicopter landing area. The location of the team first aid kit. The nominated first aider and any useful pre-existing medical conditions.
0:53 – 1:03	Have this information available to all on site and keep it in a dry, easy to access location.
1:03 – 1:06	This concludes the video on emergency planning.

Module 5: Law & Legislation

LO1 · AC 1.1, AC 1.2, AC 1.3

Timecode	Transcript
0:01 – 0:09	In this video, we will cover health and safety law and basic regulations in relation to chainsaw maintenance and crosscutting.
0:09 – 0:21	In 1974, the Health and Safety at Work Act became enforceable by law. This required that the employer and the employees have a duty of care to maintain a safe working environment,
0:21 – 0:26	follow any training given, and wear any appropriate PPE.
0:26 – 0:36	Regulations were put in place to help employers adhere to the law, of which there are many. The main regulation for chainsaw equipment is PUWER
0:36 – 1:03	the provision and use of work equipment regulations. This requires that all equipment, including chainsaws, are well maintained, are fit for purpose and operators must be trained and competent. Organisations such as the AA, the Arboriculture Association and FISA, the Forestry Industry Safety Accord and regional forestry bodies give industry guidance for tree work.
1:03 – 1:33	We also need to adhere to various regulations covering the protection of wildlife and their habitats in order to minimise environmental damage. We can choose fuelling areas away from watercourses, use spill mats, consider using battery chainsaws and use environmentally friendly oils and fuels. And you also need to think about biosecurity measures like disinfecting PPE and equipment to ensure no cross-contamination of pests and diseases.
1:33 – 1:36	This concludes the video on the law and regulations.

Module 6: Chainsaw Safety Features

LO3 · AC 3.3

Timecode	Transcript
0:01 – 0:17	In this video, we'll look at the ten main safety features on a chainsaw. These have been designed to minimise risk and protect the user. We'll make sure that all safety features are present, intact and working properly with a brief description on each.
0:21 – 0:27	To make things easier, we'll look at the safety features in different sections.
0:27 – 0:31	Firstly, we have four at the front
0:31 – 0:35	then two in the middle
0:35 – 0:39	and four at the back.
0:39 – 0:51	If you look at the saw at the front, you can see four safety features straight away.
0:51 – 0:55	The first is the chain brake and left hand guard.
0:55 – 1:05	This stops the chain rotating and helps protect the left hand when kickback occurs. The chain brake is designed to be activated by the back of the left hand.
1:05 – 1:21	It can also be activated from inertia an abrupt change in motion. Similar to how your seatbelt works. Refer to the section on kickback for more information. As kickback is a main contributing factor for chainsaw accidents.
1:21 – 1:26	Just underneath the chain brake, we have the exhaust.
1:26 – 1:32	This directs the fumes away from the operator and reduces the noise and emissions.
1:32 – 1:37	Next we have the bar and chain combination.
1:37 – 1:46	These need to be compatible with each other and the chain should have kickback reducing features.
1:46 – 1:50	The fourth and final safety feature at the front
1:50 – 2:00	is the scabbard. This protects the user when handling the chainsaw and protects other equipment from damage when in transportation.

2:00 – 2:02	In the middle of the chainsaw,
2:02 – 2:07	we have another two safety features.
2:07 – 2:12	If we turn the saw upside down, we can now clearly see the two safety features present.
2:12 – 2:19	They are one of the anti-vibration mounts and the chain catcher.
2:19 – 2:24	These anti-vibration mounts are located throughout the saw,
2:24 – 2:30	normally a mixture of springs and rubbers to help minimise the effects of vibration.
2:30 – 2:41	Hand arm vibration syndrome or HAVS can be serious and it can lead to musculoskeletal injuries and other nervous system issues.
2:41 – 2:45	The next one is the chain catcher.
2:45 – 2:49	This catches the chain when either the chain snaps or derails.
2:49 – 3:04	The chain derailing happens more than you think. The chain catcher is made of a softer metal or plastic to grip the chain as it comes off the guidebar. As it does so, the momentum of the chain whips round to the rear of the saw.
3:04 – 3:10	This is where we'll find our next safety feature. The rear or right hand guard.
3:10 – 3:21	This protects the right hand when the chain snaps or derails. Normally on top or near to the right hand guard are the safety stickers or decals.
3:21 – 3:32	These provide safety information and warnings. Icons highlighted in blue are mandatory actions you need to take, and those in yellow are warnings.
3:32 – 3:36	Next we then have the throttle
3:36 – 3:42	This two trigger mechanism stops any unwanted depression of the trigger.
3:42 – 3:45	finally we have the on off switch.
3:45 – 3:52	This needs to be operational, allowing the user to stop the engine at any given moment.
3:52 – 4:04	If one of these safety features is missing or faulty, the chainsaw should be taken out of action and repaired immediately. The saw needs to be labeled as not safe to use until it's repaired.
4:04 – 4:08	This concludes the video on the ten safety features of the chainsaw.

Module 7: Battery Chainsaws

LO3 · AC 3.2

Timecode	Transcript
0:01 – 0:05	This video covers battery chainsaws, and their charging elements.
0:05 – 0:10	Firstly, let's look at the advantages of using battery chainsaws.
0:10 – 0:25	They can be lighter, quieter, reduce vibrations, easier to control and with no emissions to the user. There's no fuel to transport and to spill when refuelling and communication on site is easier.
0:25 – 0:29	Now let's look at some hazards with the use of battery chainsaws.
0:30 – 0:47	They can be heavier and lack sufficient power. They can malfunction and short due to water ingress and carry the risk of electric shock. The batteries need to be charged, stored and disposed of correctly and can potentially combust when charging.
0:47 – 0:50	How do you maintain the battery elements?
0:51 – 0:56	Initially, start by unplugging and removing any batteries or chargers.
0:56 – 1:08	Inspect batteries and chargers for cracks and are not damaged in any way.
1:08 – 1:16	Make sure all battery tracks and contact points are dry and clean. Use a brush or airline to clean if dirty.
1:16 – 1:24	If in any doubt, retire as they can become a potential fire hazard.
1:24 – 1:28	This concludes the video on battery, chainsaws and charging elements.

Module 8: Air Filter

LO4 · AC 4.1

Timecode	Transcript
0:01 – 0:05	In this video, we'll explore the air filter
0:05 – 0:08	First of all, where is the air filter?
0:08 – 0:11	The air filter is found under the top cover
0:11 – 0:17	towards the rear of the machine.
0:17 – 0:26	The top cover is usually attached by screws, bolts, or catches and there are usually three of them
0:26 – 0:29	What does it do?
0:29 – 0:42	The air filter keeps dust and debris out of the carburettor and helps to regulate the air fuel mixture. Air filters come in various shapes, sizes, and materials, but all do the same job.
0:42 – 0:55	How do you maintain it? Air filter removal is straightforward. Some are attached with clips or screws. Others remove easily with just a twist or pull.
0:55 – 0:59	Clean the area around the air filter and carburettor.
0:59 – 1:04	Close the butterfly valve by applying the choke setting.
1:04 – 1:15	Remove the main air filter and pre filter if present, and fill the void with a clean rag or similar. This is to stop any debris falling into the carburettor when cleaning.
1:15 – 1:21	Check the air filter for any rips, tears, holes, or any discolouration.
1:21 – 1:26	Tap out any loose dust and use a soft brush or pressurised airline to clean.
1:26 – 1:50	Pressurised air must be directed from the inside out and not too close, as this may damage the filter. Some air filters separate into two parts to aid cleaning. Some air filters can be washed with warm soapy water. Leave to completely dry before refitting. If in any doubt of the condition of the air filter, replace it as it is the lungs of your machine.
1:50 – 1:56	Screw on or twist to reinstall.
1:56 – 1:58	This concludes the video on the air filter.

Module 9: Spark Plug

LO4 · AC 4.1

Timecode	Transcript
0:02 – 0:06	This video is all about the spark
0:06 – 0:16	Where is the spark plug? The spark plug is located on top of the cylinder.
0:16 – 0:18	What does it do?
0:19 – 0:24	It produces a spark to ignite the air fuel mixture in the cylinder.
0:24 – 0:26	How do you maintain it?
0:26 – 0:29	Firstly pull the HT lead off the spark plug
0:29 – 0:35	and unscrew the spark plug using the correct sized combi-spanner or socket.
0:35 – 0:42	On some models, you may have to push through a protective rubber to get to the spark plug.
0:42 – 0:45	Now we can check a few things.
0:45 – 0:53	Some spark plugs have a screw top where the lead fits. Check for tightness.
0:53 – 1:02	Examine the white ceramic for any discolouration, cracking, or damage which may cause the spark to arc.
1:02 – 1:09	Check the compression washer for cracks or wear,
1:09 – 1:14	the electrode gap is usually set at 0.5mm.
1:14 – 1:21	Check with a feeler gauge and adjust if needed.
1:21 – 1:34	If there is an accumulation of carbon or corrosion on the electrode, gently clean with a brass wire brush. If the spark plug displays any damage, then simply replace it as they are inexpensive.

1:34 – 1:36	Check the electrode colour,
1:36 – 1:45	This should be a light brown colour. This indicates the fuel air mix is correct.
1:45 – 1:50	If it is too black or dark, it could indicate a blocked air filter
1:50 – 1:55	or too much two stroke oil in the fuel.
1:55 – 2:01	If it is too grey or white, the chainsaw is running hot and could indicate a dirty fuel filter.
2:01 – 2:04	A hole in the air filter
2:04 – 2:08	Or not enough to stroke in the fuel
2:08 – 2:10	to reinstall the spark plug,
2:10 – 2:22	Screw in by hand. Being careful not to cross thread and gently tighten with a socket. Do not over tighten as this can damage the thread washer or cylinder block.
2:22 – 2:27	Remember to reinstall the HT lead.
2:27 – 2:29	This concludes the video on the spark plug.

Module 10: Cooling System

LO4 · AC 4.1

Timecode	Transcript
0:01 – 0:06	This video is about the cooling system, which uses airflow for cooling.
0:06 – 0:15	Where is the cooling system? The cooling system is comprised of the flywheel and the cylinder fins.
0:15 – 0:23	The flywheel is found underneath the recoil side cover, and the cylinder fins can found under the main top cover.
0:23 – 0:30	What does it do? Its function is to cool the main cylinder block and prevent overheating of the engine.
0:30 – 0:45	As the flywheel spins, it draws air in up, across and through the cylinder fins, allowing a constant flow of air. Some air is also fed to the air filter.
0:45 – 0:48	How do you maintain it?
0:48 – 0:55	Unscrew the four bolts on the recoil housing, on some saws you will have to remove the top cover to access one of the bolts.
0:55 – 0:59	Check for damage and clean all vents.
0:59 – 1:09	Inspect the flywheel and cylinder fins for any cracks, dents, missing fins or build up of debris and clean them with a soft brush or airline.
1:09 – 1:16	Avoid using any metal tools on the fins as this could damage them.
1:16 – 1:23	If flywheel or cylinder fins are damaged or missing, replace immediately.
1:24 – 1:31	Before putting the cover back on. See the video on the recoil assembly.
1:31 – 1:34	This concludes the video on the cooling system.

Module 11: Exhaust

LO4 · AC 4.1

Timecode	Transcript
0:00 – 0:05	In this video, we'll look at the exhaust.
0:05 – 0:07	Where is the exhaust?
0:07 – 0:11	The exhaust is generally located at the front of the machine
0:11 – 0:13	below the chain brake.
0:13 – 0:16	What does it do?
0:16 – 0:26	The exhaust reduces the noise and emissions of the engine. It is also designed so that the fumes are pushed away from the operator.
0:26 – 0:29	How do you maintain it?

0:29 – 0:29	Check
0:29 – 0:42	the bolts are tight and secure. Examine for any dents cracks or holes. Clean any debris from in and around the exhaust area with a brush or airline.
0:42 – 0:46	Some chainsaws have an external spark arrestor.
0:46 – 0:53	Periodically remove and clean with a wire brush.
0:53 – 1:01	If the exhaust is damaged, replaced with a new one and always change the gasket.
1:01 – 1:04	This concludes the video on the exhaust.

Module 12: Fuel & Oil Filters

LO4 AC 4.1 / LO3 AC 3.1

Timecode	Transcript
0:01 – 0:05	In this video, we'll cover the fuel and oil filters.
0:05 – 0:08	We'll begin with the fuel filter.
0:08 – 0:11	So where is the fuel filter?
0:11 – 0:16	The fuel filter is found inside the fuel tank attached to a black hose.
0:16 – 0:21	There are different designs, but they all have the same function.
0:21 – 0:23	What does it do?
0:23 – 0:31	The fuel filter filters out debris in the fuel from entering the carburettor and cylinder, ensuring a clean fuel supply.
0:32 – 0:41	How do you maintain it? Firstly, empty any fuel in the tank into an approved container.
0:41 – 0:46	Use a wire hook to fish out the filter. Don't use pliers or mole grips
0:46 – 0:49	as these could damage the rubber hose.
0:49 – 1:01	Inspect for any rips, holes or damage to the filter or hose. If any defects or heavy discolouration are found, replace the filter as they are inexpensive.
1:01 – 1:18	When replacing the filter, make sure you have the new filter in arm's reach. Then just pull the old one off and pop the new one
1:18 – 1:22	Now let's move on to the oil filter.
1:22 – 1:27	The oil filter is found inside the oil tank.
1:27 – 1:37	The majority of oil filters are either fixed into the main body or located on a short hose, so cannot be removed or cleaned easily.
1:37 – 1:39	What does it do?
1:39 – 1:47	The oil filter reduces debris from entering the oiling system, ensuring clean oil is fed into the bar.
1:48 – 1:50	How do you maintain it?
1:50 – 1:54	Empty any oil in the tank into an approved container.
1:54 – 1:58	Pour in the manufacturer's recommended degreaser.
1:58 – 2:11	Swill it around. Leave it for a few minutes and swill again. Then empty the degreaser. Repeat as necessary.
2:11 – 2:18	Alternatively, the filter can be accessed by removing the clutch assembly to expose the oil pump and system.
2:18 – 2:25	Refer to the additional content section and the video on the system for more help.
2:26 – 2:32	Find the rubber hose that leads into the reservoir and gently pull through the body of the chainsaw.
2:32 – 2:39	Inspect and replace both the filter and the hose if necessary.
2:39 – 2:42	This concludes the video on the fuel and filters.

Module 13: The Oiling System	LO4 · AC 4.1
Timecode	**Transcript**
0:00 – 0:02	In this video we'll look
0:02 – 0:04	at the chain oiling system.
0:04 – 0:06	First of all, why is
0:06 – 0:07	chain oil necessary.
0:07 – 0:09	And what does it do?
0:09 – 0:10	The chain is traveling roughly
0:10 – 0:12	20m a second on the guide
0:12 – 0:14	rails of the bar.
0:14 – 0:16	This is metal on metal and creates
0:16 – 0:17	a lot of heat
0:17 – 0:18	which damages the chain,
0:18 – 0:20	sprocket and bar.
0:21 – 0:23	This is where chain oil comes in.
0:25 – 0:26	to ensure the smooth
0:26 – 0:27	running of the chain.
0:27 – 0:29	If your chain and bar
0:29 – 0:30	is not being sufficiently oiled,
0:30 – 0:32	it can lead to the bar overheating
0:32 – 0:34	and showing a bluing colour.
0:35 – 0:36	the metal of the bar
0:36 – 0:37	and chain to warp
0:37 – 0:39	and become more brittle.
0:41 – 0:42	wear to the chain,
0:42 – 0:44	leading to potential snapping.
0:45 – 0:46	The bar rails will
0:46 – 0:47	develop wear grooves.
0:49 – 0:50	it will damage the
0:58 – 1:00	the underside of the chain
1:01 – 1:03	So every time you fill up with
1:03 – 1:04	fuel or battery,
1:04 – 1:06	you need to fill up with chain oil.
1:10 – 1:11	Don't be tempted to use
1:12 – 1:14	or any other nonspecific oils.
1:14 – 1:15	These won't have
1:18 – 1:20	your chain bar and oiling system.
1:21 – 1:23	options for biodegradable oil
1:23 – 1:25	if working on a sensitive site.

1:26 – 1:28	Be aware that if left in the saw
1:28 – 1:29	for long periods,
1:29 – 1:31	some of them can start to solidify
1:31 – 1:33	and gunk up the oiling system.
1:34 – 1:36	How does the oiling system work?
1:38 – 1:40	As the crank rotates, it activates
1:40 – 1:42	a small worm gear in the oil pump.
1:44 – 1:46	from the oil tank
1:48 – 1:50	and hoses to the oiling gallery,
1:53 – 1:53	fed into the guidebar
1:53 – 1:55	through the oil hole.
2:04 – 2:05	the drive links of the chain
2:05 – 2:06	pick up the oil
2:06 – 2:08	and carry it around the bar.
2:13 – 2:15	What if your chainsaw isn't oiling?
2:15 – 2:17	How do you check and maintain it?
2:18 – 2:19	The first thing to do is
2:20 – 2:22	you have chain oil in your tank.
2:22 – 2:23	Then clean your bar
2:24 – 2:26	of any buildup of debris.
2:27 – 2:28	If you don't do this,
2:28 – 2:30	and debris will block the oil
2:31 – 2:33	to the failure of the oil pump.
2:33 – 2:34	The next item to check
2:34 – 2:36	is the oil filter.
2:36 – 2:38	The majority of all filters
2:39 – 2:40	into the main body
2:40 – 2:42	or located on a short hose,
2:42 – 2:43	so cannot be removed
2:43 – 2:45	or cleaned easily.
2:45 – 2:47	Empty any oil in the tank
2:47 – 2:49	into an approved container.
2:49 – 2:50	Pour in the manufacturer's
2:50 – 2:51	recommended degreaser.
2:51 – 2:53	Swill it around.
2:53 – 2:54	Leave it for a few minutes,
2:54 – 2:56	and swill again.
2:56 – 2:58	Then empty the degreaser.
2:58 – 2:59	Repeat as necessary.

3:04 – 3:05	Alternatively, the
3:05 – 3:06	filter can be accessed
3:06 – 3:07	by removing the clutch
3:07 – 3:09	assembly to expose the oil pump
3:09 – 3:11	and system.
3:11 – 3:12	If you're unsure on
3:12 – 3:15	how to do this, refer to the video
3:15 – 3:17	on the clutch assembly.
3:18 – 3:19	You can remove the oil pump via
3:20 – 3:22	and then inspect and clean it.
3:23 – 3:25	If in any doubt, replace it
3:25 – 3:26	to get to the filter.
3:26 – 3:27	Find the rubber hose
3:27 – 3:29	that leads into the oil reservoir
3:31 – 3:33	through the body of the chainsaw.
3:34 – 3:36	Replace as necessary.
3:36 – 3:38	Also inspect the hose
3:38 – 3:40	for any rips or holes.
3:40 – 3:42	Repeat the process in reverse
3:42 – 3:44	to reinstall the system.
3:51 – 3:52	Once you have completed
3:52 – 3:53	all these steps,
3:53 – 3:54	you should have a
3:54 – 3:56	nicely oiling chain.
3:56 – 3:58	Some chainsaws have an oil
3:58 – 3:59	adjustment screw.
4:00 – 4:02	the user to increase or decrease
4:02 – 4:03	the amount of oil
4:03 – 4:05	being fed into the guide bar.
4:05 – 4:07	It is usually located
4:07 – 4:08	on the underside of the saw
4:08 – 4:09	towards the clutch drum.
4:09 – 4:11	For example,
4:11 – 4:12	you might increase the oil
4:12 – 4:13	when cutting in dry,
4:14 – 4:15	or milling timber.
4:19 – 4:21	video on the chain oiling system.

Module 14: Recoil Starter

LO4 · AC 4.1

Timecode	Transcript
0:00 – 0:06	This video is about the recoil assembly and how to loosen and tension the pull cord.
0:06 – 0:13	Well, where is it? It is found on the opposite side of the chainsaw to the chain and bar.
0:13 – 0:15	What does it do?
0:15 – 0:19	When the handle is pulled. The flywheel is engaged by small pawls and springs,
0:19 – 0:22	Either on the flywheel,
0:22 – 0:24	or the recoil housing,
0:24 – 0:28	allowing for the starting of the engine.
0:28 – 0:30	How do you maintain it?
0:30 – 0:38	Undo the four bolts on the cover. On some saws, you will have to remove the top cover to access a hidden bolt.
0:39 – 0:47	Once off, inspect the cover and mechanism for any damage or buildup of debris and clean the vents with an airline or brush.
0:47 – 0:51	How to de-tension the spring and pull cord.
0:51 – 0:55	Firstly remove the plastic insert if there is one.
0:55 – 1:01	Pull out the cord as far as you can and with the other hand, hold the mechanism with your thumb.
1:01 – 1:06	Be careful at this point, since the cord can whip back towards your hand if you lose your grip.
1:06 – 1:20	Pinch the cord just inside the cover. Pull through the entire length and hold directly vertical. Now release your thumb grip. This should allow the spring to de-tension and the cord should be slack
1:20 – 1:27	on some saws. There will be a notch that the cord has to be placed into.
1:27 – 1:35	Pull the cord back through the hole and inspect along the length of it for any damage.
1:35 – 1:41	In particular, check the knots at both ends
1:41 – 1:46	and where the cord rubs against the handle and side cover.
1:46 – 1:58	How do you re-tension the pull cord? Take a firm grip of the cord and start to wrap it clockwise around the mechanism. You need to turn the mechanism while doing this to tension the spring properly.
1:58 – 2:00	On some saws, you will need to locate the notch
2:00 – 2:23	before you start winding. Depending on how long your cord is, you will need to complete 4 to 6 turns. Hold the mechanism with your thumb and hold the handle with your other hand. Gently allow the handle to be drawn in by the spring. To check you have the correct tension, bend the handle down to the side and allow it to spring back up into position.
2:23 – 2:35	If this doesn't happen, you will need to add another wrap or two
2:35 – 2:46	To check you haven't over tightened it. Pull the cord out as far as it will go. Hold the mechanism with your thumb and twist the mechanism clockwise.
2:46 – 2:53	If you find this hard, you can pull the cord at full length and wrap around the handle.
2:53 – 2:56	It should easily move at least a quarter to half a turn more.
2:56 – 3:09	If it doesn't, you have over tightened it and will need to release a wrap or two. Once the correct tension has been achieved, simply place back onto the saw and tighten the bolts back into place.
3:09 – 3:12	This concludes the video on the recoil assembly.

Module 15: Clutch Assembly

LO4 · AC 4.1

Timecode	Transcript
0:01 – 0:05	This video is all about the clutch assembly.
0:06 – 0:08	So where is the clutch assembly?

0:08 – 0:13	The clutch assembly is located behind the chain and bar side cover.
0:13 – 0:19	The clutch assembly has the same components, whether it's in an inboard
0:19 – 0:22	or outboard configuration.
0:22 – 0:30	It consists of the sprocket,
0:30 – 0:37	the clutch drum,
0:37 – 0:41	the needle bearing,
0:41 – 0:50	and the clutch weights and springs.
0:50 – 0:53	What does the clutch assembly do?
0:53 – 1:11	The clutch assembly engages the clutch drum and sprocket. As the clutch assembly spins. The clutch weights overcome the springs and are thrown outwards by centrifugal force. Gripping the clutch drum, turning the sprocket and therefore the chain.
1:11 – 1:15	How do you remove and maintain an inboard clutch assembly?
1:15 – 1:22	This is fairly straightforward as you only need to remove the clip and washer.
1:22 – 1:28	This allows you to remove the sprocket,
1:28 – 1:30	clutch drum
1:31 – 1:33	and needle bearing.
1:33 – 1:36	Check for any damage to the springs and weights.
1:36 – 1:40	Over time, the springs can eat and wear into the weights.
1:40 – 1:46	Check the clutch drum for any cracks bluing or any damage.
1:46 – 1:59	Check the needle bearing and apply grease as necessary. On some saws, this can be done without having to remove the clutch assembly using a small grease gun pumped into the middle.
1:59 – 2:07	If you need to remove the clutch weights and springs, you will need to remove the spark plug and insert a piston stop.
2:07 – 2:21	This will allow you to undo it with a socket or spanner. This has an opposite thread, which means you have to turn it clockwise to loosen it.
2:21 – 2:29	When reinstalling the clutch drum and sprocket, make sure the oil notch lines up with the oil pump arm
2:29 – 2:31	and make sure it is properly seated.
2:31 – 2:39	If not, this can damage the oiling system.
2:39 – 2:42	Put the sprocket back on.
2:43 – 2:46	Pop the washer on
2:46 – 2:51	and use some pliers to snap the clip back into place.
2:51 – 2:56	To remove an outboard clutch assembly is similar.
2:56 – 3:01	Remove the spark plug. Insert a piston stop
3:01 – 3:10	and using a special tool, undo the clutch assembly, turning clockwise to loosen.
3:10 – 3:21	Alternatively, you can place a combi spanner in one of the grooves and hit sharply with a hammer to shock the clutch off.
3:22 – 3:36	Once loosened, remove all the components. The clutch weights, clutch, drum, sprocket and needle bearing and perform the same maintenance checks as mentioned before.
3:36 – 3:38	This concludes the video on the clutch assembly.

Module 16: Sprocket

LO4 · AC 4.3

Timecode	Transcript
0:01 – 0:04	In this video, we'll look at the sprocket.

0:04 – 0:08	First of all, where is the sprocket?
0:09 – 0:14	The sprocket is situated on or is part of the clutch drum.
0:14 – 0:19	What does it do? The sprocket drives the chain around the bar.
0:20 – 0:29	The drive links of the chain sit in the sprocket grooves, giving the chain its drive when the clutch is activated.
0:29 – 0:33	There are two types of sprocket
0:33 – 0:37	a rim sprocket and
0:37 – 0:40	a spur sprocket.
0:40 – 0:48	A rim sprocket floats on the clutch drum and is cheaper to replace but wears quicker.
0:48 – 0:56	A spur sprocket is integral with the clutch drum and lasts longer but costs a little more.
0:56 – 1:03	How do you maintain it? Inspect the sprocket for any damage, cracks or excess wear,
1:03 – 1:06	some sprockets have wear indication lines.
1:06 – 1:10	If you can't see a continuous line, the sprocket is too worn.
1:10 – 1:18	If it doesn't, check with a gauge. The wear should be no more than 0.5mm.
1:18 – 1:30	If this is the case or it is damaged, replace immediately as a worn sprocket damages the chain and the guide bar. Watch the previous video on how to remove the sprocket from the clutch assembly.
1:30 – 1:32	This concludes the video on the sprocket.

Module 17: Chain Brake

LO3 · AC 3.3

Timecode	Transcript
0:00 – 0:04	In this video, we'll be looking at the chain brake.
0:04 – 0:06	Where is the chain brake?
0:06 – 0:10	The chain brake is located at the front of the saw,
0:11 – 0:15	and consists of the left hand guard,
0:15 – 0:19	connected via a spring to the chain brake band.
0:19 – 0:25	The chain brake band is either housed in the side cover
0:25 – 0:28	or main body of the saw.
0:28 – 0:33	Depending on if you have an inboard or outboard clutch,
0:33 – 0:35	What does it do?
0:35 – 0:40	When applied, the chain brake band tightens around the clutch drum,
0:40 – 0:51	stopping it from turning the sprocket, and therefore the chain. Chain brakes are activated manually with the back of the left hand, or are triggered by inertia.
0:51 – 1:06	How do you maintain it? When working on an outboard chain brake band, the chain brake must be deactivated to remove the side cover and once off, reactivated to help release the tension in the spring,
1:06 – 1:13	Remove a few small screws and plastic cover to access the entire band.
1:13 – 1:18	Be careful when doing this as the spring can jump out.
1:18 – 1:29	Inspect the metal band for any bluing or overheating, distortion, cracks, or any damage. In particular, at the thinnest points.
1:29 – 1:34	Clean any debris with an airline or brush and replace if damaged.
1:34 – 1:40	Add grease to keep the mechanism working smoothly.
1:40 – 1:46	Reinstallation of the spring can be tricky. It's mostly brute force, when the spring is back in place
1:46 – 2:01	the brake will be in the activated position. Put the plastic cover on and screws back in, and deactivate the chain break by using the left hand guard. This will allow you to get the side cover back on

2:01 – 2:10	To remove and inspect a chain brake band positioned around an inboard clutch is much the same. Make sure the chain brake is released.
2:10 – 2:22	Remove any screws or covers to reveal the spring and chain brake mechanism.
2:22 – 2:34	Remove the band and inspect as before and clean if necessary before installing. Keep the chain brake well maintained as this will potentially save your life.
2:34 – 2:36	This concludes the video on the chain brake.

Module 18: Guidebar

LO4 · AC 4.3

Timecode	Transcript
0:01 – 0:05	In this video, we'll be covering the guidebar.
0:05 – 0:14	First of all, it might seem obvious, but where is it? The guidebar is located at the front of the chainsaw.
0:14 – 0:16	What does it
0:16 – 0:28	The guidebar provides the rails on which the chain sits, guiding the chain around the bar. It also allows the chain to circulate effectively and helps keep optimum chain cutting efficiency.
0:29 – 0:40	How do you maintain it? First thing to check for is the straightness. Is the bar bent? If so, discard and replace immediately.
0:40 – 0:45	Check the rail heights are even with a set square.
0:45 – 0:49	Then check that the rails are not pinched,
0:49 – 0:53	splayed,
0:53 – 0:58	or overly worn.
0:58 – 1:05	You can file the rails to an even height using a flat file or a bar dresser.
1:05 – 1:10	Inspect for burring, cracking, and any bluing of the metal caused by overheating
1:10 – 1:19	Check the nose sprocket, which should turn easily. Burring is where the metal wears down and folds over the side of the bar.
1:19 – 1:32	This can be filed off with a flat file filing away from the sprocket. If not removed, the burrs can lead to larger chunks being broken off the rails
1:32 – 1:38	and finally scrape out any debris sitting in the rails dragging away from the nose sprocket.
1:38 – 1:52	Clear the oil holes and check the rail heights. They should be a minimum of 4 to 6mm in depth. When reinstalling, flip the bar occasionally for a more even wear.
1:52 – 2:08	The problems you might encounter if all or some of these issues are present are, inaccurate cutting, increased vibration, reduced oiling, overheating, and increased wear on the chain, guidebar, and sprocket.
2:08 – 2:11	This concludes the video on the guidebar.

Module 19: Chain Basics

LO4 · AC 4.4

Timecode	Transcript
0:01 – 0:05	In this video, we'll look at the chainsaw chain basics.
0:05 – 0:11	A chainsaw chain is made up of different components, each having their own function.
0:11 – 0:15	Simply, we have the drive links,
0:15 – 0:17	the cutter
0:17 – 0:25	and the tie straps and rivets.
0:25 – 0:30	The drive links sit in the sprocket
0:31 – 0:41	and as it spins, they are driven around the guide bar, picking up chain oil on their way and transporting the oil around the bar.
0:41 – 0:47	The cutters sever the timber
0:47 – 0:48	the tie straps and rivets

0:48 – 0:54	hold everything together and sit on the guide bar rails.
0:54 – 0:57	The cutter does the hard work of severing the timber.
0:57 – 1:02	So let's have an in-depth look.
1:02 – 1:06	The first part to contact the timber is the depth gauge.
1:06 – 1:14	This is sloped to reduce kickback and sets the depth of cut. Regulating the size of woodchip.
1:14 – 1:20	Too high and your cutter won't cut
1:20 – 1:21	Too low and
1:21 – 1:27	this makes the chain very aggressive and dramatically increases kickback.
1:28 – 1:30	Following right behind the depth gauge
1:30 – 1:36	is the working corner.
1:36 – 1:43	The working corner is first to start severing the timber and therefore needs to be sharp.
1:43 – 1:48	The top plate and side
1:48 – 1:57	then work together to carve out a chip of wood
1:57 – 2:04	which is then carried away in the gullet
2:04 – 2:15	Where the side and top plate meet, determines how the cutter performs and is called the cutter profile.
2:15 – 2:18	The two main profiles are full chisel
2:18 – 2:23	and semi chisel.
2:23 – 2:30	The full chisel has a 90 degree angle between the side plate and top plate.
2:30 – 2:41	It is excellent at cutting softwood, but dulls quicker in harder timber and increases the risk of kickback.
2:41 – 2:47	The semi chisel has a rounded angle where the side and top plates meet.
2:47 – 2:52	It is used for hardwood and is slower cutting, but holds its edge longer
2:53 – 2:56	It is a good choice for an all round chain, especially for those starting
2:56 – 3:01	since it reduces the risk of kickback.
3:01 – 3:03	This concludes the video on Chain Basics.

Module 20: Chain Tension

LO4 · AC 4.4

Timecode	Transcript
0:00 – 0:05	This video covers the removal, refitting and tensioning of the chain.
0:06 – 0:13	Achieving the correct chain tension is important. If the chain is too loose, there is a risk of chain derailments.
0:13 – 0:17	If too tight, it can overheat the chain and bar.
0:17 – 0:23	Both of these scenarios will increase wear on the guidebar, sprocket and chain.
0:23 – 0:26	So how do you tension the chain?
0:27 – 0:38	Firstly, release the chain brake and use a combi spanner to turn the bolts anti-clockwise a few turns until loose. Do not try to tension the chain without loosening the side cover bolts first.
0:38 – 0:43	This results in damage to the chain tensioner mechanism.
0:43 – 0:49	This is found either in the saw body or the side cover.
0:49 – 0:56	Once loosened, turn the tensioner screw clockwise or anti-clockwise to achieve desired tension.
0:56 – 1:04	The correct tension should allow the chain to rotate easily, but stop when force is released.
1:04 – 1:11	Rest the nose of the bar on the work surface, then tighten the bolts back up
1:11 – 1:15	Now let's go over the process of removing and refitting the chain and guidebar.

1:15 – 1:28	When removing the chain and bar. Remember to always release the chain brake first if you don't, and try and remove the side cover, you can pull and distort the chain brake band.
1:28 – 1:33	This only applies to chainsaws with an outboard clutch assembly.
1:33 – 1:38	Once the chain brake has been released, undo the bolts and turn the chain tensioner screw anti-clockwise
1:38 – 1:46	A few turns, then loosen the bolts all the way until you can remove the side cover.
1:46 – 1:59	Bring the guidebar towards you and push towards the drive sprocket. This will allow the chain to slacken completely. Remove the chain from the nose before removing from behind the drive sprocket.
1:59 – 2:07	Repeat the same method for an inboard clutch, but remove the chain from the drive sprocket and first
2:07 – 2:10	to reassemble the chain and guidebar with an outboard clutch.
2:11 – 2:19	Make sure you put the chain behind the clutch first and it is seated in the drive sprocket.
2:19 – 2:23	Then manipulate the chain into the nose sprocket.
2:23 – 2:28	It can be easier to have the chainsaw upright when doing this.
2:28 – 2:36	If on its side, the chain can fall out the drive sprocket and groove, making things unnecessarily annoying.
2:37 – 2:44	Put the side cover back on, making sure to line up the tensioner screw with the hole in the guidebar.
2:44 – 2:51	If it's not lining up, undo the tensioner. Screw a little and move the bar back and forth until you feel it move into place.
2:51 – 2:58	Now tighten the bolts up finger tight and turn the tensioner screw until the correct tension is achieved.
2:58 – 3:04	Once this is done, firmly tighten up the bolts.
3:04 – 3:25	If you have an inboard clutch, the only difference is to install the chain around the new sprocket first, and then manipulate the chain into the drive sprocket and make sure the guidebar is located into the chain tensioner.
3:25 – 3:29	This concludes the video on chain tension removal and refitting.

Module 21: How to Identify a Chainsaw Chain

LO4 · AC 4.4

Timecode	Transcript
0:01 – 0:06	In this video, you'll learn how to identify a chainsaw chain.
0:06 – 0:16	There are many manufacturers of chain and each have their unique markings or identification numbers. We will concentrate on the three main brands Husqvarna
0:16 – 0:17	Stihl
0:17 – 0:21	and Oregon.
0:21 – 0:31	Firstly, determine what brand of chain you have. This can be found either on the tie strap or cutter link.
0:31 – 0:33	Here we have an Oregon chain
0:34 – 0:36	and this one a Husqvarna.
0:36 – 0:54	Both of these chains are identified in the same way. If you pull the chain up or out of the guidebar, you should see a number. This number then correlates to the specific chain on the chain chart. Identify what cutter profile you have a full or semi chisel,
0:54 – 0:59	And then in this example, we'll look for a Husqvarna 93
1:00 – 1:06	its corresponding file size and top plate filing angle.
1:07 – 1:18	will also tell you what pitch and gauge the chain is
1:18 – 1:22	The chain will likely have another number below the depth gauge.
1:22 – 1:32	This number can either be the depth gauge setting or the pitch number. Don't make the mistake of using this number on the chain chart.
1:32 – 1:35	Identifying a Stihl chain is slightly different.
1:35 – 1:38	Firstly, identify the number near the depth gauge,

1:39 – 1:51	In this case six. This number refers to the chain pitch. Then lift the chain up to find the number on the drive link, in this case three. This number refers to the chain gauge.
1:51 – 1:56	Together they make the number 63.
1:56 – 2:01	Now cross-reference this number to the chain
2:01 – 2:15	It will then tell you the file size, filing angle, and size of chain.
2:15 – 2:35	Sometimes the chain pitch is stamped under the depth gauge. Identify the number on the drive link as well, and find the corresponding pitch and gauge on the chart. This will give you the filing information you need.
2:35 – 2:39	This concludes the video on how to identify a chainsaw chain.

Module 22: Replacing The Chain

LO4 · AC 4.4

Timecode	Transcript
0:01 – 0:05	In this video, we'll learn the information needed to replace the chain
0:05 – 0:12	In order to replace the chain, we'll need five pieces of information, some of which can be found on the guidebar.
0:12 – 0:20	The pitch, the gauge, the number of drive links, the length of guidebar, and the cutter profile.
0:21 – 0:29	The pitch of the chain is its physical size. This is calculated by measuring the distance between any three rivets and divide by two.
0:29 – 0:36	This measurement is important, otherwise you'll chain won't fit into your nose or drive sprocket.
0:37 – 0:48	The gauge of the chain is the thickness of the drive link. This is also an important measurement since if it's too big, it won't fit inside the rails of the guidebar,
0:48 – 0:54	if too small, you'll have excessive movement within the rails.
0:54 – 1:02	The number of drive links and length of bar are essential as guidebars have slightly different shapes.
1:02 – 1:11	And finally we have the cutter profile. This depends on the type of timber you're cutting or just your personal preference.
1:11 – 1:14	This concludes the video on replacing the chain.

Module 23: Chain Sharpening

LO4 · AC 4.4

Timecode	Transcript
0:01 – 0:06	In this video, we'll learn how to sharpen a chainsaw chain.
0:06 – 0:13	Before we begin sharpening. We need gloves and eye protection.
0:13 – 0:23	Now we need to identify what chain we have. If you're unsure how to do this, revisit the video on how to identify your chain.
0:23 – 0:28	The chain shown here is a still 23.
0:28 – 0:35	which means we need a 3/16 or 4.8mm file,
0:35 – 0:42	we're filing at a 30 degree angle.
0:42 – 0:52	We then need to select the appropriate file guide. There are many types of file guide which work in different ways, but all try and help you achieve an accurate sharpen.
0:52 – 1:05	Ideally, secure your chainsaw in a workshop vice. If you're on site, use a stump vice or similar.
1:05 – 1:11	First, find the cutting tooth that's the most damaged or smallest and begin with that one.
1:11 – 1:14	Use callipers to ensure accuracy.
1:14 – 1:19	Once you have filed that tooth, all the cutters should then be filed back to that length.
1:19 – 1:31	The reason for this is to ensure a straight even cut. Reduce vibration, reduce kickback and reduce wear on the rest of the chain guidebar and sprocket.
1:31 – 1:36	Mark where you start with a pen or similar to help you keep track of where you are.
1:36 – 1:43	You can also just look and see that once you have sharpened the tooth, it will have a fresh shine to it.

1:43 – 1:51	So let's begin by placing the file in file guide into the gullet of the tooth.
1:51 – 1:57	Ensure the line runs parallel with the chain.
1:57 – 2:08	Push forward, putting pressure down and back towards the tooth.
2:08 – 2:15	Make sure you always file into and through the working corner.
2:15 – 2:18	Do not file or drag the file back against the tooth.
2:19 – 2:23	This blunts the cutter.
2:23 – 2:29	Try to keep the file level and keep the strokes consistent.
2:29 – 2:38	Be careful not to angle the file upwards, as you can inadvertently file the tie straps.
2:38 – 2:42	Don't file past the witness mark
2:42 – 2:50	There is potential for the teeth to snap off and damage the chain.
2:50 – 2:57	You can either file all the teeth on one side first, then turn the saw around to do the other side.
2:57 – 3:11	Or you can switch between hands, filing each cutter as you go. You might only need a couple of strokes on each cutter. The aim is to get consistency throughout the chain.
3:11 – 3:25	Try to keep a good side plate angle. You want to achieve a nice hooked shape like a crest of a wave.
3:25 – 3:33	If it's too hooked. This can lead to a more aggressive cut, but it will dull faster.
3:33 – 3:43	If it's too vertical or laid back. It will struggle to cut anything. The side plate angle is arguably more important than the top plate angle.
3:43 – 3:48	Once all the teeth have been sharpened evenly, we need to look at the depth gauges.
3:48 – 3:55	The depth gauges are vital as they determine how much of the working corner and cutter are allowed to sever the timber.
3:55 – 3:59	This regulates the size of chip produced.
3:59 – 4:04	Take a depth gauge setting tool of your choice
4:04 – 4:07	and place on the chain.
4:07 – 4:14	If the depth gauge stands proud, it needs to be filed.
4:14 – 4:20	Tilt the guide at an angle. This also protects the working corner you've just sharpened.
4:20 – 4:32	Push the flat file over the depth gauge a few times until it's level with the guide, and round off the edge to ensure a smooth cut.
4:32 – 4:41	Do not file over the guide, as this will gradually lower it and therefore lower the height of the depth gauge.
4:41 – 4:59	If you use a blunt, badly damaged or poorly sharpened chain, you will encounter several problems such as overheating, decreased cutting efficiency, increased kickback and vibration, uneven cuts, and increased wear on the chain, guidebar, and sprocket.
4:59 – 5:05	So keep your chain razor sharp and you will cut safer and happier,
5:05 – 5:08	This concludes the video on how to sharpen your chain.

Module 24: Kickback

LO6 · AC 6.3

Timecode	Transcript
0:01 – 0:22	In this video, we'll be covering kickback, what it is, how it happens, and what steps you can take to avoid or reduce the risk of it happening. So what's kickback? Kickback is the violent, abrupt movement of the chainsaw, usually back towards the user, when the chain moves around the top quadrant of the guidebar and nose and comes into contact with timber,
0:22 – 0:35	the chain continues to cut. At this point, it's still a pushing chain, so pushes the saw in an upwards arc and back, kicking the chainsaw towards the operator.
0:35 – 0:48	Pinch kickback is similar. The chain continues to cut, but pushes and rolls the chainsaw out of the cut into the operator. To avoid or reduce the risk of kickback. There are several things you can do.
0:48 – 0:54	First and foremost is body position. Keep your head and neck out the way.

0:54 – 1:00	See the video on work positioning for a more in-depth guide.
1:00 – 1:04	Avoid using the upper quadrant kickback zone on the guidebar.
1:04 – 1:13	Keep your chain sharpened accurately. Choose kickback reducing chain features such as ramped depth gauges,
1:13 – 1:18	bumper drive links
1:18 – 1:22	bumper tie straps,
1:22 – 1:25	keep your chainsaw in a good working order.
1:25 – 1:34	Good habits are worth forming straight from the start. It literally might save your life.
1:34 – 1:36	This concludes the video on kickback.

Module 25: Work Positioning

LO6 · AC 6.1

Timecode	Transcript
0:00 – 0:05	In this video, we'll explain the importance of good work positioning.
0:05 – 0:16	A good work position will keep you safe. Put less strain on your body. Keeping injuries to a minimum. The immediate hazards of a poor work position are cutting yourself and kickback,
0:17 – 0:22	while the longer term hazards are backache and body pain issues.
0:22 – 0:24	For the correct work position.
0:24 – 0:32	Position the chainsaw on your right hand side with your feet pointing directly in front of you, in the same direction as the chainsaw.
0:32 – 0:45	This should ensure whatever direction your feet point in your saw should follow keeping your head and neck out the way, your left thumb should always be wrapped around the handle to give you a good grip.
0:45 – 0:49	Don't rest your thumb on top.
0:50 – 0:55	Hold the chainsaw close to yourself and use your body and legs to support some of its weight.
0:55 – 1:10	Aim to keep a straight back if the timber is low or on the floor. Crouch down and use one knee to support yourself, but avoid kneeling on both knees as this can impair your escape if the timber rolls or moves towards you,
1:11 – 1:18	Head position is vital. Do not position your head over the guidebar when cutting.
1:18 – 1:22	If you get kickback, it's going straight into your head or neck.
1:23 – 1:31	If you can see the top of the chain or the outside of the guidebar, your head is in the wrong position.
1:31 – 1:38	If you can see the inside of the guidebar, your head is in the correct position.
1:38 – 1:44	Make positioning a high priority from the start and you'll form good habits that will last a lifetime.
1:44 – 1:47	This concludes the video on work positioning.

Module 26: Pre-Start Checks

LO5 · AC 5.2

Timecode	Transcript
0:01 – 0:04	In this video we'll cover the pre start checks.
0:04 – 0:12	Before you start and use your chainsaw. A number of checks need to be done to make sure the chainsaw is safe and working correctly.
0:12 – 0:19	First off, make sure your chainsaw is topped up with the appropriately mixed fuel.
0:19 – 0:25	Remember, always use a good quality two stroke oil and the ratio is 50 to 1.
0:25 – 0:27	So in a standard five litre fuel can
0:27 – 0:34	You will need 100ml of two stroke oil.
0:34 – 0:37	Always top up the chain oil when refuelling.

0:37 – 0:38	Every time.
0:38 – 0:40	No exceptions.
0:41 – 0:47	If you're using an electric chainsaw, make sure the battery is in good condition and fully charged,
0:47 – 0:52	all safety features are present and there are no obvious defects.
0:52 – 0:55	The chain is correctly tensioned
0:56 – 1:00	and the nuts clamping the side cover are tight.
1:00 – 1:06	With those initial checks done, we can begin the process of starting the chainsaw.
1:06 – 1:09	This concludes the video on the pre start checks.

Module 27: Starting The Chainsaw

LO5 · AC 5.1

Timecode	Transcript
0:00 – 0:15	This video will show you how to start your chainsaw when the engine is cold or warm. There are only a few starting procedures for all chainsaws, but check with your manufacturers make and model just to be on the safe side.
0:15 – 0:24	Firstly ensure you have completed the pre start checks. Now let's prepare the area and yourself for starting the chainsaw.
0:24 – 0:32	Start by placing your chainsaw on relatively flat ground and clear any debris away from the chain and bar.
0:32 – 0:52	Kneel down and put your right foot on the back plate with your left hand on the handle and apply the chain brake. Keep the source snug up to your left knee. This will help keep the saw stable when you pull it over. If your chainsaw has a decompression valve, now is the time to push it in. This makes pulling the starter handle easier.
0:52 – 1:00	If your saw has got a primer bubble, press it 5 to 6 times to purge any air within the fuel system.
1:00 – 1:03	Let's cover the first starting sequence.
1:03 – 1:06	The first starting sequence is full choke,
1:06 – 1:08	half revs,
1:08 – 1:11	then idle.
1:11 – 1:22	It might have an auto return stop switch. This means that when you press the switch, it turns the saw off, but automatically returns the switch to the on position.
1:23 – 1:42	Pull the switch out and up to the full choke position. Hold the starter handle tightly and pull slowly until you feel some resistance. Pull firmly and sharply directly upwards. Do this as many times as necessary until you hear the saw cough or fire up.
1:42 – 1:43	Did you hear that?
1:43 – 1:54	If you missed this fire and keep pulling, you can flood the engine. So let's hear that again.
1:54 – 2:02	Once you hear this, push the switch down to the half revs position. Pull the starter handle again until the saw begins to run.
2:02 – 2:16	Leave the chainsaw for 1 to 2 seconds and then press the trigger. This will cancel the half rev setting and bring the revs down to an idle. Your saw should be happily idling and is now warm.
2:16 – 2:19	Now let's cover the second sequence.
2:19 – 2:21	The second sequence is choke,
2:21 – 2:25	then idle.
2:25 – 2:32	Hold down the trigger and move the switch down to the choke position. Keep your thumb on the switch as you let go of the trigger.
2:32 – 2:35	This keeps the switch in the full choke position.
2:35 – 2:49	Pull the starter handle until the saw fires up and begins to run. Leave the chainsaw for 1 to 2 seconds and press the trigger. Your saw should be idling and now be warm.
2:49 – 2:55	The third sequence is a primer only. You should have a symbol like this.

2:55 – 3:00	It will also have an auto return on off switch.
3:00 – 3:14	Press the primer bubble 5 to 6 times as necessary, and pull the starter handle until the saw fires up and starts to idle.
3:14 – 3:20	Once your chainsaw is warm, there's no need to repeat the cold start method.
3:20 – 3:24	We can perform a warm start instead.
3:24 – 3:30	Make sure the switch is in the on position
3:30 – 3:38	and the chain brake is applied. Brace the chainsaw securely between your legs and pull the starter handle across your body straight upwards.
3:38 – 3:49	If it doesn't start, the saw may be still a little cold and you may have to repeat the cold start sequence again.
3:49 – 3:55	Under no circumstances should you attempt to drop start the chainsaw.
3:55 – 4:00	This is a bad habit and uncontrolled movements can lead to injuries.
4:01 – 4:09	If you have a battery operated chainsaw, make sure the battery is in good working order, fully charged and slotted securely in place.
4:09 – 4:15	Engage the chain brake and turn the saw on.
4:15 – 4:20	Now that your chainsaw is warm and ready, we can complete the pre used checks in the next video.
4:20 – 4:23	This concludes the video on how to start your chainsaw.

Module 28: Pre-Use Checks

LO5 · AC 5.2

Timecode	Transcript
0:01 – 0:06	In this video we'll go over the pre-use checks.
0:06 – 0:11	Once your chainsaw has been started and is warm, you must complete some pre-use checks.
0:11 – 0:26	Firstly, check the chain brake is working. Disengage the chain brake with your left hand fingers, keeping your thumb around the handle. To make disengaging the chain brake easier, rest the guidebar on a piece of timber or on your upper thigh.
0:26 – 0:30	This will help with fatigue in your arms.
0:30 – 0:49	Once disengaged, squeeze the throttle and bring the saw to full revs. Let go of the throttle and half a second later. Apply the chain brake. Applying the chain brake is done by rotating the left hand forwards and straightening the right arm, forcing the back handle down.
0:49 – 0:57	Do not rev the chainsaw for long periods whilst the chain brake is engaged.
0:57 – 1:01	This will burn out your clutch and chain brake band.
1:02 – 1:06	Secondly, check that the chain is being oiled.
1:06 – 1:28	Release the chain brake and hover the nose of the guidebar over a piece of timber or stump, or something similar that will not blunt your chainsaw if mistakenly touched. Squeeze the throttle and bring to full revs over the timber. After a few more seconds, you should be able to see a faint line of oil that has been flung from the end of the bar.
1:28 – 1:33	If you can't see any oil, some maintenance will be needed.
1:33 – 1:48	Thirdly, check for chain creep. This is when the chain moves at idle without you pressing the throttle. Disengage the chain brake, squeeze the throttle and allow the chainsaw to return to idle. The chain should be static and not move.
1:48 – 1:58	If you do see the chain creeping, your chain tension might need adjusting or there might be another issue and some maintenance is needed.
1:58 – 2:11	Fourth, and finally, check the off switch works. The chainsaw should cut out and stop. This is essential for the safe use of the saw. If the switch has malfunctioned and you need to turn the saw off,
2:11 – 2:23	Apply the chain brake and move the on off switch to full choke. This should flood the engine and stall it. With all four checks successful. You are now ready to use the chainsaw.
2:23 – 2:25	This concludes the video on the pre use checks.

Module 29: Cutting Basics

LO6 · AC 6.1

Timecode	Transcript
0:01 – 0:07	This video is about the basics of cutting timber using the pushing and pulling chain.
0:07 – 0:16	As you begin to use your chainsaw, you may notice the chainsaw being pulled into the cut. This is due to the cutters biting down into the timber.
0:16 – 0:24	This is called the pulling chain and it is found on the underside of the guidebar where the chain is traveling back towards the chainsaw.
0:24 – 0:31	The pushing chain is found on top of the guidebar, where the chain is traveling away from the chainsaw.
0:31 – 0:43	Using the top of the bar will push you away from the cut. Practice cutting with both the pulling and pushing chain to get a feel for how the chainsaw moves and reacts.
0:43 – 0:47	Remember to start the chain moving before entering the timber.
0:47 – 0:56	Otherwise, the chain might stall or jam in the cut. If not secured, the chain can push or pull the timber in an uncontrolled manner.
0:56 – 1:00	This is especially the case with smaller, lighter logs.
1:01 – 1:05	Do not put your foot on top of the log to steady it.
1:05 – 1:12	Instead, move the body of the saw close to the log where the felling spikes are and put the back handle on the floor.
1:13 – 1:19	Now as you start to cut, the timber will have minimal movement, allowing for safe cutting.
1:20 – 1:29	If the log's on the floor, don't try and cut all the way through, as you can easily hit the ground and blunt your chain.
1:29 – 1:33	Instead. Roll the log over
1:33 – 1:38	and match up the cuts to sever it.
1:38 – 1:50	If you're careful of the kickback zone, you can also use the top of the bar to finish the cut.
1:52 – 2:01	This is called rip cutting. It can be useful for creating planks or other products. Specialised chains are available for this type of work.
2:01 – 2:08	Notice how the chips can be longer when rip cutting. This is because you're cutting with the grain.
2:08 – 2:13	The name cross cutting is aptly named, since you are literally cutting across the grain.

Module 30: Tension & Compression

LO6 · AC 6.1, AC 6.2

Timecode	Transcript
0:01 – 0:20	In this video, we'll explain what tension and compression is. Tension and compression is found in all timber to varying degrees. For example, if a log is supported at both ends, the compression is found on the top side of the log.
0:20 – 0:37	Cut the compression up to 30% of the timber's diameter, then cut the tension side all of the way through to meet up with your compression cut. If any part of the log is overhanging or not supported, the compression is found underneath.
0:37 – 1:05	Again, always start with the compression. Cut roughly 30%. Then finish on the tension side. Meeting up with your first cut severing the timber. Sometimes tension and compression will not be obvious, so keep an eye on the kerf. This will either open or close and you can adjust your cuts if needed. If you make the wrong decision and your saw gets stuck, go to the section on removing a trapped saw.
1:05 – 1:18	To avoid getting your chainsaw stuck in the first place. Always cut the compression first. This will reduce your saw getting trapped or pinched and reduce the likelihood of the timber splitting.
1:18 – 1:22	This concludes the section on cutting timber under tension and compression.

Module 31: Releasing A Trapped Chainsaw

LO6 · AC 6.2

Timecode	Transcript
0:01 – 0:06	This video is about how to release and remove a trapped or pinched chainsaw.

0:06 – 0:15	Misjudging the tension and compression can lead to your chainsaw being trapped or pinched within the cut. Don't panic. It can be fairly common
0:15 – 0:23	If your saw does get stuck. The first thing you need to do is turn the saw off. Make sure the chain brake is off as you may be able to roll the chainsaw out
0:23 – 0:25	the cut.
0:25 – 0:30	Do not try and pull the chainsaw out with the saw still running.
0:30 – 0:39	If the saw does come free while you're doing this, you can inadvertently rev the saw, pulling it into yourself and stumble backwards.
0:40 – 0:47	If rolling the saw out the cut doesn't work. Try and lift or push the timber up or down at one end to release the saw.
0:47 – 1:03	If it's bigger timber, you can drive wedges into the kerf to open up the cut. This should release the saw and allow for removal. If the saw is still stuck and the timber is too heavy to lift or move. You will need another chainsaw to cut you free.
1:03 – 1:09	When using another chainsaw. Cut a minimum of 300mm or 1 foot away from the trapped saw.
1:09 – 1:15	Or preferably, you cut a product's length away so you don't waste any timber.
1:16 – 1:19	This concludes the video on removing a trapped or pinched chainsaw.

Module 32: Bore Cutting

LO6 · AC 6.3

Timecode	Transcript
0:00 – 0:04	This video is about performing a bore cut.
0:04 – 0:13	The bore cut is used to cut timber when it cannot be easily accessed from one side. The most common scenario is when the timber is lying close to the floor
0:13 – 0:16	and the compression is on top,
0:16 – 0:20	cutting the tension without a bore cut will put your saw in the dirt
0:21 – 0:25	and blunt your chain.
0:25 – 0:30	Let's do it properly.
0:30 – 0:40	The first method is to use the pushing chain and the top of the guidebar to make the bore. This method is useful when the timber is low and small.
0:40 – 0:43	Start by making the compression cut on top
0:43 – 0:48	20 to 30% of the timber's diameter.
0:48 – 0:53	Begin the bore below the compression cut and use the top of the guidebar.
0:53 – 1:06	The guidebar needs to be almost vertical to avoid any kickback. Cut at full revs and once the nose of the guidebar is in, smoothly move the guidebar into the horizontal position and push all the way through the timber.
1:06 – 1:09	Don't cut upwards to meet your compression cut,
1:09 – 1:16	a strip of holding timber needs to be left uncut.
1:16 – 1:33	Once you're all the way through, cut downwards to release the tension. Be careful not to exit the cut quickly and dig into the floor. Once the tension has been removed, move the saw up and cut the holding strip of timber.
1:33 – 1:40	The second method is to use the pulling chain and the bottom of the guidebar to make the bore.
1:40 – 1:41	Again,
1:41 – 1:52	start by making the compression cut on top 20 to 30% of the timber's diameter. Begin the bore below the compression cut and use the bottom of the guidebar nose.
1:52 – 1:56	Keep the guidebar nose high to avoid any kickback.
1:56 – 2:07	Cut at full revs and once the nose is in, smoothly move the guidebar into the horizontal position and bore through the timber and out the other side.
2:07 – 2:12	When you're through, cut downwards to release the tension.
2:12 – 2:19	Once the tension has been removed, move the saw up and cut the remaining strip of timber.

2:19 – 2:21	The more you practice,
2:21 – 2:24	smoother you'll get.
2:24 – 2:26	This concludes the video on bore cutting.

Module 33: Oversized & Tensioned Timber

LO6 · AC 6.4

Timecode	Transcript
0:00 – 0:06	In this video, we'll demonstrate how to cut oversize and highly tensioned timber
0:07 – 0:13	For cutting oversized timber the easiest way is to get a larger guide bar if your saw allows.
0:13 – 0:17	If that's not an option, you can use a reduction cut.
0:17 – 0:30	Position the chainsaw over the top of the log and cut the opposite side. Now bring the saw over the top and continue cutting down to finish the cut.
0:30 – 0:37	You can also cut the timber from both sides
0:37 – 0:41	or cut one side.
0:41 – 0:44	Roll the timber
0:44 – 0:47	and finish the cut.
0:47 – 1:05	Now onto highly tensioned timber. Cutting under extreme tension can be quite hazardous as the timber can split and spring towards you or trap your chainsaw. Make sure you stand on the compression side to minimise the risk of these hazards.
1:05 – 1:18	For smaller timber, you can use the toast rack method. This involves making a series of small cuts on the compression side,
1:18 – 1:26	and once the tension has eased sufficiently, then cut through the tension.
1:26 – 1:38	For larger timber. You can cut a continually widening V shape. This should safely relieve the tension.
1:38 – 1:42	This concludes the video on how to cut oversize and highly tensioned timber.

Module 34: Stacking

LO6 · AC 6.4

Timecode	Transcript
0:01 – 0:04	This video is about lifting and stacking timber
0:04 – 0:14	Back injuries from using chainsaws and lifting and moving timber can last a lifetime. So if you can, let the machine do the work for you,
0:14 – 0:21	If you have to do the work manually, keep the saw close to your body. Use your legs, not your back.
0:21 – 0:22	And never overreach.
0:23 – 0:25	Only lift what you can safely manage.
0:25 – 0:47	Otherwise, drag, roll or use lifting aids like timber tongs or hookaroons. When stacking timber, keep the stacks waist height one metre or lower. Stack logs by their species and wedge them securely if needed. Work smart lift safe and keep your back strong for the long run. This concludes the video on lifting and stacking timber.

Module 35: Additional Cuts

LO6 · AC 6.4

Timecode	Transcript
0:00 – 0:05	In this video, we'll be covering some useful additional cuts.
0:05 – 0:15	The first one we'll cover is the step cut. The step cut is useful when the tension is on top and there's risk of the chainsaw getting taken with the timber.
0:15 – 0:26	To perform a step cut. Make the first cut underneath. On the compression side of the timber. Then make the second cut on the tension side inboard of the first cut
0:26 – 0:31	and overlap the cuts.
0:31 – 0:39	Inboard is the side of least movement. So back towards where the timber is most static.

0:39 – 0:46	Don't position the tension cut outboard of the first cut.
0:46 – 0:56	You will create a small shelf which is likely to catch the chain and take the chainsaw with the timber
0:56 – 1:07	Done on the inboard side. The small shelf will leave your chainsaw safely sitting there and the timber should fall away.
1:07 – 1:11	Another cut to utilise is the box cut.
1:11 – 1:18	For demonstration purposes. The cut has been done on the end of the log. To give you a better understanding of the mechanics and what to achieve.
1:18 – 1:24	Firstly, make the compression cut underneath.
1:24 – 1:29	Then bring the chainsaw over the opposite side and make a cut.
1:29 – 1:33	Then the top.
1:33 – 1:35	Then the near side.
1:35 – 1:48	All cuts should be roughly 20 to 30% of the timber's diameter. This should leave a square or box of uncut timber in the middle. Once this has been achieved, do the final cut on the tension side.
1:49 – 2:04	If you think there is a risk of the chainsaw being taken with the timber, just make the final cut inboard of the box. This acts like a step cut and keeps your chainsaw from being taken with the timber.
2:04 – 2:06	This concludes the video on additional cuts.

SECTION 7

Assessment Bank

150 Multiple-Choice Questions · Summative Examination Pool

How the exam works

The summative examination draws a randomised 45 questions from this 150-question bank. Questions are mapped to their Learning Outcome (LO) and Assessment Criterion (AC). Learners must achieve 80% or above to pass this final examination; every module quiz separately requires 100%. Correct answers are highlighted in orange with ' . The question bank is continuously reviewed and updated via the admin panel; the version printed here reflects the bank at the time of document generation.

Q1 LO1 · AC 1.4

What is the PRIMARY purpose of chainsaw protective trousers?

A. Provide warmth in cold conditions
B. Stop or slow a chainsaw chain on contact with the leg '
C. Improve operator visibility
D. Protect against thorns and branches only

Q2 LO1 · AC 1.4

What is the FIRST action to take when a chainsaw operator sustains a deep cut?

A. Apply direct pressure to control bleeding and call 999 '
B. Remove the chainsaw chain from the wound immediately
C. Give the casualty food and water
D. Continue working and treat the injury later

Q3 LO1 · AC 1.4

How frequently must chainsaw PPE be inspected for damage?

A. Once per year
B. Once per month
C. Before every use '
D. Only when it visibly looks worn

Q4 LO2 · AC 2.1

What is the FIRST step in the 5-step risk assessment process?

A. Record your findings
B. Identify the hazards '
C. Evaluate the risks
D. Implement control measures

Q5 LO2 · AC 2.1

Under the Management of Health and Safety at Work Regulations 1999, who must carry out a risk assessment?

A. The Health and Safety Executive
B. The local authority
C. Employers and self-employed persons '
D. Only qualified safety officers

Q6 LO2 · AC 2.1

A risk assessment must be reviewed when:

A. A new calendar year begins
B. Working conditions change significantly '
C. A manager verbally requests it
D. Staff turnover occurs

Q7 LO2 · AC 2.1

In a risk assessment, what does 'risk' specifically mean?

A. The likelihood that harm will occur from a hazard '
B. The physical object that could cause harm
C. The financial cost of an accident
D. The legal penalties for non-compliance

Q8 LO2 · AC 2.2

Before starting chainsaw work, what emergency information must be established?

A. The nearest fuel station
B. The nearest hardware store
C. The site manager's home address
D. The location of emergency services and a means to contact them '

Q9 LO2 · AC 2.2

What is the recommended approach for lone workers operating a chainsaw?

A. Check in every 4 hours
B. Check in every 8 hours
C. Check in at regular agreed intervals (e.g. every 30 minutes) '
D. Check in once per day

Q10 LO2 · AC 2.2

An emergency action card should include:

A. Personal social media contacts
B. Equipment hire costs
C. Site location, grid reference, and emergency service numbers '
D. Manufacturer warranty information

Q11 LO2 · AC 2.2

What does 'point of work risk assessment' mean?

A. An assessment carried out at the actual working location before starting work '
B. An assessment completed in the office before site arrival
C. A financial assessment of the job
D. An assessment completed after work is finished

Q12 LO1 · AC 1.1, AC 1.2

Which legislation requires work equipment to be safe and suitable for its intended use?

A. Health and Safety at Work Act 1974
B. Workplace (Health, Safety and Welfare) Regulations 1992
C. Provision and Use of Work Equipment Regulations 1998 (PUWER) '
D. Manual Handling Operations Regulations 1992

Q13 LO1 · AC 1.1, AC 1.2

Who is the primary enforcing authority for health and safety in UK forestry and arboriculture?

A. The local council
B. The Forestry Commission
C. The Environment Agency
D. The Health and Safety Executive (HSE) '

Q14 LO1 · AC 1.1, AC 1.2

The Health and Safety at Work Act 1974 places a duty on employees to:

A. Take reasonable care for their own and others' health and safety '
B. Purchase their own PPE
C. Report all incidents to the police
D. Only follow instructions given in writing

Q15 LO1 · AC 1.1, AC 1.2

PUWER stands for:

A. Personal Use of Work Equipment Regulations
B. Protection and Upkeep of Work Equipment Regulations
C. Provision and Use of Work Equipment Regulations '
D. Prevention of Unsafe Work Equipment Risks

Q16 LO3 · AC 3.3

What is the primary function of the chain brake?

A. Stop the chain rotating rapidly in the event of kickback '
B. Sharpen the chain automatically
C. Prevent the engine from starting accidentally
D. Control engine speed at idle

Q17 LO3 · AC 3.3

The chain catcher on a chainsaw is designed to:

A. Collect sawdust
B. Hold the chain during sharpening
C. Reduce chain noise
D. Protect the operator if the chain breaks or derails '

Q18 LO3 · AC 3.3

What is the purpose of the anti-vibration (AV) system on a chainsaw?

A. Reduce engine noise emissions
B. Prevent chain derailment
C. Reduce vibration transmitted from the saw to the operator's hands '
D. Improve fuel efficiency

Q19 LO3 · AC 3.3

The front hand guard on a chainsaw:

A. Protects the operator's hand from the chain and activates the chain brake '
B. Acts as a handhold during transport
C. Supports the weight of the saw
D. Prevents fuel from splashing the operator

Q20 LO4 · AC 4.1

Why is it critical to clean or replace a blocked chainsaw air filter?

A. A blocked filter makes the saw louder
B. A blocked filter improves chain speed
C. A blocked filter has no effect on performance
D. A blocked filter causes a rich mixture, poor performance and overheating '

Q21 LO4 · AC 4.1

How should a foam air filter be cleaned?

A. Washed in warm soapy water, dried fully before re-fitting '
B. Washed in petrol and left to dry
C. Scraped clean with a metal tool
D. Replaced immediately without attempting to clean

Q22 LO4 · AC 4.1

When inspecting the air filter, you should also check:

A. The bar sprocket nose
B. The spark plug gap only
C. The filter housing and gasket for damage or leaks '
D. The chain tension only

Q23 LO4 · AC 4.1

Operating with a damaged or missing air filter will most likely cause:

A. Improved engine performance
B. Lower chain speed only
C. Reduced vibration only
D. Increased engine wear due to ingested debris '

Q24 LO4 · AC 4.1

What should you assess when inspecting a chainsaw spark plug?

A. Electrode condition, gap, and look for fouling or damage '
B. Replace it after every use regardless of condition
C. Only inspect if the saw fails to start
D. Clean it with water and refit

Q25 LO4 · AC 4.1

A black, sooty spark plug electrode typically indicates:

A. A lean fuel mixture (too much air)
B. A rich fuel mixture or excess oil in the combustion chamber '
C. Correct engine operation
D. Severe overheating

Q26 LO4 · AC 4.1

What tool is needed to correctly set the spark plug electrode gap?

A. A torque wrench
B. A vernier calliper
C. A multimeter
D. A feeler gauge '

Q27 LO4 · AC 4.1

What is the risk of overtightening a spark plug during installation?

A. The plug may fall out during operation
B. The engine will run too rich
C. The cylinder head threads can be stripped or the plug body cracked '
D. The chain will derail

Q28 LO4 · AC 4.1

Why must cooling fins on a chainsaw cylinder be kept clean?

A. To improve the appearance of the saw
B. To reduce chain noise
C. To improve fuel efficiency only
D. To allow adequate airflow and prevent engine overheating '

Q29 LO4 · AC 4.1

What should be used to clean clogged cooling fins?

A. A soft brush or plastic tool to dislodge debris without damaging fins '
B. A metal scraper
C. High-pressure water directly on the engine
D. Petrol sprayed directly into the fins

Q30 LO4 · AC 4.1

Where does cooling air typically enter most chainsaw designs?

A. Through the exhaust port
B. Through the chain tensioner slot
C. Through the flywheel and air intake shroud '
D. Through the bar mounting holes

Q31 LO4 · AC 4.1

Operating with blocked cooling fins can result in:

A. Better fuel economy
B. Increased chain speed
C. Reduced vibration levels
D. Piston seizure and major engine damage '

Q32 LO4 · AC 4.1

What hazard does a blocked spark arrestor in the exhaust present?

A. Increased fuel consumption only
B. Reduced chain speed
C. Fire risk from ejected hot carbon particles '
D. Loss of chain lubrication

Q33 LO4 · AC 4.1

Signs of a failing chainsaw exhaust system include:

A. Quieter engine operation
B. Improved fuel economy
C. Reduced chain vibration
D. Exhaust leaks causing burn risk and increased operator noise exposure '

Q34 LO4 · AC 4.1

When should the spark arrestor be cleaned or replaced?

A. Per the manufacturer's schedule or when visibly blocked '
B. It is a sealed unit and never serviced
C. Only when the saw fails to start
D. After every single use

Q35 LO4 · AC 4.1

Chainsaw exhaust fumes are particularly dangerous in enclosed or confined spaces because:

A. They are brightly coloured and have a strong smell
B. They cause skin irritation only
C. Carbon monoxide is colourless and odourless and can reach lethal concentrations '
D. They corrode metal parts rapidly

Q36 LO4 · AC 4.1, AC 3.1

What is the purpose of the chainsaw fuel filter?

A. To mix fuel and bar oil
B. To store reserve fuel
C. To heat the fuel before combustion
D. To prevent debris entering the fuel system and damaging the carburettor '

Q37 LO4 · AC 4.1, AC 3.1

Bar and chain oil should have what key property to be effective?

A. Sufficient tackiness to adhere to the chain and bar during operation '
B. Be very thin to flow freely at all temperatures
C. Be identical in specification to engine oil
D. Be water-based for environmental compliance

Q38 LO4 · AC 4.1, AC 3.1

A partially blocked fuel filter will typically cause:

A. Improved fuel economy
B. Overfuelling and a rich mixture
C. Fuel starvation, causing the engine to cut out under load '
D. No noticeable change in performance

Q39 LO4 · AC 4.1, AC 3.1

How often should chainsaw fuel and bar oil filters be inspected?

A. Only during the annual service
B. Only when a fuel problem is noticed
C. Every 1,000 hours of use
D. Per the manufacturer's maintenance schedule '

Q40 LO4 · AC 4.1

What must be confirmed before starting a chainsaw using the recoil starter?

A. Chain brake is engaged, saw is secure, and area is clear of people '
B. Chain is loose for easy starting
C. Throttle is fully open
D. Bar oil reservoir is empty

Q41 LO4 · AC 4.1

If the recoil starter cord does not retract after pulling, this most likely indicates:

A. The engine is flooded
B. A broken or damaged recoil spring '
C. The chain brake is disengaged
D. Low fuel level

Q42 LO4 · AC 4.1

What is the correct ground-starting position for a chainsaw?

A. Hold the saw between your knees with the bar pointing forward
B. Balance the saw on the chain brake guard
C. Place the rear handle under your right foot, hold the front handle, bar clear of the ground '
D. Hold the saw by the bar with one hand

Q43 LO4 · AC 4.1

The chain brake must be engaged during the starting procedure:

A. Only after the engine has warmed up
B. Never — it prevents the engine from starting
C. Only when starting on steep slopes
D. At all times throughout the starting procedure '

Q44 LO4 · AC 4.1

When does the centrifugal clutch on a chainsaw typically engage the chain?

A. At idle speed only
B. Immediately on engine start
C. At a higher RPM as engine speed increases above idle '
D. Only when the throttle trigger is fully released

Q45 LO4 · AC 4.1

A worn clutch drum will most commonly result in:

A. The chain running at idle — a serious safety hazard '
B. Improved chain engagement at lower speeds
C. Reduced fuel consumption
D. Increased cutting speed

Q46 LO4 · AC 4.1

What is typically required to safely remove a chainsaw clutch assembly?

A. A standard socket set only
B. Mole grips on the flywheel
C. Only a large screwdriver
D. A clutch removal tool with the correct thread direction and a piston stop '

Q47 LO4 · AC 4.1

The clutch springs are responsible for:

- **A. Holding the clutch pads retracted at idle so the chain does not move '**
- B. Returning the throttle trigger to idle position
- C. Tensioning the guide bar
- D. Driving the oil pump

Q48 LO4 · AC 4.3

Why must a worn drive sprocket be replaced?

- A. To improve the appearance of the saw
- B. A worn sprocket reduces vibration
- C. Worn teeth improve chain engagement
- **D. To decrease wear on the chain and guidebar '**

Q49 LO4 · AC 4.3

When fitting a new chain, what else should always be inspected for wear?

- **A. The drive sprocket, as chain and sprocket wear together '**
- B. The air filter
- C. The spark plug
- D. The fuel filter

Q50 LO4 · AC 4.3

What is the maximum recommended sprocket wear before it must be replaced?

- A. 0.1mm
- B. 1.0mm
- **C. 0.5mm or as specified by the manufacturer '**
- D. 2.0mm

Q51 LO4 · AC 4.3

A rim sprocket design allows:

- A. Greater cutting accuracy
- B. Higher idle speeds
- C. Direct drive to the chain without a clutch
- **D. The rim sprocket can be replaced without replacing the clutch drum as well '**

Q52 LO3 · AC 3.3

The inertia-activated chain brake operates by:

- A. The operator pressing a dedicated button
- B. The front hand guard being pressed by the operator's wrist
- **C. A flyweight mechanism detecting rapid saw body rotation and triggering automatically '**
- D. Releasing the throttle trigger

Q53 LO3 · AC 3.3

How should the chain brake be tested in the workshop environment?

- A. Rev to full throttle and deliberately trigger kickback
- B. Check it in the workshop with the bar removed
- C. It does not require testing before use
- **D. Reassemble the saw and activate the chain brake and try to spin the chain manually '**

Q54 LO3 · AC 3.3

If the chain continues rotating when the chain brake is activated, you must:

- **A. Stop work immediately — do not use the saw until the brake is repaired '**
- B. Continue working but exercise extra care
- C. Increase the engine idle speed
- D. Tighten the guide bar nuts

Q55 LO3 · AC 3.3

What should be checked when maintaining the chain brake?

A. Oil the brake band lightly after each use to prevent corrosion
B. The brake band must be clean and free of oil or grease — replace if worn, glazed, or below minimum thickness '
C. The brake spring should be removed during long-term storage
D. Tighten the front hand guard pivot to increase braking force

Q56 LO4 · AC 4.3

Why should the guide bar be regularly rotated 180 degrees?

A. To improve chain speed
B. To change the cutting direction
C. To allow even wear on both sides of the bar rail '
D. To clear the bar oil feed holes

Q57 LO4 · AC 4.3

What should be checked in the guide bar groove during inspection?

A. Fuel tank level
B. Bar length only
C. Drive link count
D. Groove depth, rail burrs, and bar oil holes for blockage '

Q58 LO4 · AC 4.3

A guide bar with a pinched (closed or damaged) rail groove causes:

A. Reduced chain lubrication and premature bar and chain wear '
B. Improved cutting accuracy
C. Faster chain speed
D. Reduced vibration

Q59 LO4 · AC 4.3

Bar oil feed holes must be kept clear to ensure:

A. Correct chain tension is maintained
B. Correct chain brake function
C. Adequate lubrication of the chain and bar throughout cutting '
D. Engine cooling airflow

Q60 LO4 · AC 4.4

Drive links on a chainsaw chain serve to:

A. Cut through the wood
B. Reduce vibration transmission
C. Measure the chain length
D. Engage with the drive sprocket and guide bar groove to drive and guide the chain '

Q61 LO4 · AC 4.4

What is a semi-chisel chain profile good for?

A. Cutting timber under tension
B. Cutting mostly softwood with increased kickback
C. Cutting mostly hardwood with reduced kickback '
D. Cutting metal and other foreign objects

Q62 LO4 · AC 4.4

The depth gauge (raker) on a chain cutter controls:

A. Chain speed
B. Chain tension
C. The depth of cut each cutter tooth takes into the wood '
D. Bar lubrication flow

Q63 LO4 · AC 4.4

Which part of the chainsaw chain actually cuts the wood?

A. The tie straps
B. The drive links
C. The left and right cutter teeth '
D. The bumper drive links

Q64 LO4 · AC 4.4

How is correct chain tension identified on a standard chainsaw?

A. The chain hangs loosely below the bar
B. The chain cannot be moved by hand at all
C. The chain pulls slightly away from the bar but snaps back with drive links remaining in the groove '
D. Pliers are needed to move the chain along the bar

Q65 LO4 · AC 4.4

Why is operating with a chain that is too loose dangerous?

A. It reduces fuel consumption
B. It improves cutting performance
C. It reduces operating noise
D. A loose chain can derail from the bar, creating a serious injury risk '

Q66 LO4 · AC 4.4

When should chain tension be checked?

A. Before use and at regular intervals — new chains need especially frequent re-tensioning '
B. Only at the start of the working day
C. Only when the chain appears loose
D. Once per week only

Q67 LO4 · AC 4.4

A chain that is too tight will cause:

A. Improved cutting speed
B. Excessive wear on bar, chain, and sprocket, and potentially dangerous bar heating '
C. Reduced kickback risk
D. Better lubrication distribution

Q68 LO4 · AC 4.4

How do you identify a Stihl chainsaw chain?

A. Check the colour of the tie straps and cutter teeth
B. Count the number of cutters and divide by two
C. Find the number below the depth gauge and on the drive link, then cross-reference with the chain chart '
D. Look for the Stihl logo stamped on the guide bar

Q69 LO4 · AC 4.4

Where on a chainsaw chain is the identification number stamped?

A. On the tie straps
B. On the depth gauge only
C. On the drive link or on the cutting tooth below the depth gauge '
D. On the cutter tooth face only

Q70 LO4 · AC 4.4

What does the chain identification number allow you to do?

A. Set the correct chain tension
B. Determine the bar oil viscosity
C. Identify the pitch, gauge and which file size and filing angle to use '
D. Calculate the correct bar oil viscosity

Q71 LO4 · AC 4.4

How do you identify a Husqvarna chainsaw chain?

A. By the number of cutting teeth on the top of the chain
B. By the colour of the depth gauge
C. By the chain weight in grams
D. Find the part number stamped on the drive link and cross-reference with the chain identification chart '

Q72 LO4 · AC 4.4

What is the pitch of a chainsaw chain?

A. The overall length of the chain in inches
B. The width of the drive link that fits the guide bar groove
C. The average distance between three consecutive rivets divided by two '
D. The angle of the cutting tooth top plate

Q73 LO4 · AC 4.4

What is the gauge of a chainsaw chain?

A. The number of drive links in the chain
B. The thickness of the drive link that fits into the guide bar groove '
C. The distance between consecutive rivets
D. The height of the depth gauge above the cutter

Q74 LO4 · AC 4.4

What are the two main cutter profiles used on chainsaw chains?

A. Round and square
B. Semi-chisel and full-chisel '
C. Micro and macro chisel
D. Left and right profile only

Q75 LO4 · AC 4.4

When selecting a replacement chain, which three measurements must match the original?

A. Bar length, engine displacement, and chain weight
B. Chain colour, manufacturer, and number of cutters
C. Pitch, gauge, and number of drive links '
D. Cutter height, depth gauge setting, and chain age

Q76 LO4 · AC 4.4

When sharpening a cutter, the file should be held at:

A. The manufacturer-specified horizontal angle (typically 25-35 degrees) '
B. 90 degrees straight across the chain
C. 45 degrees downward into the cutter
D. Parallel to the guide bar

Q77 LO4 · AC 4.4

The depth gauge (raker) on a chain cutter:

A. Guides the file during sharpening
B. Measures chain tension directly
C. Indicates when the chain needs full replacement
D. Limits how deeply each cutter bites into the wood, controlling chip thickness '

Q78 LO4 · AC 4.4

If depth gauges become too high relative to the cutters after repeated sharpening:

A. The chain cuts more aggressively
B. Chain speed increases significantly
C. The chain cuts poorly because the cutters cannot engage the wood adequately '
D. There is no effect on cutting performance

Q79 LO4 · AC 4.4

After sharpening, all cutters on the chain must be:

A. Different lengths to compensate for uneven wood hardness
B. The same length and sharpened to the same angle for smooth, efficient cutting '
C. Longer on one side than the other
D. Set to varying depth gauge heights

Q80 LO6 · AC 6.4

What is the maximum recommended height for a freestanding timber stack without additional support?

A. 1.0 metre, as specified in the site risk assessment '
B. 3.0 metres
C. There is no recommended maximum height
D. 1.5 metres

Q81 LO6 · AC 6.4

When stacking multiple different species, what should you do?

A. Stack them randomly to save time
B. Arrange and separate species into their respective stacks '
C. Mix species to improve stability
D. Only stack one species per day

Q82 LO6 · AC 6.4

The stacking work area should be:

A. Uneven ground to give better grip to stacked timber
B. As close to felling operations as possible regardless of slope
C. On stable, clear ground away from slope run-out zones '
D. Directly beside any active felling work

Q83 LO6 · AC 6.4

Before approaching an existing timber stack for further work, you must:

A. Approach quickly to check for hazards
B. Place additional timber on top to test stability
C. Ignore the stack and focus on other nearby work
D. Visually assess the stack for signs of instability before approaching '

Q84 LO6 · AC 6.1

The recommended cutting stance for cross-cutting timber is:

A. Feet together, facing the cut directly
B. Kneeling beside the log
C. Balanced stance with feet shoulder-width apart, good footing, body weight evenly distributed '
D. Standing directly over the log

Q85 LO6 · AC 6.1

Why must the operator never stand directly in line with the bar when cross-cutting?

A. It reduces cutting efficiency
B. It is harder to see the cut line accurately
C. It increases vibration exposure to the legs
D. In the event of kickback, the bar swings towards the operator's face and upper body '

Q86 LO6 · AC 6.1

When should the operator plan and check their escape route?

A. Before starting any cut '
B. After the cut is completed
C. Only when working on steep slopes
D. Only when working in proximity to other people

Q87 LO6 · AC 6.1

Why do you need to keep your thumb wrapped around the handle?

A. To reduce hand fatigue during long cuts
B. To provide a firm secure grip '
C. To activate the throttle lock more easily
D. To improve vibration dampening

Q88 LO5 · AC 5.2

Which of the following is NOT typically part of a pre-start chainsaw check?

A. Checking fuel and bar oil levels
B. Inspecting chain brake function
C. Adjusting the carburettor idle speed '
D. Checking chain tension and sharpness

Q89 LO5 · AC 5.2

If fuel is leaking from the saw during a pre-start check, you should:

A. Stop — do not start the saw and arrange repair before use '
B. Continue but check it again at lunch
C. Use the saw quickly before the leak worsens
D. Wipe the fuel away and proceed normally

Q90 LO5 · AC 5.2

Pre-start checks should be carried out:

A. Once per week
B. Only when the operator changes
C. Before each work session and after the saw has been left unattended '
D. Only at the start of a new contract

Q91 LO5 · AC 5.2

During a pre-start chain inspection, you are checking for:

A. Number of cutting teeth only
B. Chain colour and manufacturer markings
C. Chain speed at idle
D. Sharpness, correct tension, drive links in groove, and no damaged links '

Q92 LO5 · AC 5.1

Why must the chain brake be engaged before starting a chainsaw?

A. It is not necessary to engage the brake for starting
B. To increase compression for easier starting
C. To warm the engine before operational use
D. To prevent chain rotation during the starting procedure, reducing injury risk '

Q93 LO5 · AC 5.1

The correct ground-starting position for a chainsaw is:

A. Place rear handle under right foot, hold front handle with bar clear of ground '
B. Hold front handle, place saw between the knees
C. Hang saw by the front guard and pull cord upward
D. Start with the bar held vertically

Q94 LO5 · AC 5.1

A flooded chainsaw engine is typically caused by:

A. Too little fuel reaching the engine
B. A blocked air filter
C. Excess fuel in the engine from incorrect choke or throttle use during starting '
D. Low engine compression

Q95 LO5 · AC 5.1

After starting a cold engine, why must the choke be returned to the run position?

A. The choke improves fuel efficiency during warm-up
B. Leaving the choke on causes an over-rich mixture that stalls or damages the engine '
C. The choke controls chain speed at idle
D. The choke only affects the initial starting phase

Q96 LO5 · AC 5.2

The throttle trigger interlock must be confirmed to:

A. Open the throttle smoothly when fully pressed
B. Rev the engine to maximum speed
C. Prevent throttle engagement unless the interlock is simultaneously depressed '
D. Allow the chain to move at idle speed

Q97 LO5 · AC 5.2

How is bar oil reaching the chain confirmed during a pre-use check?

A. Confirm the bar oil reservoir is completely empty before refilling
B. Remove the oil cap during operation to observe flow
C. Run the chain dry for the first 10 minutes and then check
D. Hold saw over light-coloured material and check for oil spray while revving '

Q98 LO5 · AC 5.2

Why should you check the chain is static whilst idling?

A. To confirm the chain brake is not engaged
B. To ensure the chain is not moving unintentionally '
C. To measure idle speed accuracy
D. To check bar oil is reaching the chain

Q99 LO5 · AC 5.2

If the chain brake does NOT stop the chain when tested during a pre-use check, you must:

A. Continue working but remain extra vigilant
B. Tighten the guide bar nuts and retest
C. Take the saw out of service immediately and arrange for repair '
D. Engage the brake manually with your hand as a workaround

Q100 LO6 · AC 6.1

What is the difference between a pushing chain and a pulling chain?

A. A pushing chain cuts on the upstroke; a pulling chain cuts on the downstroke
B. The pulling chain (bottom of bar) draws the saw into the cut; the pushing chain (top of bar) pushes the saw back towards the operator '
C. Pushing chains are used on petrol saws; pulling chains on battery saws
D. There is no difference — all chainsaw chains work identically

Q101 LO6 · AC 6.1

Why is it important to secure timber before cross-cutting?

A. To improve the visual accuracy of the cut line
B. To prevent the saw from overheating during the cut
C. To reduce the need for bumper spike contact
D. To ensure the timber cannot roll or move unexpectedly, which could trap the bar or endanger the operator '

Q102 LO6 · AC 6.1

What happens if the chain contacts the ground or a foreign object during cutting?

A. The saw engine stalls automatically as a safety measure
B. The chain brake engages to protect the operator
C. The chain will immediately become blunt or damaged, reducing cutting efficiency and increasing hazard '
D. Chain speed increases due to reduced resistance

Q103 LO6 · AC 6.1

When making a standard downward cross-cut, which part of the bar should you use?

A. The nose of the bar for maximum reach
B. The upper edge of the bar (pushing chain)
C. Either edge — it makes no difference
D. The lower edge of the bar (pulling chain) to draw the saw through the timber in a controlled manner '

Q104 LO6 · AC 6.1, AC 6.2

When a log is in compression on its upper face (supported at both ends), the safe cutting technique is:

A. Cut straight through from the top in a single pass
B. Cut the compression side first (from the top), then finish the cut from below '
C. Cut from the end of the log inward
D. Do not attempt to cut logs in compression

Q105 LO6 · AC 6.1, AC 6.2

When a log is in tension on its upper face (cantilevered, supported at one end), the safe technique is:

A. Cut the top (tension side) first, then finish from below
B. Cut straight through from the top in a single pass
C. Cut the compression side first (from below), then finish the cut from the top '
D. Always cut from the side to avoid all risk

Q106 LO6 · AC 6.1, AC 6.2

If a bar becomes pinched in a compression cut, what is the likely outcome?

A. The chain cuts faster to free itself
B. The chain brake activates automatically
C. The saw will safely shut down
D. The wood grips the bar, potentially stalling the saw or causing loss of control '

Q107 LO6 · AC 6.1, AC 6.2

Before any cross-cut, the operator must assess:

A. Where tension and compression zones are to determine the safe cutting sequence '
B. The colour and species of the timber only
C. Only whether the timber is wet or dry
D. Only the diameter of the timber

Q108 LO6 · AC 6.2

If the chainsaw bar becomes pinched during a cut, the FIRST action is:

A. Pull the saw repeatedly until it comes free
B. Rev the engine to maximum to force the bar free
C. Remove the chain from the bar while it remains pinched
D. Turn the saw off and assess the situation '

Q109 LO6 · AC 6.2

A safe method to release a pinched bar without power tools is:

A. Lever the kerf open using a wedge or felling lever inserted into the cut '
B. Strike the log firmly with the chainsaw body
C. Cut from a completely different direction at full speed without further assessment
D. Stand on the log and pull the saw upward with both hands

Q110 LO6 · AC 6.2

When using a felling wedge to release a trapped bar, it should be inserted:

A. Into the kerf on the compression side to open the cut '
B. Into the kerf on the tension side
C. Using the chainsaw bar itself to drive the wedge in
D. While the chain is still running at full speed

Q111 LO6 · AC 6.2

If the engine stalls while the bar is pinched in timber, you should:

A. Restart the engine immediately and force the cut through
B. Use bare hands to manually open the kerf
C. Make the saw safe and use hand tools or a second saw to release it safely '
D. Leave the saw in the timber and continue other work

Q112 LO6 · AC 6.3

Bore cutting is performed using:

A. The top of the bar in a controlled downward motion
B. The full length of the bar in a rapid thrust
C. The lower part of the bar nose, entering the wood to minimise kickback risk '
D. The chain at idle speed only

Q113 LO6 · AC 6.3

Why is bore cutting considered a higher-risk technique?

A. It consumes significantly more fuel
B. It requires a longer bar than standard cross-cutting
C. It is only a risk in wet or icy conditions
D. It uses the bar nose area where kickback is most likely to be initiated '

Q114 LO6 · AC 6.3

Before performing a bore cut, the operator must:

A. Assess wood tension and compression and confirm a clear escape route '
B. Loosen the chain for greater flexibility
C. Set the engine to idle speed for safety
D. Ensure the bar is shorter than 35cm

Q115 LO6 · AC 6.3

Bore cutting is most commonly used for:

A. Routine cross-cutting of felled timber
B. Sharpening the chain in the field
C. Cutting timber where access is limited from one side '
D. Felling trees with a diameter greater than bar length

Q116 LO6 · AC 6.4

When timber diameter exceeds the guide bar length, the correct cross-cutting technique is:

A. Fit a longer bar before attempting the cut
B. Cut at an angle to reach the centre of the log
C. Do not attempt — the timber cannot be safely cut
D. Cut from one side, then reposition and cut from the opposite side '

Q117 LO6 · AC 6.4

The key additional hazard when working with large diameter timber is:

A. Roll hazard — large logs can roll unexpectedly during or after cutting '
B. The increased weight of the chainsaw
C. Increased noise levels from the larger log
D. Excessive sawdust production

Q118 LO6 · AC 6.4

Tensioned timber creates a specific risk of:

A. The wood burning during the cutting process
B. The chain becoming magnetised
C. Sudden and unpredictable movement of log sections endangering the operator '
D. Fuel contamination from the timber

Q119 LO6 · AC 6.4

Before cutting oversized or tensioned timber, the operator must:

- **A. Thoroughly assess the timber, identify tension/compression zones, plan all cuts, and confirm escape route '**
- B. Begin cutting immediately to release the tension safely
- C. Ask a bystander to watch from a safe distance
- D. Wait 10 minutes after the timber has come to rest before cutting

Q120 LO3 · AC 3.2

What is the primary advantage of a battery-powered chainsaw over a petrol chainsaw?

- A. Greater cutting power for heavy timber
- B. Longer run time without stopping
- C. Cheaper to purchase initially
- **D. Zero exhaust emissions and lower noise levels, making it suitable for indoor or urban environments '**

Q121 LO3 · AC 3.2

How should a lithium-ion chainsaw battery be stored when not in use for an extended period?

- A. Fully discharged in a sealed plastic bag
- B. Fully charged in direct sunlight to maintain cell voltage
- **C. At approximately 40–60% charge in a cool, dry place away from extreme temperatures '**
- D. Submerged in oil to prevent corrosion

Q122 LO3 · AC 3.2

Which of the following is a key limitation of battery-powered chainsaws compared to petrol models?

- A. They cannot be fitted with a chain brake
- B. They require more PPE than petrol chainsaws
- **C. Run time is limited by battery capacity, requiring charged spare batteries for prolonged use '**
- D. They are heavier due to the engine block

Q123 LO3 · AC 3.2

What safety check is unique to a battery-powered chainsaw before use?

- **A. Verify the battery charge level and inspect the battery and terminals for damage before fitting '**
- B. Check the spark plug gap
- C. Check the fuel-to-oil mix ratio
- D. Test the recoil starter mechanism

Q124 LO1 · AC 1.1

What condition is caused by prolonged exposure to hand-arm vibration from chainsaws?

- A. Carpal tunnel syndrome only
- **B. Hand-Arm Vibration Syndrome (HAVS) '**
- C. Repetitive strain injury (RSI)
- D. Arthritis

Q125 LO1 · AC 1.1

Which of the following is an early symptom of Hand-Arm Vibration Syndrome (HAVS) that should be reported to an employer?

- A. Improved grip strength after extended use
- **B. Tingling, numbness or whitening of fingers, particularly in cold conditions '**
- C. Mild muscle soreness in the upper back only
- D. Increased sensitivity to heat in the palms

Q126 LO1 · AC 1.1

What is the MAIN risk to hearing from chainsaw operation?

- A. Temporary hearing loss only, which always recovers
- **B. Permanent noise-induced hearing loss '**
- C. Tinnitus that resolves after rest
- D. Pressure damage from exhaust fumes

Q127 LO1 · AC 1.1

Chainsaw exhaust fumes contain which dangerous gas that can build up in poorly ventilated areas?

A. Carbon dioxide (CO ,)
B. Nitrogen dioxide (NO ,)
C. Carbon monoxide (CO) '
D. Sulphur dioxide (SO ,)

Q128 LO1 · AC 1.4

What standard must a safety helmet worn during chainsaw work meet in the UK?

A. BS EN 388
B. BS EN 397 '
C. BS EN 471
D. BS EN 166

Q129 LO1 · AC 1.4

When must hearing protection be worn during chainsaw operation?

A. Only when working within 10 metres of other people
B. Only on tasks lasting over 30 minutes
C. At all times when the saw is running — chainsaw noise routinely exceeds safe exposure limits '
D. Only when instructed by a supervisor on site

Q130 LO2 · AC 2.2

Under RIDDOR 2013, which type of accident at work must be reported to the HSE?

A. Any bruise or scrape
B. Over-three-day injuries only
C. Specified injuries, dangerous occurrences, and deaths '
D. Minor cuts requiring no medical treatment

Q131 LO2 · AC 2.2

What type of fire extinguisher is safest to use on a chainsaw fuel fire?

A. Water
B. CO , or dry powder '
C. Foam only
D. Wet chemical

Q132 LO2 · AC 2.2

If a chainsaw operator collapses due to suspected carbon monoxide poisoning, what is the FIRST action?

A. Give the casualty water
B. Move the casualty to fresh air immediately '
C. Apply a tourniquet
D. Administer pure oxygen from a canister

Q133 LO2 · AC 2.2

What information should be written on a site emergency card before work begins?

A. Operator's personal insurance details
B. Grid reference or address, nearest hospital, and emergency contact numbers '
C. The operator's next of kin only
D. The local authority planning reference

Q134 LO3 · AC 3.3

What does the right-hand rear guard (back hand guard) on a chainsaw protect the operator from?

A. Fuel spray during refuelling
B. Injury to the right hand if the chain breaks or derails from the guide bar '
C. Exhaust heat on the throttle hand
D. Accidental contact with the front hand guard

Q135 LO3 · AC 3.3

Which chain brake activation mechanism fires automatically during kickback WITHOUT requiring physical contact with the front hand guard?

A. The throttle interlock mechanism
B. The rear hand guard pressure sensor
C. The inertia-activated brake, triggered by rapid rotation of the saw body '
D. The decompression valve releasing

Q136 LO3 · AC 3.3

How often should the chain brake function be checked as part of routine chainsaw operation?

A. Once per week before the first job of the week
B. Before every work session as part of pre-use checks '
C. Only after a suspected kickback incident has occurred
D. During the annual service only

Q137 LO3 · AC 3.3

If anti-vibration (AV) mounts are found to be cracked or hardened during a chainsaw inspection, the correct action is:

A. Continue using the saw but limit each session to 30 minutes
B. Apply silicone sealant to the cracks and continue monitoring
C. Do not use the saw — arrange replacement of AV mounts before further operation '
D. Only replace the mounts during the next scheduled annual service

Q138 LO4 · AC 4.4

What does the depth gauge (raker) on a chainsaw chain control?

A. The angle at which the file is held during sharpening
B. The speed at which the chain runs around the bar
C. The depth of cut — how much material each cutter tooth removes with each pass '
D. The tension of the chain on the guide bar

Q139 LO4 · AC 4.4

If depth gauges are set too low on a chainsaw chain, what is the most likely effect?

A. The chain will not cut
B. Increased vibration and kickback risk '
C. The chain will slip off the bar
D. Reduced fuel consumption

Q140 LO4 · AC 4.4

How should chain tension be checked on a chainsaw?

A. The chain should sag visibly below the bar
B. The chain should sit flush with the bar groove and snap back when pulled '
C. The chain should be tight enough that it cannot be moved by hand
D. Chain tension should only be checked when the chain is cold

Q141 LO4 · AC 4.4

What does the drive link on a chainsaw chain do?

A. Connects the cutting teeth to the tie straps
B. Fits into the bar groove and is driven by the sprocket '
C. Provides the depth gauge function
D. Attaches the chain to the clutch drum

Q142 LO4 · AC 4.4

Why must the chainsaw bar groove be cleaned regularly?

A. To improve the appearance of the bar
B. To prevent sawdust build-up that can cause the chain to jam or the bar to overheat '
C. To allow the chain to run faster
D. To prevent rust forming on the bar tip

Q143 LO4 · AC 4.4

In which direction should a correctly fitted chainsaw chain travel along the top of the bar?

- **A. Away from the operator (towards the bar tip) '**
- B. Towards the operator
- C. Either direction is acceptable
- D. The direction depends on wood species

Q144 LO5 · AC 5.1

As part of a pre-start inspection, how should the operator confirm the bar oil tank has been filled?

- A. Remove the bar and tip the saw to check for oil flow
- B. Check the oil cap is tight — no further check required
- **C. Check the oil tank level visually or with a dipstick and fill before starting '**
- D. Run the saw for 30 seconds first to prime the oiling system

Q145 LO5 · AC 5.1

During a pre-start guide bar inspection, which condition would require the bar to be taken out of service?

- A. Minor surface rust on the bar rails
- B. Sawdust packed in the oil feed hole (cleaned before use)
- **C. A cracked or bent bar, or rails worn beyond manufacturer specification '**
- D. A bar sprocket nose making occasional clicking noises

Q146 LO5 · AC 5.2

When performing a warm start on a chainsaw that has recently been running, the choke should be:

- A. Set to half-choke to enrich the warm fuel mixture
- B. Fully closed until the engine fires on the first pull
- **C. Left in the open (run) position — no choke is needed for a warm engine '**
- D. Adjusted to the manufacturer's warm-start position on the carburettor

Q147 LO5 · AC 5.2

During a pre-use check, why must the operator confirm that all controls return to their neutral position when released?

- A. To confirm fuel is reaching the carburettor correctly
- **B. To prevent the chain running unintentionally, reducing injury risk during operation '**
- C. To check the chain brake is correctly disengaged
- D. To verify engine temperature is within normal range

Q148 LO6 · AC 6.2

A log is resting on two supports with the cut point in the middle. Where is the compression zone, and how should the cut be started?

- A. Compression at the bottom — start cutting from below and cut upward
- **B. Compression at the top — start from below (tension side) so the kerf opens rather than pinches '**
- C. No compression zone exists when both ends are supported
- D. Compression at the top — start from the top to cut through it first

Q149 LO6 · AC 6.2

A log is cantilevered — supported at one end only, with the cut near the free overhanging end. Where is the compression zone?

- A. At the top of the log — start from above and cut downward through the tension zone last
- **B. At the bottom of the log — start from above (tension side) to avoid bar pinch from below '**
- C. There is no compression in an overhanging log
- D. At the top of the log — start from below so compression holds the kerf open

Q150 LO1 · AC 1.1, AC 1.2

Under the Manual Handling Operations Regulations 1992 (MHOR), where manual handling involving a risk of injury cannot be avoided, what must an employer do?

- A. Issue all workers with a manual handling certificate
- **B. Make a suitable assessment of the manual handling task and take appropriate steps to reduce the risk of injury '**
- C. Prohibit manual handling activities entirely on site

D. Ensure a written permit to lift is issued before any log is moved

SECTION 8

Supplementary Oral & Practical Mock Questions

77 Questions · Formative Practice Only — Not Formally Assessed

Important — these questions do not contribute to the pass/fail outcome

These 77 questions are oral and practical preparation aids. They do NOT form part of the summative examination, do NOT appear on the certificate, and are NOT formally assessed. Within the app, learners practise them via an AI-assisted voice or text feature that provides formative feedback.

M1 Oral / Practical · Formative Only

What are the 5 steps to risk assessment?

Key points:

- Step 1 — Identify the hazards
  Keywords: identif, hazard, danger, find the, spot the, look for, look around, things that can hurt, see any danger, look for dangers, spot dangers, see dangers, find risks, notice hazards, watch out for, be aware of, identify risks, spot risks, look around for, check for
- Step 2 — Decide who might be harmed and how
  Keywords: who might, harmed, harm, injur, affected, at risk, decide who, who might get hurt, who gets hurt, who is at risk, who could be hurt, who is in danger, affected people, people at risk, vulnerable people, who might get injured, injury risk
- Step 3 — Evaluate the risks and decide on precautions
  Keywords: evaluat, precaution, control measure, assess the risk, weigh, decide on, how bad, how to stop it, how to prevent, how risky, how dangerous, what could go wrong, chance of injury, likelihood, probability, seriousness, severity, how bad is it, think about risk
- Step 4 — Record the findings and implement them
  Keywords: record, implement, document, write, log, findings, write down what you found, write it down, document it, record it, log it, note it, keep a record, put in writing, make a note
- Step 5 — Review and update the assessment
  Keywords: review, update, revise, monitor, revisit, check again, check back later, still working, check it again, check again later, look again, go over it again, update it, keep it current, review regularly, check periodically

M2 Oral / Practical · Formative Only

Give me at least three hazards with the site and their control measures.

Key points:

- Members of public — control: exclusion zone, warning signs, lookout person
  Keywords: public, bystander, people, pedestrian, exclusion zone, other person, member of public, others nearby, lookout, banks person, banksman, people walking by, keep them away, exclusion, clear area, keep people back, nobody nearby, zone
- bystanders — control: exclusion zone, warning signs, lookout person
  Keywords: public, bystander, people, pedestrian, exclusion zone, other person, member of public, others nearby, lookout, banks person, banksman, people walking by, keep them away, exclusion, clear area, keep people back, nobody nearby, zone
- Overhead hazards — power lines — control: safe clearance, helmet, survey
  Keywords: overhead, power line, electric, cable, hung up, branch above, overhead line, deadwood, widow maker, dead wood, pylon, utility line, power cable
- dead wood — control: safe clearance, helmet, survey
  Keywords: overhead, power line, electric, cable, hung up, branch above, overhead line, deadwood, widow maker, dead wood, pylon, utility line, power cable
- hung-up branches — control: safe clearance, helmet, survey
  Keywords: overhead, power line, electric, cable, hung up, branch above, overhead line, deadwood, widow maker, dead wood, pylon, utility line, power cable
- Uneven, slippery — control: grip boots, escape route, uphill side
  Keywords: uneven, slippery, slope, terrain, muddy, wet, steep, ground condition, unstable ground, grip, footing, slippery mud, uneven ground, watch where you step, escape, get away, run, clear path, exit
- steep terrain — control: grip boots, escape route, uphill side
  Keywords: uneven, slippery, slope, terrain, muddy, wet, steep, ground condition, unstable ground, grip, footing, slippery mud, uneven ground, watch where you step, escape, get away, run, clear path, exit
- Underground services — buried pipes — control: utility map, CAT scan, hand dig
  Keywords: underground, buried, pipe, service, cable below, buried cable, dig, cat scanner, utility map, buried service
- cables — control: utility map, CAT scan, hand dig
  Keywords: underground, buried, pipe, service, cable below, buried cable, dig, cat scanner, utility map, buried service
- Adverse weather — wind — control: monitor forecast, stop work
  Keywords: weather, wind, rain, lightning, visibility, frost, ice, storm, snow, gale, weather condition, bad weather, heavy rain, strong winds, stop working
- rain — control: monitor forecast, stop work
  Keywords: weather, wind, rain, lightning, visibility, frost, ice, storm, snow, gale, weather condition, bad weather, heavy rain, strong winds, stop working
- lightning — control: monitor forecast, stop work
  Keywords: weather, wind, rain, lightning, visibility, frost, ice, storm, snow, gale, weather condition, bad weather, heavy rain, strong winds, stop working
- poor visibility — control: monitor forecast, stop work
  Keywords: weather, wind, rain, lightning, visibility, frost, ice, storm, snow, gale, weather condition, bad weather, heavy rain, strong winds, stop working

- Vehicular — control: road signs, cones, traffic management, hi-vis
  Keywords: traffic, vehicle, car, lorry, truck, road, passing vehicle, machinery, moving vehicle, road user
- site traffic — control: road signs, cones, traffic management, hi-vis
  Keywords: traffic, vehicle, car, lorry, truck, road, passing vehicle, machinery, moving vehicle, road user
- Livestock, cattle, sheep — control: clear area, liaise with landowner
  Keywords: livestock, cattle, sheep, animal, dog, horse, cow, farm animal, landowner
- aggressive animals — control: clear area, liaise with landowner
  Keywords: livestock, cattle, sheep, animal, dog, horse, cow, farm animal, landowner
- Stinging insects — bees — control: avoid nests, carry epipen if allergic
  Keywords: bee, wasp, hornet, insect, sting, nest, swarm, hornet nest, wasp nest, bugs
- wasps — control: avoid nests, carry epipen if allergic
  Keywords: bee, wasp, hornet, insect, sting, nest, swarm, hornet nest, wasp nest, bugs
- hornets — control: avoid nests, carry epipen if allergic
  Keywords: bee, wasp, hornet, insect, sting, nest, swarm, hornet nest, wasp nest, bugs
- Metal, wire — control: metal detector, visual inspection
  Keywords: metal, nail, wire, embedded, fence post, staple, metal in tree, barbed wire, hidden metal
- nails embedded in trees — control: metal detector, visual inspection
  Keywords: metal, nail, wire, embedded, fence post, staple, metal in tree, barbed wire, hidden metal
- Holes, trenches, ditches, water — control: mark hazards, avoid edges
  Keywords: hole, trench, ditch, pond, river, water, well, drop, cliff, edge, unstable soil, land slip, hidden holes
- unstable ground — control: mark hazards, avoid edges
  Keywords: hole, trench, ditch, pond, river, water, well, drop, cliff, edge, unstable soil, land slip, hidden holes
- Tripping hazards — stumps — control: clear site, good housekeeping
  Keywords: trip, stumps, rock, bramble, debris, clutter, housekeeping, tangle, branch on ground, tripping over
- rocks — control: clear site, good housekeeping
  Keywords: trip, stumps, rock, bramble, debris, clutter, housekeeping, tangle, branch on ground, tripping over
- brambles — control: clear site, good housekeeping
  Keywords: trip, stumps, rock, bramble, debris, clutter, housekeeping, tangle, branch on ground, tripping over
- debris — control: clear site, good housekeeping
  Keywords: trip, stumps, rock, bramble, debris, clutter, housekeeping, tangle, branch on ground, tripping over
- Fire risk — dry tinder — control: clear debris, carry extinguisher
  Keywords: fire, tinder, dry, spark, fire risk, extinguisher, flammable debris, wildfire, combustible
- sparks — control: clear debris, carry extinguisher
  Keywords: fire, tinder, dry, spark, fire risk, extinguisher, flammable debris, wildfire, combustible
- Nearby structures, fences, walls — control: felling direction away from structures
  Keywords: structure, fence, wall, building, property, shed, barn, roof, nearby building, which way, direction, fall direction, lean direction
- property — control: felling direction away from structures
  Keywords: structure, fence, wall, building, property, shed, barn, roof, nearby building, which way, direction, fall direction, lean direction
- Dust, spores — control: RPE, dust mask
  Keywords: dust, pore, pollen, spore, mould, respiratory, rpe, dust mask, airborne
- pollen — control: RPE, dust mask
  Keywords: dust, pore, pollen, spore, mould, respiratory, rpe, dust mask, airborne
- Restricted escape routes — control: clear two escape paths at 45 degrees
  Keywords: escape route, escape path, exit route, escape, retreat, evacuation route, 45 degree, escape, get away, run, clear path, exit
- Sun — control: drink water, wear hat, take breaks
  Keywords: too hot, sun, sunstroke, heat exhaustion, overheating, drink water, wear hat, hot sun
- heat exposure — control: drink water, wear hat, take breaks
  Keywords: too hot, sun, sunstroke, heat exhaustion, overheating, drink water, wear hat, hot sun

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M3

Oral / Practical · Formative Only

Give me at least three hazards with the chainsaw and their control measures.

Key points:

- Kickback — sudden upward/backward motion of bar — control: two-handed grip, chain brake, no tip cutting
  Keywords: kickback, kick back, kick-back, bar nose, tip contact, kickback zone, saw kicking back, jumps back, bar jumping, two hands, both hands, firm grip, hold firmly
- Cuts from the moving chain — control: chainsaw PPE, trousers, gloves, boots
  Keywords: cut, chain contact, bar contact, laceration, chainsaw trousers, cut resistant, blade contact, ppe, protective clothing, sharp chain cutting you, tough pants, chainsaw trousers
- Fuel spillage — control: cool engine before refuel, no smoking, move 3m before start
  Keywords: fuel, spillage, fire, ignit, flammable, petrol, fuel spill, refuel, fuel leak, fuel cap, two stroke, spilling gas, catch fire
- fire risk — control: cool engine before refuel, no smoking, move 3m before start
  Keywords: fuel, spillage, fire, ignit, flammable, petrol, fuel spill, refuel, fuel leak, fuel cap, two stroke, spilling gas, catch fire
- Exhaust fumes — control: ventilated area, never operate indoors
  Keywords: exhaust, fume, emission, carbon monoxide, co poison, exhaust gas, ventilation, indoors, inhalation, smoke from the engine, fresh air
- carbon monoxide — control: ventilated area, never operate indoors
  Keywords: exhaust, fume, emission, carbon monoxide, co poison, exhaust gas, ventilation, indoors, inhalation, smoke from the engine, fresh air
- Burns from hot engine parts — control: gloves, cool before handling, avoid muffler contact
  Keywords: burn, hot surface, hot exhaust, muffler, hot engine, blister, scald, burned on the hot engine, don't touch hot parts
- Vibration causing HAVS — control: anti-vibration gloves, limit exposure time, warm hands
  Keywords: vibrat, havs, hand arm vibration, white finger, vwf, raynaud, vibration exposure, av glove, shaking that hurts your hands, shake your hands

- Noise causing hearing damage — control: ear defenders, limit daily exposure duration
  Keywords: noise, hearing, deaf, ear, ear defend, decibel, hearing damage, hearing loss, loud, hearing protection, really loud noise, ear muffs, ear protection, ear muffs, hearing protection, ears
- Flying debris — control: face shield, safety glasses
  Keywords: debris, sawdust, eye, flying, wood chip, projectile, face shield, visor, glasses, face protection, wood chips flying into your eyes
- sawdust causing eye injury — control: face shield, safety glasses
  Keywords: debris, sawdust, eye, flying, wood chip, projectile, face shield, visor, glasses, face protection, wood chips flying into your eyes
- Chain break — control: tension checks, inspect chain links, chain catcher
  Keywords: chain break, chain snap, derail, chain off, chain throw, broken chain, chain catcher, chain tension, chain breaking and flying off
- derailment — control: tension checks, inspect chain links, chain catcher
  Keywords: chain break, chain snap, derail, chain off, chain throw, broken chain, chain catcher, chain tension, chain breaking and flying off
- Chain brake failure — control: pre-start check, functional test, remove from service if faulty
  Keywords: brake fail, brake malfunction, brake not work, brake defect, brake failure, chain brake, inertia brake, front guard
- Accidental starting — control: start on ground, throttle lock check, no drop start
  Keywords: accidental start, unintentional, drop start, throttle stick, throttle fault, starting, choke, cold start, saw starting by accident, safety lock
- throttle fault — control: start on ground, throttle lock check, no drop start
  Keywords: accidental start, unintentional, drop start, throttle stick, throttle fault, starting, choke, cold start, saw starting by accident, safety lock
- Leaking fuel cap — control: inspect caps and O-rings, tighten before use
  Keywords: leak, fuel cap, oil cap, o ring, chain oil, fuel leak, oil leak, bar oil
- chain oil — control: inspect caps and O-rings, tighten before use
  Keywords: leak, fuel cap, oil cap, o ring, chain oil, fuel leak, oil leak, bar oil
- Loose — control: inspect chassis before use
  Keywords: loose, bolt, nut, hand guard, front guard, rear guard, missing, chassis, casing
- missing nuts, bolts — control: inspect chassis before use
  Keywords: loose, bolt, nut, hand guard, front guard, rear guard, missing, chassis, casing
- hand guards — control: inspect chassis before use
  Keywords: loose, bolt, nut, hand guard, front guard, rear guard, missing, chassis, casing

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M4 Oral / Practical · Formative Only

Give me at least three hazards with the job to be done and the control measures.

Key points:

- Kickback during cutting — control: correct technique, chain brake, low-kickback chain
  Keywords: kickback, kick back, kick-back, cutting technique
- Falling timber, snags, widow makers — control: exclusion zone, check overhead
  Keywords: falling, snag, widow maker, dead wood, deadwood, hung up, log roll, falling branch, falling timber, overhead dead, leaning tree, heavy branches falling, hard hat, branches falling on your head, exclusion, clear area, keep people back, nobody nearby, zone
- overhead dead wood — control: exclusion zone, check overhead
  Keywords: falling, snag, widow maker, dead wood, deadwood, hung up, log roll, falling branch, falling timber, overhead dead, leaning tree, heavy branches falling, hard hat, branches falling on your head, exclusion, clear area, keep people back, nobody nearby, zone
- Fatigue — control: regular breaks, check-in system, buddy system, GPS beacon
  Keywords: fatigue, tired, lone, alone, exhaustion, working alone, breaks, check in, buddy, sole operator, lone worker, getting tired and making mistakes, take lots of breaks
- lone working — control: regular breaks, check-in system, buddy system, GPS beacon
  Keywords: fatigue, tired, lone, alone, exhaustion, working alone, breaks, check in, buddy, sole operator, lone worker, getting tired and making mistakes, take lots of breaks
- Manual handling of heavy logs — control: log jacks, winches, team lift, machinery
  Keywords: manual handling, heavy, lifting, log jack, timber, heavy log, winch, forwarder, team lift, log weight, hurting your back lifting, bend your knees
- Trapped bar — control: assess tension, use wedges, correct felling technique
  Keywords: trapped, pinch, barber chair, barber-chair, compression, tension, wedge, felling wedge, back cut, hinge, getting stuck under a falling tree
- barber-chairing — control: assess tension, use wedges, correct felling technique
  Keywords: trapped, pinch, barber chair, barber-chair, compression, tension, wedge, felling wedge, back cut, hinge, getting stuck under a falling tree
- Rolling — control: stand uphill, chock logs, never below rolling timber
  Keywords: roll, rolling log, shifting log, moving timber, log move, chock, uphill, timber roll, wood springing back
- shifting logs — control: stand uphill, chock logs, never below rolling timber
  Keywords: roll, rolling log, shifting log, moving timber, log move, chock, uphill, timber roll, wood springing back
- Struck by falling timber from colleague's work — control: communication, exclusion zones, signals
  Keywords: struck, colleague, co-worker, another worker, someone else, communication, signal, hand signal, working too close to someone else, exclusion, clear area, keep people back, nobody nearby, zone
- Slipping while carrying a running chainsaw — control: engage chain brake when moving, shut off over 3m
  Keywords: slip, carrying saw, running saw, moving with saw, chain brake when walking, carrying chainsaw, trip with saw, slipping while cutting
- Tree hung-up in adjacent trees — control: never work under, use winch or machinery to pull down
  Keywords: hung up, hung tree, leaner, lodged, adjacent tree, stuck in tree, tree lean, hung in
- Root plate rebound after felling — control: assess weight, cut from safe angle, stand clear
  Keywords: root plate, root ball, rebound, spring back, root spring, base, rootplate
- Operator fatigue and loss of concentration — control: mandatory breaks, rotate tasks, hydration
  Keywords: concentration, fatigue, tired, loss of focus, exhaustion, hydration, rotate task, break
- Inadequate training — control: hold valid certification, use supervision
  Keywords: training, competence, qualification, certificate, nptc, lantra, untrained, supervision, skills, not knowing what to do

- lack of competence — control: hold valid certification, use supervision
  Keywords: training, competence, qualification, certificate, nptc, lantra, untrained, supervision, skills, not knowing what to do
- Dust, noise — control: RPE, hearing protection, limit exposure
  Keywords: dust, noise, vibration from task, havs, hand arm, white finger, hearing damage, rpe, respiratory
- vibration from the task — control: RPE, hearing protection, limit exposure
  Keywords: dust, noise, vibration from task, havs, hand arm, white finger, hearing damage, rpe, respiratory
- Working at height — control: climbing harness, ropes, anchors, work at height regulations
  Keywords: height, climb, elevated, platform, aerial, harness, rope, work at height, high up, tree climb
- Biological hazards — ticks — control: inspect area, repellent, first aid
  Keywords: tick, stinging plant, poison, nettle, sting, biological, insect bite, bramble, thorn, repellent
- stinging plants — control: inspect area, repellent, first aid
  Keywords: tick, stinging plant, poison, nettle, sting, biological, insect bite, bramble, thorn, repellent
- poisonous plants — control: inspect area, repellent, first aid
  Keywords: tick, stinging plant, poison, nettle, sting, biological, insect bite, bramble, thorn, repellent

M5

Oral / Practical · Formative Only

What emergency information do I need to know for the site?

Key points:

- Site location — grid reference
  Keywords: grid ref, grid reference, coordinates, exactly where you are on a map, where you are on a map, map reference, location grid
- Site location — address
  Keywords: address, site address, postcode, street address, where the site is, site location address
- Site location — what3words
  Keywords: what three words, what3words, three words location, words location, what three word location
- Designated meeting / muster point
  Keywords: meeting place, muster point, rendezvous, rally point, assembly point, meeting point
- Muster / assembly point
  Keywords: meeting place, muster point, rendezvous, rally point, assembly point, meeting point
- Nearest access point
  Keywords: access point, street, access route, entry point, nearest road, how to get in, how to get out of the woods
- Nearest access point or road
  Keywords: access point, street, access route, entry point, nearest road, how to get in, how to get out of the woods
- Suitable helicopter landing area
  Keywords: helicopter, landing, heli, air ambu, landing spot, landing area
- Nearest hospital, A&E
  Keywords: doctor, hospital, accident and emergency, a&e, phone number, nearest medical, a and e, closest hospital, phone number for ambulance
- doctor's number
  Keywords: doctor, hospital, accident and emergency, a&e, phone number, nearest medical, a and e, closest hospital, phone number for ambulance
- Works manager
  Keywords: works manager, supervisor contact, manager, emergency contact, contact details, who to call if someone gets hurt
- supervisor contact details
  Keywords: works manager, supervisor contact, manager, emergency contact, contact details, who to call if someone gets hurt
- Your own mobile phone number
  Keywords: own number, mobile number, your number, personal contact, your mobile, my number
- Whether there is phone signal on site
  Keywords: phone signal, signal, mobile signal, no signal, reception, connectivity, if there is phone signal
- Who knows you are working there — check-in system
  Keywords: who knows, check in, someone knows, lone worker, check-in, knows you are working, logged in
- Location of fire extinguisher and first aid kit
  Keywords: fire extinguisher, extinguisher, first aid kit, first aid, first aid box, fire kit

M6

Oral / Practical · Formative Only

What is the Health and Safety at Work Act 1974 all about?

So what is the Health and Safety at Work Act all about?

Key points:

- Employees must take reasonable care of their own and others' safety
  Keywords: reasonable care, care of others, own safety, other people, duty of care, workers must be careful not to hurt themselves, care of yourself, not hurt yourself, be careful, look after yourself, look out for others, watch out, stay safe, careful, not hurt anyone, dont hurt yourself, keep safe, mind yourself, take care, careful at work, safe, careful, care, responsibility, duty
- Follow training and instructions received
  Keywords: follow training, follow your training, trained, act on training, training received, instructions, follow the safety rules, follow rules, listen to training, pay attention, do what you are told, obey instructions, do as trained, trained properly, know what to do, follow instructions, trained well
- Do not misuse
  Keywords: misuse, interfere, tamper, not misuse, do not interfere, safety equipment, don't do silly things that are dangerous
- interfere with safety equipment
  Keywords: misuse, interfere, tamper, not misuse, do not interfere, safety equipment, don't do silly things that are dangerous

- Employers must provide a safe workplace and safe systems of work
  Keywords: employer, safe workplace, safe system, safe place, employer duty, provide safe, bosses must keep workers safe, bosses keep their workers safe, safe tools to use, teach workers how to be safe, work place safe, keep work place safe, safe environment, safe at work, protect workers, safe conditions, look after workers
- Everyone must work together to prevent accidents
  Keywords: work together, cooperation, everyone responsible, joint responsibility, everyone has to, everyone's duty, everyone must work together to stop accidents
- Employees must report hazards and dangerous situations
  Keywords: report hazard, tell someone, report danger, notify, inform, if something dangerous, report it, must tell someone, dangerous situation

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M7 Oral / Practical · Formative Only

Under PUWER 1998, what does it say about the equipment?

Under the Provision and Use of Work Equipment Regulations — PUWER — what does it say about the equipment?

Key points:

- Equipment must be maintained in an efficient state and good repair
  Keywords: maintained, maintenance, kept in good, serviceable, good repair, efficient state, checked and fixed often, must be checked, not broken, in good condition, looked after, kept working, in good nick, well maintained, regularly checked, serviced, working order, good working order
- Equipment must be fit for purpose and suitable for the task
  Keywords: fit for purpose, suitable, appropriate, right tool, adequate, fit for, right tools for the job, tools must be the right ones, right for the job, good for the job, proper tool, right equipment, correct tool, appropriate tool, works for this, meant for this, designed for this, right tool, fit for purpose, appropriate, correct tool, suitable
- Must only be used by trained and competent operators
  Keywords: trained, competent, qualified, trained operator, only used by, certified, only people who know how, taught how to use, only qualified people, people who know how, trained people, competent users, certified users, licensed operators, qualified to use
- Equipment must have appropriate safeguards and controls
  Keywords: safeguard, guard, control, safety device, protection fitted, safety guards on them, must have safety guards, guards, safety guards, protective devices, safety mechanisms, safety features, protections fitted
- Equipment must be safe to use
  Keywords: safe to use, safe equipment, must be safe, safe for use, not dangerous, safe, wont hurt you, safe to handle, safe to operate, user safe, operator safe

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M8 Oral / Practical · Formative Only

Who provides industry guidance for tree work?

Key points:

- FISA — Forest Industry Safety Accord
  Keywords: fisa, forest industry safety accord, forest industry, forest industry safety, safety accord, fisa safety, forest safety group, fisa group
- AFAG — Arboricultural Forestry Advisory Group
  Keywords: afag, arboricultural forestry advisory, advisory group, tree work advisory, forestry advice group, arboricultural advisory
- Forestry Commission
  Keywords: forestry commission, natural resources wales, forestry england, forestry body
- Natural Resources Wales
  Keywords: forestry commission, natural resources wales, forestry england, forestry body
- Forestry England
  Keywords: forestry commission, natural resources wales, forestry england, forestry body
- Arboricultural Association (AA)
  Keywords: arboricultural association, arb association, aa, arboricultural assoc, tree surgeon association
- HSE — Health and Safety Executive
  Keywords: hse, health and safety executive, health and safety, government safety, government people who check on safety, safety inspector, safety official, safety authority, safety regulator, workplace inspector, government safety people, health and safety people
- Industry safety organisations
  Keywords: industry guidance, industry group, safety organisation, groups of experts, expert group, groups that make, safety inspectors, safety body, industry bodies, professional bodies, trade bodies, safety groups, expert committees, working groups, safety committees
- advisory groups
  Keywords: industry guidance, industry group, safety organisation, groups of experts, expert group, groups that make, safety inspectors, safety body

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M9 Oral / Practical · Formative Only

Why is it important to maintain the saw to the manufacturer's specifications?

Key points:

- Ensures the machine is safe to use and reduces the risk of accidents
  Keywords: safe to use, safety, safe operation, operator safety, reduces risk, prevent accident, keeps the saw working safely, stops accidents from happening, safe, safety, keep safe, keep it safe, dangerous, prevent injury, stay safe, be safe, safer
- Reduces machinery downtime and costly repairs
  Keywords: downtime, repair, breakdown, out of service, reliability, less repair, stops the saw from breaking down, costly repairs, cost, repair, expensive, downtime, breakdown, bills, less repair
- Maintains performance and cutting efficiency
  Keywords: performance, efficient, efficiency, cutting ability, optimum, operate correctly, cutting performance, cuts better, easier to start, runs smoothly, cuts well, good cut, works well, cutting well, work well

- Maintains the machine's longevity
  Keywords: longevity, last longer, lifespan, life of the saw, extends life, makes the saw last a long time, save money, last, lifespan, long life, last long
- It is what the manufacturer recommends
  Keywords: manufacturer, manufacturer says, made it say, manufacturer recommendations, manufacturer guidance, what the people who made it say, maker says, maker recommends, company says, builders say, factory says, instruction book, what it says in the book, manual says, what the book says, guidance, instructions, follow, told to, required

M10

Oral / Practical · Formative Only

The manufacturer has installed safety features on the saw — run through where they are and what they do.

The manufacturer has installed safety features on the saw. Just run through where they are and what they do.

Key points:

- Combined chain brake and front hand guard — stops chain on kickback, protects hand
  Keywords: chain brake, front hand guard, hand guard, brake stop, stops the chain, front guard, chain brake stops chain if it jumps back, kickback chain brake
- Throttle trigger lockout — prevents accidental throttle operation
  Keywords: throttle lockout, trigger lockout, accidental throttle, throttle safety, lock out, throttle trigger, safety trigger, stops you pressing the gas by accident, accidental press
- Chain catcher — catches chain if it breaks
  Keywords: chain catcher, catch the chain, derailed chain, chain off, chain catcher pin, catches the chain if it breaks
- Chain derailment
  Keywords: chain catcher, catch the chain, derailed chain, chain off, chain catcher pin, catches the chain if it breaks
- Rear hand guard
  Keywords: rear guard, rear hand, rear chain, back guard, back hand guard, breakage guard
- chain breakage guard — protects the rear hand
  Keywords: rear guard, rear hand, rear chain, back guard, back hand guard, breakage guard
- Anti-vibration mounts — reduces vibration to the hands
  Keywords: anti-vibration, anti vibration, vibration mount, reduce vibration, dampen vibration, av system, shaky mounts, stops your hands from shaking too much
- On/off switch — stops the engine
  Keywords: on off switch, on/off, kill switch, stop switch, engine stop switch, stops engine, turns the saw off quickly, turns saw off
- Safety decals — mandatory information (PPE symbols)
  Keywords: safety decal, decal, symbol, label, warning label, mandatory information, ear defender, eye protection symbol
- Low-kickback chain — reduces kickback characteristics
  Keywords: low kickback, kickback chain, anti-kickback chain, safety chain, low kick back chain
- Exhaust
  Keywords: exhaust, silencer, muffler, noise reduc, emission, silences, muffler makes saw less noisy
- silencer — reduces noise and emissions
  Keywords: exhaust, silencer, muffler, noise reduc, emission, silences, muffler makes saw less noisy
- Bar cover
  Keywords: bar cover, guide bar cover, scabbard, cover protect, chain cover, protective cover, covers the sharp chain when you carry it
- scabbard — protects bar and chain during transport
  Keywords: bar cover, guide bar cover, scabbard, cover protect, chain cover, protective cover, covers the sharp chain when you carry it
- Spiked bumper
  Keywords: spiked bumper, spike, bumper spike, dogs, felling dogs, grip the wood, grip wood when cutting, helps grip
- Spiked bumper
  Keywords: spiked bumper, spike, bumper spike, dogs, felling dogs, grip the wood, grip wood when cutting, helps grip
- spike bumper — helps grip the wood when cutting
  Keywords: spiked bumper, spike, bumper spike, dogs, felling dogs, grip the wood, grip wood when cutting, helps grip

M11

Oral / Practical · Formative Only

If you find one of the safety features is broken or missing, what will you do?

Key points:

- Stop work immediately and take the saw out of service
  Keywords: stop work, cease work, take out of service, do not use, withdraw, stop using, out of service, do not use the saw at all, don't use it, dont use it, put down, leave it, dont touch it, pack it up, stop immediately, cease using, halt work, quit using
- Tag it / report it to your supervisor
  Keywords: tag, report, supervisor, employer, inform, label, notify, report it, tell your boss, put a sign on it saying it is broken, tell someone, tell the boss, inform supervisor, let them know, report broken, flag it, mark it
- Report to employer or supervisor
  Keywords: tag, report, supervisor, employer, inform, label, notify, report it, tell your boss, put a sign on it saying it is broken, tell employer, inform boss, let manager know, notify work, report to work
- Do not use until repaired
  Keywords: do not use, not until repaired, until fixed, competent person, repaired, replaced, wait until completely fixed, never pretend it's okay
- replaced by a competent person
  Keywords: do not use, not until repaired, until fixed, competent person, repaired, replaced, wait until completely fixed, never pretend it's okay
- Write it in the maintenance log
  Keywords: write down, maintenance log, record, log it, document, write down what is broken, write in a book
- record it
  Keywords: write down, maintenance log, record, log it, document, write down what is broken, write in a book

M12

Oral / Practical · Formative Only

What are the advantages of using battery chainsaws?

Key points:

- Quieter operation — reduced noise levels
  Keywords: quieter, reduced noise, less noise, lower noise, quiet, less sound, much quieter to use, quieter operation, much quieter, less noisy, not as loud, quiet saw, silent saw, low noise, quiet, less noise, silent, quieter, noise
- Zero exhaust emissions — no fumes
  Keywords: no emission, zero emission, no exhaust, no fumes, clean, emission free, no fume, don't make bad smoke, no bad smoke, no smoke, no bad smell, clean air, environmentally friendly, eco friendly, green, no pollution, no fumes, clean, green, eco, no emissions, no smoke
- Less vibration transmitted to the operator
  Keywords: less vibration, reduced vibration, lower vibration, vibration reduction, less vibrat, don't shake your hands as much, less shaking, less vibration, vibration, hands, comfortable
- No fuel to carry — no fuel spillage risk
  Keywords: no fuel, fuel transport, spillage, no petrol, no spill, no fuel need, don't need fuel, don't need smelly gas, no fuel needed, no fuel, no petrol, no gas, fuel free
- Better communication — can hear others on site
  Keywords: communication, talk to, speak to, hear each other, clearer comms, communicate, can hear people talking, hear people better, better communication, communication, hear, talk, comms
- Less maintenance required
  Keywords: less maintenance, lower maintenance, fewer service, easier maintenance, easier to maintain, easier to clean and look after, easy maintenance, low maintenance, simple
- Can be used indoors
  Keywords: indoor, urban, sensitive, inside, enclosed, public area, inside a shed, indoor, inside, urban
- in urban / sensitive environments
  Keywords: indoor, urban, sensitive, inside, enclosed, public area, inside a shed
- Easy to start — just a button
  Keywords: easy to start, push button, just push a button, button start, easier to start, button to start, easy start, press button, simple start
- Lighter to hold
  Keywords: lighter, not as heavy, light weight, less heavy, easier to carry, less weight, not heavy, light, lighter, easy carry, not heavy
- Easy to carry / lightweight
  Keywords: lighter, not as heavy, light weight, less heavy, easier to carry, less weight, not heavy

M13

Oral / Practical · Formative Only

What are the disadvantages of using battery chainsaws?

Key points:

- Limited run time — battery capacity restricts working time
  Keywords: limited run, battery life, run time, capacity, run out, limited time, batteries run out quickly, run out of battery, batteries drain, batteries die quick, doesnt last long, short battery, runs out fast, limited power, battery drains, battery life, runs out, short battery, limited time
- Less power available for heavy
  Keywords: less power, not as powerful, heavy timber, large diameter, power limitation, not strong enough for huge trees, sometimes not strong enough, less power, weaker, not powerful, small trees only
- large-diameter timber
  Keywords: less power, not as powerful, heavy timber, large diameter, power limitation, not strong enough for huge trees, sometimes not strong enough
- Battery compatibility and misalignment risks
  Keywords: incompatible, compatibility, wrong battery, incorrect battery, misalign, not compatible, compatibility, wrong battery, battery issue
- Machine can be live when battery is fitted — no exhaust sound warning
  Keywords: live, battery in, always on, no warning, silent, no engine noise, energised, so quiet you might not hear them running
- Risk of electric shock
  Keywords: electric shock, short circuit, charging risk, electrocution, fire when charging, overheating battery, water can break them and give shock, electric shock from water
- short circuit when charging
  Keywords: electric shock, short circuit, charging risk, electrocution, fire when charging, overheating battery, water can break them and give shock, electric shock from water
- Cold weather reduces battery performance
  Keywords: cold, temperature, cold weather, winter, cold affect, battery cold, might not work well in very cold, cold weather performance
- Battery storage and disposal considerations
  Keywords: storage, disposal, store battery, dispos, battery storage, expensive batteries, batteries expensive, expensive to buy
- Need to bring extra batteries — charging not possible in the field
  Keywords: extra batteries, spare batteries, can't charge in the woods, can't charge in middle of woods, need to charge beforehand, remember to charge night before, need to bring extra
- Heavier batteries add weight to the saw
  Keywords: batteries heavy, heavy to carry, battery weight, adds weight, heavy battery pack

M14

Oral / Practical · Formative Only

What maintenance might you do with the batteries, the saw and charger?

Key points:

- Inspect battery for damage, cracks
  Keywords: inspect battery, check battery, battery damage, crack, deform, battery condition, check for cracks or broken, look for cracks

- deformation
  Keywords: inspect battery, check battery, battery damage, crack, deform, battery condition, check for cracks or broken, look for cracks
- Clean battery contacts and guide tracks
  Keywords: clean contacts, battery contacts, guide track, terminal, clean terminal, track clean, keep metal bits clean so they connect well
- Store battery at correct charge level in a cool, dry place
  Keywords: store battery, storage, correct charge, 40, 50, 60, cool dry, storage charge, keep in dry place, store somewhere dry, not too hot or cold
- Inspect charger leads and connections for damage
  Keywords: charger lead, charger cable, charger connection, charger damage, inspect charger, charger cord not chewed, cord not chewed or broken
- Clean the saw's air filter and cooling system
  Keywords: air filter, cooling system, clean saw, filter clean, cooling clean, clean saw so sawdust doesn't get in battery hole
- Check battery compartment for damage
  Keywords: compartment, battery compartment, housing, battery bay, compartment clean, clean the battery hole, wipe dust off with dry cloth
- Clean debris from battery compartment
  Keywords: compartment, battery compartment, housing, battery bay, compartment clean, clean the battery hole, wipe dust off with dry cloth
- Do not overcharge
  Keywords: overcharge, don't overcharge, not leave plugged in, don't leave plugged in too long, after they are full, overcharging
- leave plugged in after fully charged
  Keywords: overcharge, don't overcharge, not leave plugged in, don't leave plugged in too long, after they are full, overcharging
- Do not drop batteries — protect from impact
  Keywords: don't drop, drop batteries, protect from impact, avoid dropping, don't drop on ground

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M15 Oral / Practical · Formative Only

Can you take your top cover off your chainsaw please.

M16 Oral / Practical · Formative Only

So what's this?

Key points:

- Air filter
  Keywords: air filter, filter, air cleaner, little sponge, sponge filter, foam filter, paper filter, cleans the air, sponge thing, filter thing, sponge bit, dirty sponge, foam bit, cleaner thing

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M17 Oral / Practical · Formative Only

What does it do? (air filter)

What does it do?

Key points:

- Prevents debris and dust entering the carburettor
  Keywords: prevent debris, stops dirt, dirt out, debris, carburettor, stops dust, clean air, air fuel, filters the air, cleans the air before it goes in, stops dirt getting into engine, mask for the chainsaw, keeps dirt out, filters air, cleans air going in, stops stuff getting in, protects engine from dirt
- Carburettor / engine
  Keywords: prevent debris, stops dirt, dirt out, debris, carburettor, stops dust, clean air, air fuel, filters the air, cleans the air before it goes in, stops dirt getting into engine, mask for the chainsaw

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M18 Oral / Practical · Formative Only

How do you maintain it? (air filter)

And how do you maintain it?

Key points:

- Remove and clean — tap out, wash
  Keywords: clean, tap out, wash, blow out, remove debris, clean filter, wash in warm water, warm water with soap, blow the dust off, tap gently, brush softly, air gun, air line, remove, remove the filter, take it out
- blow out debris
  Keywords: clean, tap out, wash, blow out, remove debris, clean filter, wash in warm water, warm water with soap, blow the dust off, tap gently, brush softly, air gun, air line, remove, remove the filter, take it out
- Inspect the filter housing and gasket for damage
  Keywords: inspect, inspect housing, gasket, housing, housing damage, filter housing, look for holes in it, check for damage, check it
- Replace the filter if damaged
  Keywords: replace, new filter, fit new, replace if, damaged filter, buy a new one, too dirty to clean
- Damaged or worn filter
  Keywords: replace, new filter, fit new, replace if, damaged filter, buy a new one, too dirty to clean
- Ensure completely dry before refitting
  Keywords: completely dry, dry before, dry it out, let it dry, must be dry, make sure it is dry, dry

M19

Oral / Practical · Formative Only

So what's this? (spark plug)

So what's this?

Key points:

- Spark plug
  Keywords: spark plug, plug, sparking plug, little metal stick, metal stick with white top, goes into the engine, makes the spark

M20

Oral / Practical · Formative Only

What does it do? (spark plug)

What does it do?

Key points:

- Ignites the fuel/air mixture to fire the engine
  Keywords: ignite, ignition, fires, combustion, spark, fire the engine, fuel mix, makes a tiny spark, lights the gas on fire, mini explosions, like a match, makes the engine go bang, burn the fuel

M21

Oral / Practical · Formative Only

What are you looking for and how would you maintain it? (spark plug)

Can you tell me what you're looking for and how you'd maintain it?

Key points:

- Check electrode condition and gap — use a feeler gauge
  Keywords: electrode, gap, feeler gauge, check gap, electrode gap, electrode condition, tiny gap at the bottom, right size gap
- Look for fouling — carbon deposits, sooty
  Keywords: fouling, carbon, sooty, oily, deposit, black, contamination, black soot, clean off black soot
- oily residue
  Keywords: fouling, carbon, sooty, oily, deposit, black, contamination, black soot, clean off black soot
- Check for cracks
  Keywords: crack, damage, ceramic, broken, physical damage, white part cracked, look for cracks
- physical damage to the ceramic
  Keywords: crack, damage, ceramic, broken, physical damage, white part cracked, look for cracks
- Clean
  Keywords: clean, replace, clean plug, new plug, fit new, put a new one in, wire brush, little wire brush
- replace as necessary
  Keywords: clean, replace, clean plug, new plug, fit new, put a new one in, wire brush, little wire brush

M22

Oral / Practical · Formative Only

In conjunction with the flywheel, what do these cylinder fins help do?

Key points:

- Cool the engine — draw air over the cylinder to prevent overheating
  Keywords: cool, cooling, prevent overheat, overheat, airflow, air over, temperature, help the engine cool down, let the heat escape, stops the saw from getting too hot, spread the heat out, keeps engine at good temperature, stop it getting hot, keep cool, prevent getting too hot, blow air, fan effect, cooling fins, heat escape, air cooling, temperature control

M23

Oral / Practical · Formative Only

How do you maintain the cooling system?

So how do you maintain the cooling system?

Key points:

- Remove debris from the cylinder fins — brush
  Keywords: remove debris, clean fins, clean the fins, debris from fins, brush, compressed air, fin clean, cylinder fin, brush all sawdust out of little gaps, sawdust out of the gaps, blow air through it, tap out stuck dirt, clean
- compressed air
  Keywords: remove debris, clean fins, clean the fins, debris from fins, brush, compressed air, fin clean, cylinder fin, brush all sawdust out of little gaps, sawdust out of the gaps, blow air through it, tap out stuck dirt, clean
- Check and clear the flywheel housing and air intake
  Keywords: flywheel, air intake, housing, clear intake, flywheel housing, nothing blocking the air, inspect, check intake, check the housing, inspect the housing
- Keep clean regularly — don't let mud
  Keywords: keep clean, regularly clean, clean regularly, don't let mud dry on it, wipe clean when finished, check often, not clogged up, regular clean, keep it clean
- debris build up
  Keywords: keep clean, regularly clean, clean regularly, don't let mud dry on it, wipe clean when finished, check often, not clogged up, regular clean, keep it clean

M24

Oral / Practical · Formative Only

So what's this? (exhaust)

So what's this?

Key points:

- Exhaust
  Keywords: exhaust, silencer, muffler, exhaust system, where the smoke comes out, metal box on the front, little chimney
- silencer
  Keywords: exhaust, silencer, muffler, exhaust system, where the smoke comes out, metal box on the front, little chimney
- muffler
  Keywords: exhaust, silencer, muffler, exhaust system, where the smoke comes out, metal box on the front, little chimney

M25

Oral / Practical · Formative Only

What does it do? (exhaust)

What does it do?

Key points:

- Reduces noise and directs exhaust gases
  Keywords: reduce noise, quieter, exhaust gas, emission, noise reduction, direct gas, muffles, blows the bad smoke away, makes loud engine noise quieter, directs hot air away, hot gases away, like exhaust pipe on car, smoke pipe, fume pipe, gas pipe, smoke outlet, blows smoke, blows fumes, directs gas
- emissions away from the operator
  Keywords: reduce noise, quieter, exhaust gas, emission, noise reduction, direct gas, muffles, blows the bad smoke away, makes loud engine noise quieter, directs hot air away, hot gases away, like exhaust pipe on car
- Spark arrestor — contains spark arrestor screen to prevent fire
  Keywords: spark arrestor, spark arrest, prevent fire, fire prevention, spark screen, little screen inside it, net to catch sparks, traps burning carbon, stops sparks flying

M26

Oral / Practical · Formative Only

How do you maintain it? (exhaust)

And how do you maintain it?

Key points:

- Check fixings — nuts and bolts are secure
  Keywords: nuts, bolts, fixings, secure, tighten, check bolts, fixing check, check the screws holding it are tight, screws tight, not rattling, check
- Check
  Keywords: spark arrestor, spark arrester, arrestor, clean arrestor, carbon build, clean the little spark catching net, brush the net, carbon mesh, clean, clean it
- clean the spark arrestor
  Keywords: spark arrestor, spark arrester, arrestor, clean arrestor, carbon build, clean the little spark catching net, brush the net, carbon mesh, clean, clean it
- Check for cracks
  Keywords: crack, damage, body crack, check damage, inspect exhaust, look for cracks or holes, cracks or holes, rusted through, inspect, inspect it
- damage to the body
  Keywords: crack, damage, body crack, check damage, inspect exhaust, look for cracks or holes, cracks or holes, rusted through, inspect, inspect it
- Let cool before handling — do not touch when hot
  Keywords: cool down, let cool, don't touch when hot, wait until cold, completely cold, allow to cool, don't touch until cold

M27

Oral / Practical · Formative Only

What is in here that you need to maintain and how would you do it? (fuel tank)

What is in here that you need to maintain and how would you do it?

Key points:

- Fuel filter — located in the fuel tank
  Keywords: fuel filter, filter in fuel, fuel tank filter, in the tank, fuel pick-up, little fuel filter inside gas tank, inside the gas tank, filter
- Remove, inspect and clean
  Keywords: remove filter, replace fuel filter, clean filter, new filter, change filter, use a little hook to pull it out, hook to pull it out, change once a year, if dirty or blocked, can't clean it just replace, remove, inspect, replace, clean, change
- replace the fuel filter
  Keywords: remove filter, replace fuel filter, clean filter, new filter, change filter, use a little hook to pull it out, hook to pull it out, change once a year, if dirty or blocked, can't clean it just replace, remove, inspect, replace, clean, change

M28

Oral / Practical · Formative Only

What is in here that you need to maintain and how would you do it? (oil tank)

So what is in here that you need to maintain and how would you do it?

Key points:

- Oil filter — located in the bar oil tank
  Keywords: oil filter, filter in oil, bar oil tank, oil tank filter, oil pick-up, oil filter inside oil tank, inside the oil tank, filter
- Remove, inspect and clean
  Keywords: remove oil filter, replace oil filter, clean oil filter, new oil filter, change oil filter, hook it out like gas filter, sometimes wash in clean gas, sticky and dirty put new one in, chain gets enough slippery oil, remove, inspect, replace, clean, change
- replace the oil filter
  Keywords: remove oil filter, replace oil filter, clean oil filter, new oil filter, change oil filter, hook it out like gas filter, sometimes wash in clean gas, sticky and dirty put new one in, chain gets enough slippery oil, remove, inspect, replace, clean, change

M29

Oral / Practical · Formative Only

This is the recoil housing — can you take it off and de-tension the pull cord please.

This is the recoil housing, can you take it off and de-tension the pull cord please.

M30

Oral / Practical · Formative Only

Where would you expect to find damage? (recoil starter)

Where would you expect to find damage?

Key points:

- Pull cord / rope — look for fraying, wear
  Keywords: pull cord, rope, cord, fray, wear on cord, knotting, cord damage, string you pull, fuzzy or breaking, string, frayed cord, broken cord
- Fraying or knotting
  Keywords: pull cord, rope, cord, fray, wear on cord, knotting, cord damage, string you pull, fuzzy or breaking, string, frayed cord, broken cord
- Recoil spring / coil spring — check for breakage
  Keywords: recoil spring, coil spring, spring, broken spring, spring damage, metal spring inside, spring might be snapped
- Spring distortion
  Keywords: recoil spring, coil spring, spring, broken spring, spring damage, metal spring inside, spring might be snapped
- Pawl / ratchet — check for wear
  Keywords: pawl, ratchet, dog, catch, mechanism, ratchet wear, little plastic teeth, worn down teeth
- Pawl / ratchet breakage
  Keywords: pawl, ratchet, dog, catch, mechanism, ratchet wear, little plastic teeth, worn down teeth
- Housing — check for cracks
  Keywords: housing crack, plastic cover, cover cracked, casing damage, plastic cracked, hole in cover, cracked housing
- damage to the casing
  Keywords: housing crack, plastic cover, cover cracked, casing damage, plastic cracked, hole in cover, cracked housing

M31

Oral / Practical · Formative Only

Now can you take off the chain and bar side cover please.

M32

Oral / Practical · Formative Only

So what's this? (sprocket)

So what's this?

Key points:

- Sprocket
  Keywords: sprocket, drive sprocket, clutch sprocket, chain sprocket, little metal star, metal gear, gear, metal star, star thing, star wheel, cog, gear wheel, chain wheel, drive wheel
- drive sprocket
  Keywords: sprocket, drive sprocket, clutch sprocket, chain sprocket, little metal star, metal gear, gear, metal star

M33

Oral / Practical · Formative Only

What does it do? (sprocket)

What does it do?

Key points:

- Drives the chain along the guidebar
  Keywords: drives the chain, drive chain, moves the chain, chain drive, carries chain, pulls the chain around, transfers power to chain, grips the bottom parts of chain, makes the chain spin, drives cutting teeth into wood

M34

Oral / Practical · Formative Only

How do you check for any wear or damage? (sprocket)

How do you check for any wear or damage?

Key points:

- Check for hooked
  Keywords: hooked, wolf teeth, hook, curved teeth, sharp point, worn tooth, teeth shape, look at metal teeth for deep cuts, deep cuts in teeth, inspect, inspect the teeth, check the teeth, check
- wolf-teeth — look for sharp, curved points
  Keywords: hooked, wolf teeth, hook, curved teeth, sharp point, worn tooth, teeth shape, look at metal teeth for deep cuts, deep cuts in teeth, inspect, inspect the teeth, check the teeth, check
- Check the needle bearing
  Keywords: needle bearing, bearing, drum, needle cage, bearing wear, see if it wobbles, wobbles when you touch it
- drum for wear
  Keywords: needle bearing, bearing, drum, needle cage, bearing wear, see if it wobbles, wobbles when you touch it
- Check depth of tooth wear — replace if excessively worn
  Keywords: depth of wear, tooth depth, excessive wear, measure wear, replace sprocket, grooves worn into metal, grooves too deep, if grooves are deeper than a tiny bit, replace if worn, replace
- Listen for clanking
  Keywords: clanking, rattling, noise, listen for, clank, rattle, listen
- rattling noises
  Keywords: clanking, rattling, noise, listen for, clank, rattle, listen

M35

Oral / Practical · Formative Only

If you need to replace the sprocket, how are you going to do that?

Key points:

- Use a clutch removal tool to remove the clutch
  Keywords: clutch removal, removal tool, remove clutch, clutch tool, spanner, take the clutch off first, special tool to unscrew clutch, clutch unscrews backwards
- Remove the needle bearing and spacers carefully
  Keywords: needle bearing, needle cage, bearing, spacer, remove bearing, little roller bearing, bearing inside
- Fit the new sprocket and reassemble correctly
  Keywords: fit new, new sprocket, replace sprocket, reassemble, refit, slide new sprocket onto shaft, screw clutch back on
- Torque to manufacturer's specification
  Keywords: torque, specification, torque spec, correct torque, tighten to spec, tighten correctly

M36

Oral / Practical · Formative Only

Now on to the chain brake — where would you find it on your saw?

Key points:

- Front of the saw — the front hand guard
  Keywords: front, front hand guard, front guard, front of saw, front handle, left hand guard, built into the side cover, big plastic guard, in front of your top hand, right in front of your hand
- front handle area
  Keywords: front, front hand guard, front guard, front of saw, front handle, left hand guard, built into the side cover, big plastic guard, in front of your top hand, right in front of your hand

M37

Oral / Practical · Formative Only

What does the chain brake do?

What does it do?

Key points:

- Stops the chain rapidly — activated manually
  Keywords: stops the chain, halt chain, chain stops, inertia, kickback, activated, stops chain spinning instantly, stops chain from spinning, protects you if saw kicks back, very strong handbrake for saw, life-saving safety feature, safety brake, hand guard brake, chain stopper, emergency stop, kickback brake, automatic brake, stops chain spinning
- by inertia on kickback
  Keywords: stops the chain, halt chain, chain stops, inertia, kickback, activated, stops chain spinning instantly, stops chain from spinning, protects you if saw kicks back, very strong handbrake for saw, life-saving safety feature
- The brake band clamps around the clutch drum to arrest the chain
  Keywords: brake band, clutch drum, band clamp, drum, arrest, clamp

M38

Oral / Practical · Formative Only

How do you check the chain brake for any wear or damage?

How do you check for any wear or damage?

Key points:

- Check the brake band for wear — check thickness
  Keywords: brake band, band wear, band thickness, band condition, check band, metal band not worn too thin, look at the metal band, band too thin, inspect, check
- Check the front hand guard for damage
  Keywords: front hand guard, guard damage, guard crack, hand guard condition, plastic handle not cracked, check plastic handle, inspect the guard
- Guard cracks
  Keywords: front hand guard, guard damage, guard crack, hand guard condition, plastic handle not cracked, check plastic handle, inspect the guard
- Test the brake activation — manual and inertia
  Keywords: test brake, activate brake, brake test, manual activation, inertia test, test activation, push handle forward to hear loud click, loud click, try to pull chain when brake on, shouldn't move, test it, test
- Check the spring inside is strong and not worn
  Keywords: spring inside, spring strong, spring condition, check spring, spring not worn

M39 Oral / Practical · Formative Only

This is your guidebar — what signs of wear or damage are you looking for?

Key points:

- Uneven
  Keywords: uneven rail, burred, rail condition, burr, rail wear, uneven wear, rail bur, sharp metal edges, burrs on rails, look for sharp metal edges called burrs, one side of rail higher
- burred rails
  Keywords: uneven rail, burred, rail condition, burr, rail wear, uneven wear, rail bur, sharp metal edges, burrs on rails, look for sharp metal edges called burrs, one side of rail higher
- Groove depth — worn groove
  Keywords: groove depth, groove wear, worn groove, groove, groove check, groove in middle squashed, groove squashed
- Bar straightness — bent bar
  Keywords: straightness, straight, bent, bent bar, bar straight, bar bent or twisted, look if bar is bent
- Sprocket nose wear
  Keywords: sprocket nose, nose sprocket, nose wear, tip, nose damage, tip wheel broken or stuck, check the tip wheel
- Nose / tip damage
  Keywords: sprocket nose, nose sprocket, nose wear, tip, nose damage, tip wheel broken or stuck, check the tip wheel
- Overheating marks
  Keywords: overheat, heat mark, discolouration, bluing, burn mark, overheated bar, blue marks, look for blue marks meaning too hot
- heat discolouration
  Keywords: overheat, heat mark, discolouration, bluing, burn mark, overheated bar, blue marks, look for blue marks meaning too hot
- Oil hole blocked with dirt
  Keywords: oil hole, oil hole blocked, blocked oil hole, oil port blocked, oil hole with dirt, blocked with dirt
- Groove too wide
  Keywords: groove too wide, chain loose, chain sitting loosely, groove wider than chain, chain fits loosely, chain doesn't fit
- chain sitting loosely
  Keywords: groove too wide, chain loose, chain sitting loosely, groove wider than chain, chain fits loosely, chain doesn't fit

M40 Oral / Practical · Formative Only

If the bar shows excessive wear or damage, what problems might you get?

So if the bar shows excessive wear or damage, what problems might you get?

Key points:

- Chainsaw will not cut in a straight line — it will drift
  Keywords: not cut straight, drift, deviate, wander, pull to one side, doesn't cut straight, saw will not cut in straight line, cut in a curve, cuts crooked, curved cut, bent cut, doesnt cut straight, wanders off line, veers to side
- Overheating of the bar and poor lubrication
  Keywords: overheat, heat up, poor lubrication, dry chain, lubrication problem, hot bar, bar will get very hot, bar very hot and smoke
- Chain derailment
  Keywords: chain off, derail, accelerated wear, chain wear, sprocket wear, wear out faster, chain might jump off bar completely, chain jumps off, chain ruined quickly
- accelerated chain / sprocket wear
  Keywords: chain off, derail, accelerated wear, chain wear, sprocket wear, wear out faster, chain might jump off bar completely, chain jumps off, chain ruined quickly
- Reduced cutting performance — saw works harder
  Keywords: reduced performance, slower, slower cut, harder to cut, more effort, saw will cut very slowly, harder to push through wood
- Increased vibration and danger
  Keywords: vibrate, shake, vibration, more dangerous, much more dangerous, unsafe, dangerous to use, vibrate and shake hands

M41 Oral / Practical · Formative Only

How are you going to maintain your bar?

Key points:

- Clean the groove and clear the oil holes
  Keywords: clean groove, clear oil hole, oil hole, groove clean, clear the groove, oil port, poke oil hole with wire, wire to keep clear, clean dirt out of middle groove, clean, clean the bar, clean it

- Remove burrs from the rails with a flat file
  Keywords: remove burr, deburr, flat file, file the rail, burr removal, dress rails, use flat file to scrape off burrs, scrape off sharp metal burrs, file, remove burrs, file it
- Turn
  Keywords: turn the bar, reverse bar, flip bar, rotate bar, bar turn, even wear, turn bar upside down, turn it over so it wears evenly, flip it over
- reverse the bar to even out wear
  Keywords: turn the bar, reverse bar, flip bar, rotate bar, bar turn, even wear, turn bar upside down, turn it over so it wears evenly, flip it over
- Grease the nose sprocket
  Keywords: grease nose, bar nose, nose sprocket grease, lubricate nose, grease tip, put grease in little hole at tip, grease tip if it has a wheel
- Wipe clean and store dry to prevent rust
  Keywords: wipe clean, store dry, prevent rust, clean with rag, wipe with rag, dry storage, store somewhere dry

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M42 Oral / Practical · Formative Only

Can you reassemble your saw please.

M43 Oral / Practical · Formative Only

When you're ready, can you clamp your saw into a vice please.

M44 Oral / Practical · Formative Only

How do you identify your chain?

Key points:

- Pitch — the measurement of the chain drive links
  Keywords: pitch, chain pitch, how far apart the rivets are, rivets, distance between links
- Gauge — the width of the drive link
  Keywords: gauge, chain gauge, drive link gauge, how thick the bottom parts are, thickness of drive links
- Number of drive links
  Keywords: drive link, number of link, drive links, link count, count how many bottom parts, count the drive links
- Markings on the drive link
  Keywords: marking, stamp, chain box, packaging, manufacturer mark, stamped on, numbers stamped on the side, look up the saw model in manual, match to numbers on guide bar
- chain packaging
  Keywords: marking, stamp, chain box, packaging, manufacturer mark, stamped on, numbers stamped on the side, look up the saw model in manual, match to numbers on guide bar

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M45 Oral / Practical · Formative Only

What information do you need to replace your chain?

Key points:

- Pitch of the chain
  Keywords: pitch, pitch of chain, the pitch
- Gauge of the chain
  Keywords: gauge, gauge of chain, the gauge
- Number of drive links
  Keywords: drive link, number of link, exactly how many drive links, drive link count
- Length
  Keywords: bar length, guide bar length, length of bar, bar size, bar inches, how long the guide bar is, how long the bar is, bar length inches
- size of the guidebar
  Keywords: bar length, guide bar length, length of bar, bar size, bar inches, how long the guide bar is, how long the bar is, bar length inches
- Cutter type
  Keywords: cutter type, chain type, type of chain, cutter profile, what shape of cutting teeth
- chain type
  Keywords: cutter type, chain type, type of chain, cutter profile, what shape of cutting teeth
- Brand of your chainsaw
  Keywords: brand, make, brand of chainsaw, chainsaw make, manufacturer name
- Model number of your chainsaw
  Keywords: model, model number, model of chainsaw, saw model
- Part number from the old chain
  Keywords: part number, old chain number, chain part number, number from old chain, number from the box

M46 Oral / Practical · Formative Only

What cutter profile do you have on your chain?

Key points:

- Chisel / full chisel
  Keywords: chisel, semi-chisel, semi chisel, full chisel, round, cutter type, rounded corners, square corners, sharp right angle, little curve
- semi-chisel (whichever applies)
  Keywords: chisel, semi-chisel, semi chisel, full chisel, round, cutter type, rounded corners, square corners, sharp right angle, little curve

M47 Oral / Practical · Formative Only

What is the other main type of cutter profile?

Key points:

- Chisel
  Keywords: chisel, semi-chisel, semi chisel, full chisel, other type, other profile, if you have semi-chisel the other is full-chisel, one has sharp square corners, one has rounded corners
- semi-chisel — whichever was not given in the previous answer
  Keywords: chisel, semi-chisel, semi chisel, full chisel, other type, other profile, if you have semi-chisel the other is full-chisel, one has sharp square corners, one has rounded corners

M48 Oral / Practical · Formative Only

And what are their different uses?

Key points:

- Full chisel — fast cutting, hard
  Keywords: full chisel, chisel, fast, hard wood, frozen wood, professional, faster cut, cuts very fast in clean soft wood, faster in clean wood, cuts fast, quick cutting, fast cutter, aggressive cutting, professional cutter
- frozen wood, professional use
  Keywords: full chisel, chisel, fast, hard wood, frozen wood, professional, faster cut, cuts very fast in clean soft wood, faster in clean wood
- Semi-chisel — softer
  Keywords: semi-chisel, semi chisel, dirty wood, stays sharp, beginner, softer, less kickback, better for dirty or hard wood, muddy wood, stays sharp longer in dirt, safer for beginners, for dirty wood, stays sharp longer, rounded teeth, curved teeth, less aggressive
- dirty wood, stays sharper longer, better for beginners
  Keywords: semi-chisel, semi chisel, dirty wood, stays sharp, beginner, softer, less kickback, better for dirty or hard wood, muddy wood, stays sharp longer in dirt

M49 Oral / Practical · Formative Only

What top plate angle are you filing at?

Key points:

- State the correct top plate filing angle (e.g. 25°, 30°
  Keywords: degree, angle, 25, 30, 35, top plate angle, filing angle, 25 or 30 degree, hold the file at, slightly slanted
- 35°) for the chain
  Keywords: degree, angle, 25, 30, 35, top plate angle, filing angle, 25 or 30 degree, hold the file at, slightly slanted

M50 Oral / Practical · Formative Only

And what file size are you using?

Key points:

- State the correct file diameter for the chain (e.g. 4mm, 3/16", 13/64")
  Keywords: millimetre, mm, file size, diameter, round file, 3/16, 4mm, 13/64, size of file, 5/32, 7/32, depends on pitch, depends on chain size

M51 Oral / Practical · Formative Only

So where are you going to start sharpening?

Key points:

- Start at the shortest / most damaged cutter,
  Keywords: shortest, most damaged, marked, start at, first cutter, mark a cutter, shortest cutter, most worn down, worst tooth, mark the first tooth, marker pen
- mark a starting cutter
  Keywords: shortest, most damaged, marked, start at, first cutter, mark a cutter, shortest cutter, most worn down, worst tooth, mark the first tooth, marker pen

M52

Oral / Practical · Formative Only

So what's this? (depth gauge / raker)

So what's this?

Key points:

- Depth gauge
  Keywords: depth gauge, raker, depth limiter, gauge, little bump of metal, little shark fin, shark fin, bump in front of tooth, depth gauge, raker, gauge setting, depth setting
- Depth gauge / raker
  Keywords: depth gauge, raker, depth limiter, gauge, little bump of metal, little shark fin, shark fin, bump in front of tooth, depth gauge, raker, gauge setting, depth setting

M53

Oral / Practical · Formative Only

What does it do? (depth gauge)

What does it do?

Key points:

- Controls the depth of cut — limits how much wood the cutter takes in each pass
  Keywords: depth of cut, controls depth, limits cut, how much wood, bite, chip size, controls bite, decides how deep the tooth can bite, bumper to stop tooth going too deep, controls size of wood chips, controls how big a bite, stops cutting too deep, limits bite, controls bite size, how deep the tooth goes, depth limiter, shallower cut

M54

Oral / Practical · Formative Only

Why does the depth gauge need to be set correctly, and what's the danger in setting it too low?

And why does it need to be set correctly, and what's the danger in setting it too low?

Key points:

- Controls chip size and cutting efficiency — optimum cutting speed
  Keywords: chip size, cutting efficiency, optimum speed, cutting speed, optimum cutting, if too high won't cut, makes dust instead of chips, just makes dust
- Set too low: increased kickback risk — chain grabs aggressively into the wood
  Keywords: too low, kickback, grab, aggressive, chain grab, danger too low, excessive kickback, saw very jumpy and dangerous, tooth bites too much wood at once, very jumpy and dangerous
- Too low also strains the engine
  Keywords: strains engine, engine strain, huge strain on engine, engine can break, too much strain, wears engine out

M55

Oral / Practical · Formative Only

Why do you want to keep all your cutters the same size and the angles all the same?

Key points:

- Ensures the saw cuts in a straight line — balanced load on all cutters
  Keywords: straight line, balanced, equal load, cuts straight, even load, balance, cuts in perfectly straight line, if one side longer saw cuts in curve, doesnt wander, doesnt curve, cuts evenly, no curve, no drift, straight cut, even cut, balanced cut
- Reduces vibration and wear — even distribution of cutting force
  Keywords: reduce vibration, less vibration, even wear, equal wear, smooth operation, reduce wear, stops saw from vibrating, every tooth does same amount of work
- Prolongs chain life
  Keywords: chain life, last longer, chain lasts longer, prolongs life, chain lasts much longer, extends chain life
- Safer and more predictable to use
  Keywords: safer, predictable, much safer, safe, predictable cutting, much safer to use, stops guide bar wearing on one side

M56

Oral / Practical · Formative Only

What problems might you get if you use a really worn or badly sharpened chain?

Key points:

- Increased risk of kickback
  Keywords: kickback, kick back, kick-back, increased kickback, saw might kick back, more kickback, kickback more easily
- Chain will not cut in a straight line — drifts to one side
  Keywords: drift, not cut straight, wander, pull to one side, doesn't cut straight, cuts in a curve, cuts sideways
- Overheating of the bar and chain
  Keywords: overheat, heat, bar overheat, chain overheat, excessive heat, engine will get very hot
- Increased operator effort
  Keywords: operator effort, more effort, fatigue, vibration, harder to cut, strain on operator, push really hard, have to push really hard, get very tired
- vibration and fatigue
  Keywords: operator effort, more effort, fatigue, vibration, harder to cut, strain on operator, push really hard, have to push really hard, get very tired
- Produces dust instead of chips — sign of poor sharpness
  Keywords: makes dust, fine dust, dust instead of chips, dust not chips, no wood chips, fine saw dust

M57

Oral / Practical · Formative Only

How do you dispose of your contaminated chainsaw waste and your litter?

Key points:

- Collect oil-contaminated waste separately — take to a licensed waste facility
  Keywords: contaminated waste, oil waste, licensed facility, licensed tip, waste facility, waste oil, separate waste, old dirty oil must go to special recycling, special recycling place, take to recycling
- Never bury
  Keywords: never bury, do not bury, not burn, do not burn, bury, incinerate, do not pour oil on ground, do not pour on ground, don't pour in river
- burn contaminated waste
  Keywords: never bury, do not bury, not burn, do not burn, bury, incinerate, do not pour oil on ground, do not pour on ground, don't pour in river
- Take all litter away from site — leave no trace
  Keywords: take litter, leave no trace, clean up site, remove litter, take away, take it home, never leave oily rags in woods, take everything away with you, keep woods clean

M58

Oral / Practical · Formative Only

What is the safe working distance between operators when cross cutting?

Key points:

- Two tree lengths
  Keywords: two tree length, two times, two tree, height of tree, tallest tree, two lengths, exclusion zone, twice the height of the tree, twice tree length, minimum 5 metres, 5 metres, 5m, twice the tree height, double the tree, two times the height, safe distance, far apart, well apart, plenty of space, keep well back
- two times the height of the tallest tree being worked on (minimum)
  Keywords: two tree length, two times, two tree, height of tree, tallest tree, two lengths, exclusion zone, twice the height of the tree, twice tree length, minimum 5 metres, 5 metres, 5m

M59

Oral / Practical · Formative Only

What bio-security measures might you put in place?

Key points:

- Clean and disinfect tools, equipment and footwear before and after entering a site
  Keywords: clean tools, disinfect, clean boots, footwear, clean equipment, clean and disinfect, wash equipment, clean all mud and sawdust off boots, wash your chainsaw and tools, clean before going to new forest
- Avoid moving potentially infected plant material off site
  Keywords: infected material, infected, plant material, move material, spread disease, not move, don't move sick wood to healthy forest, don't move infected wood
- Check for specific diseases
  Keywords: ash dieback, phytophthora, disease, pathogen, specific disease, chalara, tell an expert if you see very sick tree, check the rules for the forest
- pathogens relevant to the site (e.g. ash dieback, Phytophthora)
  Keywords: ash dieback, phytophthora, disease, pathogen, specific disease, chalara, tell an expert if you see very sick tree, check the rules for the forest
- Park and access on hard ground to reduce spread
  Keywords: hard ground, hard road, park on hard road, not in muddy woods, park on hard surface

M60

Oral / Practical · Formative Only

What environmental factors do you need to consider?

Key points:

- Wildlife and protected species — nesting birds
  Keywords: wildlife, protected species, nesting bird, bat, badger, ecology, nesting season, look out for birds nesting, birds nesting in the trees
- Bats
  Keywords: wildlife, protected species, nesting bird, bat, badger, ecology, nesting season, look out for birds nesting, birds nesting in the trees
- Badgers
  Keywords: wildlife, protected species, nesting bird, bat, badger, ecology, nesting season, look out for birds nesting, birds nesting in the trees
- Watercourses — prevent oil
  Keywords: watercourse, stream, pond, river, pollution, oil spill, fuel spill near water, don't let oil or gas spill into rivers, into rivers or streams, oil into river
- fuel pollution of streams and ponds
  Keywords: watercourse, stream, pond, river, pollution, oil spill, fuel spill near water, don't let oil or gas spill into rivers, into rivers or streams, oil into river
- Surrounding vegetation — avoid unnecessary damage to roots
  Keywords: vegetation, ground flora, roots, surrounding trees, ground damage, flora, rare plants, protect rare plants
- ground flora
  Keywords: vegetation, ground flora, roots, surrounding trees, ground damage, flora, rare plants, protect rare plants

- Ground conditions — minimise soil compaction and rutting
  Keywords: soil compaction, rutting, ground condition, compaction, soil damage, ground damage, don't ruin the ground, heavy trucks in mud
- Weather — high winds
  Keywords: weather, wind, high wind, rain, visibility, weather conditions, noise near houses, too much noise, noise near people
- Rain / heavy rain
  Keywords: weather, wind, high wind, rain, visibility, weather conditions, noise near houses, too much noise, noise near people
- visibility
  Keywords: weather, wind, high wind, rain, visibility, weather conditions, noise near houses, too much noise, noise near people
- Follow rules and regulations about when and what to cut
  Keywords: rules, regulations, allowed to cut, permission, felling licence, follow the rules, follow rules about when allowed to cut

M61

Oral / Practical · Formative Only

In this picture, where is the compression found?

Key points:

- On the upper, on top
  Keywords: top, upper, above, top side, upper side, on top, compression is on top, squashed on top, squashed together on top, top is being squashed, top part, top bit, compressed on top
- top side of the timber (for a supported log)
  Keywords: top, upper, above, top side, upper side, on top, compression is on top, squashed on top, squashed together on top, top is being squashed

M62

Oral / Practical · Formative Only

And in this picture, where is the compression?

Key points:

- On the underside / bottom of the timber (for a cantilevered
  Keywords: bottom, underside, below, under, lower side, underneath, compression underneath, squashed on the bottom, bottom is being squashed, bottom is squashed, on bottom, bottom part, bottom bit, squashed underneath
- hanging log)
  Keywords: bottom, underside, below, under, lower side, underneath, compression underneath, squashed on the bottom, bottom is being squashed, bottom is squashed

M63

Oral / Practical · Formative Only

What are you going to do if you get your saw stuck or pinched?

What are you going to do if you get your saw stuck, or pinched?

Key points:

- Do not force the saw — switch it off
  Keywords: do not force, switch off, turn off, don't force, stop the saw, kill switch, turn the chainsaw off immediately, first turn it off, switch it off immediately, dont force, dont pull, dont tug, dont yank, leave it alone, dont struggle, dont fight it
- Use a felling wedge
  Keywords: wedge, felling wedge, bar, lever, open the cut, use a wedge, hammer a plastic wedge, plastic wedge, use a strong stick as a lever, lift the wood
- bar to open the cut
  Keywords: wedge, felling wedge, bar, lever, open the cut, use a wedge, hammer a plastic wedge, plastic wedge, use a strong stick as a lever, lift the wood
- Assess the tension before attempting to free the saw
  Keywords: assess tension, check tension, tension, assess, identify tension, before freeing, read the tension in the wood
- Never leave the saw running unattended in the timber
  Keywords: never leave running, do not leave running, unattended, not leave, saw running in wood, never try to cut saw out while still running
- Beware of log rolling
  Keywords: log rolling, rolling onto you, careful log doesn't roll, when it comes free, log roll, timber moving, log moves
- moving when freed
  Keywords: log rolling, rolling onto you, careful log doesn't roll, when it comes free, log roll, timber moving, log moves

M64

Oral / Practical · Formative Only

How are you going to cross cut a piece of timber that's slightly larger than your guidebar?

Key points:

- Cut from one side then reposition and cut from the opposite side to meet in the middle
  Keywords: one side, opposite side, cut from both sides, reposition, both sides, meet in middle, first cut down from one side, then walk around, walk around to other side, cut from both sides
- Ensure the two cuts align — use a straight edge
  Keywords: align, score, mark, straight edge, guide, ensure cuts meet, line saw up with first cut, line up carefully
- score the timber first
  Keywords: align, score, mark, straight edge, guide, ensure cuts meet, line saw up with first cut, line up carefully
- Or use a bore cut to plunge through the timber
  Keywords: bore cut, bore cutting, plunge cut, push the tip straight through, bore cut technique, bore through

M65

Oral / Practical · Formative Only

How are you going to cut timber that's under high tension?

Key points:

- Identify compression and tension zones before cutting
  Keywords: identify compression, identify tension, compression zone, tension zone, assess before cutting, high tension means wood bent like bow
- Overbuck from the top (compression side)
  Keywords: overbuck, underbuck, top cut, undercut, cut from above, cut from below, shallow cut on compression side first, relief cut on squashed side, main cut from tension side
- underbuck from below (tension side) as appropriate
  Keywords: overbuck, underbuck, top cut, undercut, cut from above, cut from below, shallow cut on compression side first, relief cut on squashed side, main cut from tension side
- Make a relief cut first to release tension
  Keywords: relief cut, release tension, relieve tension, first cut, relief notch, small shallow cut first, small cut first
- Be prepared for sudden movement of the timber
  Keywords: sudden movement, timber movement, movement, snap, jump, spring back, move suddenly, stand well back, be ready to move quickly, wood snaps suddenly

M66

Oral / Practical · Formative Only

When would you use a bore cut when cross cutting timber?

Key points:

- Timber diameter is larger than the guidebar — cannot cut through in one pass
  Keywords: larger than, bigger than, longer bar, too large, diameter larger, can't cut through, one pass, timber too big for bar, log bigger than guidebar
- Timber under tension — to prevent the bar being pinched
  Keywords: tension, pinch, trapped, under tension, prevent pinch, stop bar getting trapped, avoid trapping, heavy tension, stop wood from splitting wildly, barber-chairing
- Trapped bar / pinch
  Keywords: tension, pinch, trapped, under tension, prevent pinch, stop bar getting trapped, avoid trapping, heavy tension, stop wood from splitting wildly, barber-chairing
- Working in confined space — to avoid the kickback zone at the nose
  Keywords: confined, tight space, kickback zone, nose, tip of bar, restricted, nose kickback, avoid kickback at nose, confined space
- Timber on
  Keywords: near ground, on the ground, floor, minimal access, on ground, near or on ground, floor is in the way, tension cut blocked, cant access tension, can't access tension, access near ground, lying near ground, close to ground, access to tensioned side
- near the ground — floor blocks access to the tension side; minimal access
  Keywords: near ground, on the ground, floor, minimal access, on ground, near or on ground, floor is in the way, tension cut blocked, cant access tension, can't access tension, access near ground, lying near ground, close to ground, access to tensioned side

M67

Oral / Practical · Formative Only

Once you've cut your timber, how are you going to move it safely?

Key points:

- Use appropriate tools — cant hook, timber jack,
  Keywords: cant hook, timber jack, cant dog, peavey, mechanical aid, log moving tool, timber tongs, log hook, special tools, use tools, use equipment, mechanical aids, log tools, moving tools, dont lift by hand
- cant dog
  Keywords: cant hook, timber jack, cant dog, peavey, mechanical aid, log moving tool, timber tongs, log hook, special tools
- Assess the weight — team lift if too heavy, correct manual handling posture
  Keywords: assess weight, team lift, manual handling, correct posture, back straight, not too heavy, think before you lift, don't just grab it, ask a friend to help, bend your knees
- Clear a safe path before moving timber
  Keywords: clear path, safe path, clear before moving, clear a path to walk, clear safe path, look before carrying
- Wear gloves to protect from splinters
  Keywords: gloves, wear gloves, tough gloves, protect from splinters, chainsaw gloves

M68

Oral / Practical · Formative Only

How do you make moving and working more ergonomic?

Key points:

- Use the correct tools and mechanical aids to reduce manual effort
  Keywords: correct tools, mechanical aid, cant hook, reduce manual, use equipment, right tool, use special tools
- Plan your movements — minimise unnecessary handling and twisting
  Keywords: plan movements, minimise handling, avoid twisting, plan, unnecessary lifting, minimise, don't twist your back, turn your feet instead of twisting spine, keep back straight
- Take regular breaks to avoid fatigue
  Keywords: take breaks, regular break, rest, avoid fatigue, fatigue, rest period, stretch muscles, listen to your body
- Keep the load close to your body and use your legs
  Keywords: close to body, use your legs, leg muscles, bend knees, knees bent, legs do the work, let strong leg muscles do the work

M69 Oral / Practical · Formative Only

How are you going to make sure your timber stacks are stable?

Key points:

- Use stacking stakes
  Keywords: stacking stake, binder, stake, secure stack, binding, stacking binder, cross the logs over each other at corners, interlock, lock them in, stakes, binders, straps, tie them, secure logs, hold stack together
- binders — secure the stack
  Keywords: stacking stake, binder, stake, secure stack, binding, stacking binder, cross the logs over each other at corners, interlock, lock them in
- Stack on firm
  Keywords: firm ground, level ground, flat ground, stable ground, solid base, build stack on flat level ground, level flat ground
- level ground
  Keywords: firm ground, level ground, flat ground, stable ground, solid base, build stack on flat level ground, level flat ground
- Do not stack too high — keep within a safe and manageable height
  Keywords: not too high, safe height, manageable height, height limit, stack height, don't stack too high, might fall over, fall over
- Use bearers at the base and heaviest logs at the bottom
  Keywords: bearers, base, bearer, base logs, strong straight logs at bottom, heaviest logs at bottom, big heavy logs at bottom
- Lean the stack slightly inwards and ensure it won't roll
  Keywords: leans inward, lean inwards, inwards not outwards, won't roll away, won't roll, slightly inwards, stable lean

M70 Oral / Practical · Formative Only

What else do you need to think about when stacking lots of different timber?

Key points:

- Sort and stack by species
  Keywords: sort, species, length, size, identification, same length, group by, sort logs by size, separate different types, sort by size so they stack neatly, group by type, same wood together, organise by kind, separate types, keep same together
- Sort by length / size
  Keywords: sort, species, length, size, identification, same length, group by, sort logs by size, separate different types, sort by size so they stack neatly
- size for ease of identification
  Keywords: sort, species, length, size, identification, same length, group by, sort logs by size, separate different types, sort by size so they stack neatly
- Bio-security — do not mix potentially infected species
  Keywords: bio-security, biosecurity, infected, mix species, different sites, disease spread
- Different sites / locations
  Keywords: bio-security, biosecurity, infected, mix species, different sites, disease spread
- Access for machinery — leave sufficient space for vehicles
  Keywords: access, machinery, vehicle, space, room for, equipment access, forwarder, don't block any gates or roads, think about how to get wood out again, getting wood out again
- Equipment / machinery access
  Keywords: access, machinery, vehicle, space, room for, equipment access, forwarder, don't block any gates or roads, think about how to get wood out again, getting wood out again
- Stability considerations — heavier
  Keywords: stability, heavy at bottom, base, heavier, longer at base, stable stack, heavy thick logs at bottom, smaller lighter branches on top
- longer pieces at the base
  Keywords: stability, heavy at bottom, base, heavier, longer at base, stable stack, heavy thick logs at bottom, smaller lighter branches on top
- Leave gaps for airflow — allows timber to dry
  Keywords: gaps, airflow, dry out, ventilation, leave gaps, air to blow through, gaps for air, wood dries out
- Cover the top to protect from rain
  Keywords: cover, rain, cover top, protect from rain, cover the top to keep rain off, protect from weather

M71 Oral / Practical · Formative Only

Before you start your saw, what checks do you need to do?

Key points:

- Check fuel level and bar oil level
  Keywords: fuel, fuel level, oil level, bar oil, check fuel, check oil, enough gas and chain oil, fuel and oil in the tanks, enough petrol, enough gas, gas in tank, petrol in tank, oil in tank, bar oil full, chain oil, lubrication oil, fuel and oil
- Check bar and chain condition — tension
  Keywords: bar, chain, tension, sharp, lubrication, bar condition, chain condition, chain tension, chain is tightened correctly, check chain tension, chain tight, chain loose, bar ok, bar good, chain good, chain sharp, check sharpness
- Check chain sharpness
  Keywords: bar, chain, tension, sharp, lubrication, bar condition, chain condition, chain tension, chain is tightened correctly, check chain tension
- lubrication
  Keywords: bar, chain, tension, sharp, lubrication, bar condition, chain condition, chain tension, chain is tightened correctly, check chain tension
- Check all nuts and bolts are tight
  Keywords: nuts, bolts, tight, secure, fixings, tighten, not loose, loose screws, check for loose screws, everything tight

- Check all safety features are present and working — chain brake, guards
  Keywords: safety features, chain brake, guards, present, working, functional, chain brake handle clicks, chain brake clicks forward
- Check PPE is correct and in good condition
  Keywords: ppe, personal protective, chainsaw trousers, helmet, visor, gloves, boots, protective equipment, wearing all your safety clothes
- Check the area is clear of hazards and bystanders
  Keywords: area clear, exclusion zone, bystander, hazard free, clear area, people clear, site clear, look around to make sure area is safe, no one standing too close
- Check the air filter is clean
  Keywords: air filter clean, filter clean, clean air filter, air filter check, filter checked

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M72

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What functional checks do you need to perform before you actually use your saw?

Key points:

- Start the saw and test the chain brake — manual activation and inertia
  Keywords: start the saw, test chain brake, chain brake, manual, inertia, test brake, push chain brake forward, gently rev engine with brake on, chain must not move, check brake works, test the brake, brake check, make sure brake works, check safety brake, try the brake, brake test
- Check the chain does not move at idle speed
  Keywords: idle, chain doesn't move, chain not move, idle speed, chain at idle, not moving at idle, idles nicely without chain moving, chain still at idle
- Test the throttle trigger lockout
  Keywords: throttle lockout, trigger lockout, throttle trigger, lockout, trigger test, press safety trigger lock, can't press gas without it, safety trigger check
- Check the chain oiler is working — bar oil reaching the chain
  Keywords: chain oiler, oiler, oil reaching, bar oil, oil flow, oiling system, oil throwing, hold over clean wood and rev to see oil spray, oil spray off chain, oil sprays off
- Check the anti-vibration system is functioning
  Keywords: anti-vibration, vibration, av system, vibration system, anti vibration, vibration check
- Check the stop switch turns the engine off instantly
  Keywords: stop switch, kill switch, turns engine off instantly, engine off instantly, check stop switch, switch off test
- Check for unusual noises
  Keywords: unusual noise, strange noise, rattle, rattling, strange rattling, funny noise, no strange noises
- Check for rattles or unusual noises
  Keywords: unusual noise, strange noise, rattle, rattling, strange rattling, funny noise, no strange noises
- Perform a test cut to ensure everything feels correct
  Keywords: test cut, tiny test cut, try a cut, test cut feels right, test cut in wood, small test cut

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M73

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What bit of the bar do you need to avoid cutting with?

Key points:

- Upper quadrant
  Keywords: upper quadrant, top quarter, nose, tip, kickback zone, upper part of tip, danger zone, top of the bar nose, upper nose, bar nose, avoid the tip, rounded tip, the tip of the bar, kickback danger zone, upper part of the rounded tip, upper part of the tip, top of the nose, top of the rounded tip, nose contact
- nose of the bar — the kickback danger zone
  Keywords: upper quadrant, top quarter, nose, tip, kickback zone, upper part of tip, danger zone, top of the bar nose, upper nose, bar nose, avoid the tip, rounded tip, the tip of the bar, kickback danger zone, upper part of the rounded tip, upper part of the tip, top of the nose, top of the rounded tip, nose contact

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M74

Oral / Practical · Formative Only

What do the Manual Handling Operations Regulations 1992 (MHOR) require you to do before manually moving heavy timber on site?

What do the Manual Handling Operations Regulations 1992 require you to do before manually moving heavy timber on site?

Key points:

- Avoid manual handling where reasonably practicable — use mechanical aids
  Keywords: avoid, mechanical aid, cant hook, timber jack, avoid manual handling, use tools, use equipment, reasonably practicable, avoid it if you can, use mechanical aids, dont lift by hand
- Assess the load — weight, shape, distance to be moved
  Keywords: assess, assessment, assess the load, assess weight, weight, shape, distance, assess the task, assess the risk, look at it first, think about it
- Reduce the risk of injury — correct posture, bend knees, back straight
  Keywords: reduce risk, correct posture, bend knees, back straight, knees bent, posture, reduce injury, safe posture, lift safely, safe lifting, back upright
- Team lift if the load is too heavy for one person
  Keywords: team lift, two person lift, ask for help, team, two people, get help, dont lift alone, lift together
- Never twist — turn your feet instead
  Keywords: never twist, dont twist, turn feet, feet not spine, no twisting, pivot your feet, avoid twisting

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M75

Oral / Practical · Formative Only

What PPE do you need when using and operating a chainsaw?

Key points:

- Safety helmet
  Keywords: helmet, hard hat, safety helmet, head protection, chainsaw helmet, forestry helmet, strong safety helmet, head from falling branches, helmet to protect head
- hard hat — head protection from falling branches
  Keywords: helmet, hard hat, safety helmet, head protection, chainsaw helmet, forestry helmet, strong safety helmet, head from falling branches, helmet to protect head
- Ear defenders
  Keywords: ear defenders, earplugs, hearing protection, ear protection, ear muffs, earmuffs, ear defenders or earplugs, stop the loud noise, protect hearing, ear, ear protection, ear muffs, hearing protection, ears
- earplugs — hearing protection
  Keywords: ear defenders, earplugs, hearing protection, ear protection, ear muffs, earmuffs, ear defenders or earplugs, stop the loud noise, protect hearing, ear
- Safety glasses
  Keywords: safety glasses, visor, mesh visor, face shield, eye protection, goggles, face protection, safety glasses or mesh visor, stop wood chips hitting eyes
- visor / mesh visor — eye protection
  Keywords: safety glasses, visor, mesh visor, face shield, eye protection, goggles, face protection, safety glasses or mesh visor, stop wood chips hitting eyes
- Chainsaw trousers
  Keywords: trousers, chainsaw trousers, chainsaw pants, protective trousers, leg protection, cut-resistant trousers, kevlar trousers, chainsaw chaps, special trousers with fibres, fibres to jam the saw, special chainsaw trousers
- chaps — leg protection with cut-resistant fibres
  Keywords: trousers, chainsaw trousers, chainsaw pants, protective trousers, leg protection, cut-resistant trousers, kevlar trousers, chainsaw chaps, special trousers with fibres, fibres to jam the saw, special chainsaw trousers
- Chainsaw boots — foot protection with steel toecap and ankle protection
  Keywords: boots, chainsaw boots, safety boots, steel toe boots, protective boots, chainsaw footwear, tough chainsaw boots, steel toes and good grip, steel toe caps, foot protection
- Chainsaw gloves — hand and wrist protection
  Keywords: gloves, chainsaw gloves, hand protection, protective gloves, strong gloves, extra protection on left hand, gloves with protection
- Hi-visibility clothing — so others can see you
  Keywords: hi-vis, high visibility, bright clothing, fluorescent, visible, hi vis jacket, bright colors, wear bright colors, can be seen in the woods
- Helmet with visor and ear defenders combined
  Keywords: combined helmet, helmet with visor, helmet visor and ear, all in one helmet, forestry helmet with visor, combined protection, helmet, hard hat, head protection, visor

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M76

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What do the Manual Handling Operations Regulations 1992 require employers to do?

What do the Manual Handling Operations Regulations 1992 — MHOR — require employers to do?

Key points:

- Avoid hazardous manual handling operations where reasonably practicable
  Keywords: avoid, avoid manual handling, avoid lifting, reasonably practicable, avoid where possible, mechanise, don't lift if you don't have to, eliminate manual handling, avoid the manual handling, get machinery to do it, use equipment instead
- Assess the risk of injury from manual handling that cannot be avoided
  Keywords: assess, risk assessment, assess the risk, assess injury, evaluate, assess manual handling risk, check how heavy it is, look at the risk, assess before lifting, assess the load, risk assess, evaluate risk, check risk
- Reduce the risk of injury to the lowest reasonably practicable level
  Keywords: reduce risk, reduce injury, lower risk, minimise risk, reduce the risk, control measures, reduce harm, make it safer, lower the risk of injury, lower chance of getting hurt, take steps to reduce, mitigate, risk reduction
- Use mechanical aids such as log jacks, winches, or machinery where possible
  Keywords: mechanical, log jack, winch, forwarder, machinery, mechanical aid, equipment, use tools, use machines, use a machine, lifting equipment, tractor, crane, use log jack, avoid heavy lifting by using equipment
- Consider the task, load, environment, and individual capability (TILE)
  Keywords: tile, task, load, environment, individual, person, capability, tile framework, task load environment individual, consider the load weight, consider the person doing it, consider where you are lifting, think about all factors

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M77

Oral / Practical · Formative Only

How do the Manual Handling Operations Regulations 1992 apply to log handling and timber stacking?

How do the Manual Handling Operations Regulations 1992 apply specifically to log handling and timber stacking on site?

Key points:

- Assess log weight and size before manually lifting — use log jacks or winches for heavy timber
  Keywords: assess, log weight, log size, weight, heavy, log jack, winch, check weight, assess before lifting, heavy logs, assess the log, check how heavy, too heavy to lift by hand, mechanical aid, use log jack, use winch
- Stand uphill of logs when stacking to prevent rolling injury
  Keywords: uphill, stand uphill, roll, rolling log, prevent rolling, uphill side, timber rolling, log roll, stand on the uphill side, logs can roll, chock logs, stop logs rolling

- Use correct lifting technique — bend knees, straight back, load close to body
  Keywords: bend knees, straight back, close to body, correct technique, lift correctly, proper lift, back straight, knees bent, keep load close, body close, manual handling technique, how to lift correctly, proper manual handling, correct posture
- Do not lift loads that are too heavy alone — team lift or use machinery
  Keywords: team lift, two people, two person lift, do not lift alone, ask for help, team, machinery, another person, buddy, two of you, too heavy for one person, get help lifting, never lift alone if too heavy
- Stacking logs at a safe height to reduce overreach and crush risk
  Keywords: stacking height, safe height, stack safely, stack height, do not stack too high, logs falling, crush risk, collapse, safe stack, stack at waist height, not too high, stacking at the correct height, log stack

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With all checks done and everything in good condition and working correctly, you are ready to undertake the practical assessment.

For the cutting part of the assessment you will need to demonstrate multiple cuts under tension and compression and perform bore cuts.

Good luck.

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